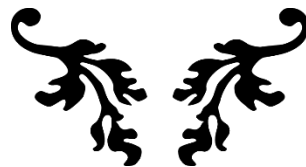




LJ University
University with a Difference



NOTES OF NETWORK SECURITY & MANAGEMENT



LJ Polytechnic

**Prepared and Compiled by
CE & IT Department**

STUDENT ADVISORY

Dear Students,

Please be informed that the notes provided by the institute offer a concise presentation of the syllabus. While these notes are helpful for an overview and quick revision, We would strongly suggest that you refer to the prescribed textbooks / Reference book for a comprehensive understanding and thorough preparation of all exams and writing in the examination.

Best regards,

LJ Polytechnic.

પ્રિય વિદ્યાર્થીઓ,

તમને જાણ કરવામા આવે છે કે સંસ્થા દ્વારા પ્રદાન કરવામાં આવેલી નોંધો અભ્યાસક્રમની સંક્ષિપ્ત પ્રસ્તુતિ આપે છે. આ નોંધો વિહંગાવલોકન અને ઝડપી પુનરાવર્તન માટે મદદરૂપ હોઈ શકે છે તેમ છતાં, અમે ભારપૂર્વક સૂચન કરીએ છીએ કે વિદ્યાર્થી તમામ પરીક્ષાઓ અને પરીક્ષામાં લેખનની વ્યાપક સમજણ અને સંપૂર્ણ તૈયારી માટે માત્ર સૂચવેલા પાઠ્યપુસ્તકો/સંદર્ભ પુસ્તકનો સંદર્ભ લો.

એલજે પોલિટેકનિક.

UNIT-1

INTRODUCTION TO SECURITY MECHANISMS

1.1 VARIOUS SECURITY TERMS

1.1.1 Introduction

In today's digital world, computer networks form the backbone of communication, data sharing, and business operations. With this growing reliance on interconnected systems comes an increasing need to protect them against threats and ensure their efficient management. **Network Security and Management** is a crucial field that addresses both the protection and the control of network infrastructure.

Network Security focuses on protecting data during transmission and safeguarding the infrastructure from unauthorized access, misuse, or damage. It involves principles like confidentiality, integrity, authentication, and availability, implemented through firewalls, encryption, intrusion detection systems, and security protocols.

Network Management, on the other hand, ensures the smooth functioning of networks through tasks such as configuration, monitoring, performance tuning, fault detection, and troubleshooting. It is essential for maintaining optimal network performance, minimizing downtime, and supporting organizational needs.

Computer Security: The generic name for the collection of tools designed to protect data and prevent hackers is Computer Security.

Network Security: Network Security refers to the measures taken by any enterprise or organization to secure its computer network and data using both hardware and software systems. Network security measures are needed to protect data during their transmission.

Internet Security: Internet security refers to security designed to protect systems and the activities of employees and other users while connected to the internet, web browsers, web apps, websites and networks. Internet security solutions protect users and corporate assets from cybersecurity attacks and threats.

Threat: A potential for violation of security, which exists when there is a circumstance, capability, action, or event that could breach security and cause harm. That is, a threat is a possible danger that might exploit a vulnerability.

Attack: An attack is an information security threat that involves an attempt to obtain, alter, destroy, remove, implant or reveal information without authorized access or permission.

1.1.2 Virus

A virus is a type of code that enters the system along with any file or program and carries out malfunctions in the system. The virus-affected program will be a replica of the existing program. They enter the system through any file and when the file runs, parallelly the virus also runs in the background.

There are many ways in which the virus gets into the system. Some of them are through mail attachments, by clicking inappropriate advertisements and by downloading any software or files from unauthorized websites.

The main objective of viruses is to spread them along different hosts. They steal the personal data and other credentials of the system. Various types of viruses are explained as follows:

- 1) **Parasitic Virus:** The traditional and still most common form of virus. A parasitic virus attaches itself to executable files and replicates, when the infected program is executed, by finding other executable files to infect.
- 2) **Memory-Resident Virus:** Lodges in main memory as part of a resident system program. From that point on, the virus infects every program that executes.
- 3) **Boot Sector Virus:** Infects a master boot record or boot record and spreads when a system is booted from the disk containing the virus.
- 4) **Stealth Virus:** A form of virus explicitly designed to hide itself from detection by antivirus software.
- 5) **Polymorphic Virus:** A virus that mutates with every infection, making detection by the "signature" of the virus impossible.
- 6) **Metamorphic Virus:** As with a polymorphic virus, a metamorphic virus mutates with every infection. The difference is that a metamorphic virus rewrites itself completely at each iteration, increasing the difficulty of detection. Metamorphic viruses may change their behavior as well as their appearance.

How can a computer be protected from viruses?

- 1) **Install Antivirus or Anti-Malware Software**
 - Use reliable antivirus software (e.g., Bitdefender, Norton, Kaspersky, Windows Defender).
 - Keep it updated and run regular system scans.
- 2) **Keep Your Operating System and Software Updated**
 - Install security patches and updates as soon as they are available.
 - Vulnerabilities in outdated software are a major entry point for viruses.
- 3) **Use a Firewall**
 - Enable the built-in firewall on your system or use a third-party one.
 - A firewall monitors network traffic and blocks suspicious activity.
- 4) **Avoid Suspicious Links and Attachments**
 - Don't open email attachments or click links from unknown or untrusted sources.
 - Phishing emails are a common virus delivery method.
- 5) **Download Only from Trusted Sources**
 - Use official websites and trusted platforms (e.g., Microsoft Store, Apple App Store).
 - Avoid downloading cracked software or pirated files.
- 6) **Use Strong Passwords and Enable Two-Factor Authentication (2FA)**
 - Prevent unauthorized access that could lead to virus installation.

7) Backup Important Data Regularly

- Use external drives or cloud backups (e.g., Google Drive, OneDrive).
- In case of infection (e.g., ransomware), you can restore your data safely.

8) Disable Auto-Run for External Devices

- Prevent viruses on USB drives from automatically running when connected

9) Be Cautious with Public Wi-Fi

- Use a VPN when using unsecured networks to protect data and prevent intrusion.

10) Educate Yourself and Others

- Awareness of social engineering tactics and virus threats greatly reduces risk.

1.1.3 Intruder

An intruder (also called a hacker) is an individual who performs security attacks on another's domain in a networked computing environment. The intruder may attempt to read privileged data (like password cracking), perform unauthorized modification of data or disrupt normal functioning of data. There are three types of intruders:

- 1) **Masquerader:** An individual who is not authorized to use the computer and who penetrates a system's access controls to exploit a legitimate user's account. The masquerader is likely to be an outsider
- 2) **Misfeasor:** A legitimate user who accesses data, programs, or resources for which such access is not authorized or who is authorized for such access but misuses his or her privileges. Misfeasor is an insider.
- 3) **Clandestine User:** An individual who seizes supervisory control of the system and uses this control to evade auditing and access controls or to suppress audit collection. Clandestine users can be either insiders or outsiders.

1.2 SECURITY BASICS

1.2.1 Pillars of Information Security

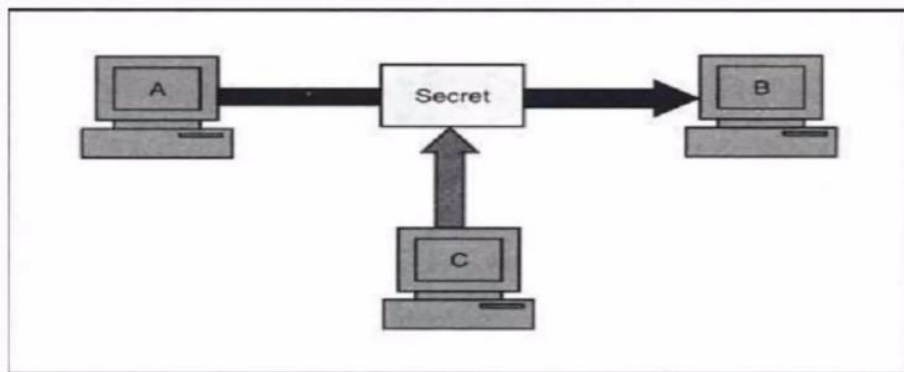
It is necessary to protect information from being stolen, compromised or attacked. The CIA triad is one of the most important models that is designed to guide policies for information security within an organization. **CIA** stands for **Confidentiality, Integrity and Availability**. These are considered as three pillars of Information Security.



1) Confidentiality:

Confidentiality means that only authorized individuals/systems can view sensitive or classified information. The data being sent over the network should not be accessed by unauthorized individuals. The attacker may try to capture the data using different tools available on the Internet and gain access to your information.

For Example: User A sends a message to User B. Another User C gets access to this message which is not desired and therefore defeats the purpose of confidentiality. This type of attack is called **Interception** and we can say that confidentiality of the message is lost.



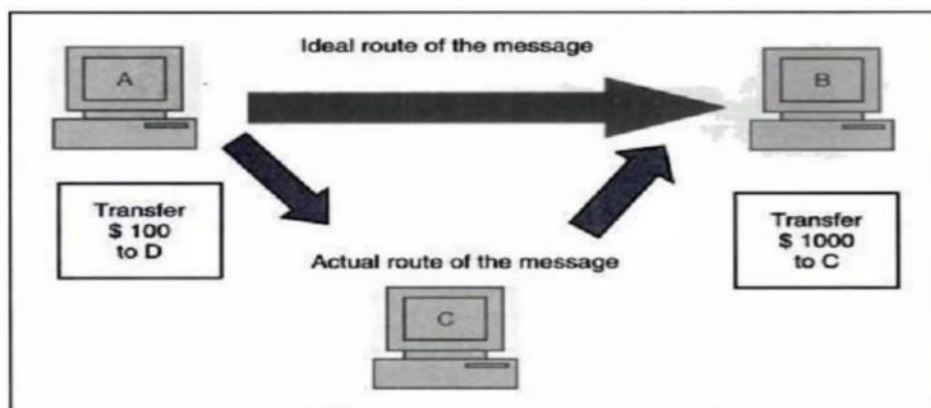
(Figure: Loss of Confidentiality)

To fight against confidentiality breaches, you can classify and label restricted data, enable access control policies, encrypt data, and use multi-factor authentication (MFA) systems. It is also advisable to ensure that all in the organization have the training and knowledge they need to recognize the dangers and avoid them.

2) Integrity:

Integrity makes sure that data has not been modified.

For Example, User A wants to send a message to User B. User C somehow manages to access the data of User A changes its contents and sends a changed message to User B. Users A and B have no idea that the contents of the message were changed. This type of attack is called **Modification** and we can say that the integrity of the message is lost.



(Figure: Loss of Integrity)

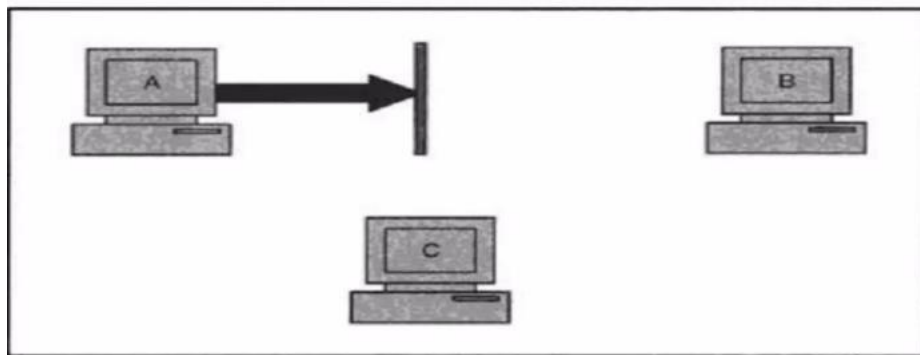
To protect the integrity of your data, you can use hashing, encryption, digital certificates, or digital signatures. For websites, you can employ trustworthy certificate authorities (CAs) that verify the authenticity of your website so visitors know they are getting the site they intended to visit.

3) Availability:

Availability means that the network should be readily available to its users.

This applies to systems and data. To ensure availability, the network administrator should maintain hardware, make regular upgrades, have a plan for fail-over and prevent bottlenecks in a network.

For Example: Due to the intentional actions of unauthorized User C, an authorized User A may not be able to contact server computer B. This type of attack is called **Interruption**. Thus, proper measures should be taken to prevent such attacks.



(Figure: Loss of Availability)

To ensure availability, organizations can use redundant networks, servers and applications. These can be programmed to become available when the primary system has been disrupted or broken. You can also enhance availability by staying on top of upgrades to software packages and security systems. Backups and full disaster recovery plans also help a company regain availability soon after a negative event.

Example: Online Banking System

1) Confidentiality

- **Scenario:** When a user logs into their bank account, data such as login credentials and account details are encrypted using SSL/TLS.
- **Purpose:** Prevents eavesdroppers (like hackers on public Wi-Fi) from reading sensitive information.

2) Integrity

- **Scenario:** When you transfer ₹10,000 or \$500 to someone, checksums and hashing are used to ensure the data is not altered in transit.
- **Purpose:** Ensures the transferred amount isn't changed by a man-in-the-middle attack. You send ₹10,000, and ₹10,000 is received—not more or less.

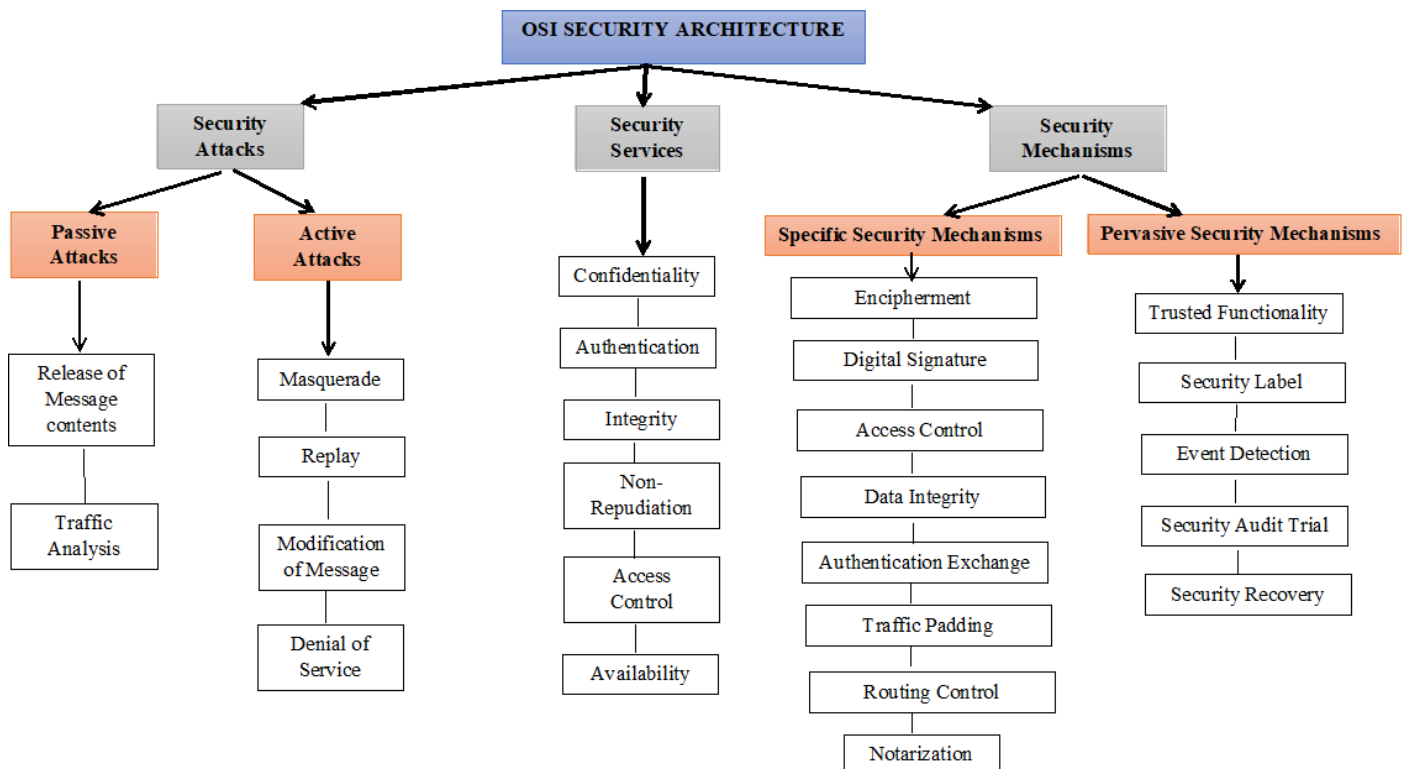
3) Availability

- **Scenario:** Banks implement redundant servers, load balancing, and DDoS protection to keep their website and app always accessible to users.
- **Purpose:** Ensures that customers can access their accounts at any time—even during high traffic or attempted cyberattacks.

1.2.2 The OSI Security Architecture

The OSI (Open Systems Interconnection) security architecture provides a systematic framework for defining security attacks, security mechanisms and security services. As this architecture was developed as an international standard, various vendors have developed security features for their products and services that relate to this structured definition of services and mechanisms.

- **Security Attacks:** Any action that compromises the security of information owned by an organization.
- **Security Mechanisms:** A process (or a device incorporating such a process) that is designed to detect, prevent or recover from a security attack.
- **Security Services:** A processing or communication service that enhances the security of the data processing systems and the information transfers of an organization. The services are intended to counter security attacks, and they make use of one or more security mechanisms to provide the service.



Security Attacks:

Security Attacks are of two types:

- 1) **Passive Attack**
- 2) **Active Attack**

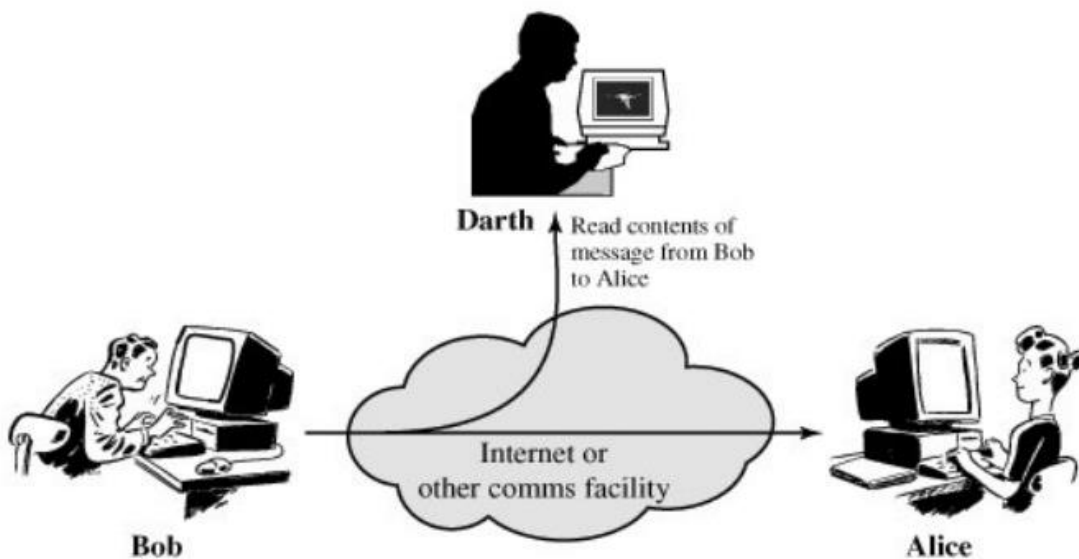
Passive Attack:

In a passive attack, the attacker attempts to learn or make use of information from the system but does not affect system resources. The goal of the opponent/attacker is to obtain information that is transmitted. Passive attacks are of two types:

- a) **Release of message contents (Eavesdropping)**
- b) **Traffic analysis**

a) Release of Message Contents (Eavesdropping):

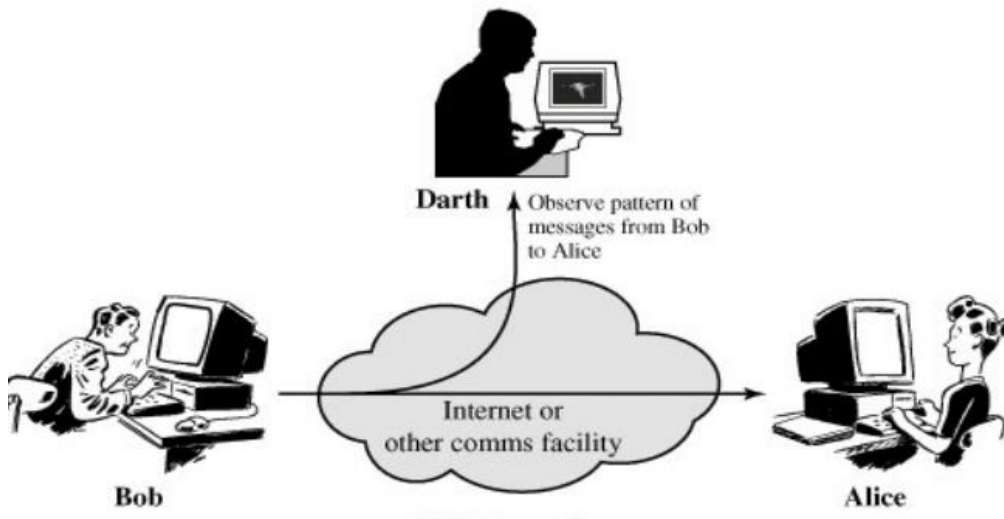
It involves the attacker intercepting and listening to communications between two or more parties without their knowledge or consent. This attack can be performed using a variety of techniques, such as packet sniffing, or man-in-the-middle attacks.



b) Traffic Analysis:

This involves the attacker analyzing network traffic patterns and metadata to gather information about the system, network, or device. Here the intruder can't read the message but only understand the pattern and length of encryption. Traffic analysis can be performed using a variety of techniques, such as network

low analysis, or protocol analysis.



Passive attacks are very difficult to detect because they do not involve any alteration of the data. Typically, the message traffic is sent and received in a normal fashion and neither the sender nor receiver is aware that a third party has read the messages or observed the traffic pattern. However, it is feasible to prevent the success of these attacks, usually using encryption. Thus, the emphasis in dealing with passive attacks is on prevention rather than detection.

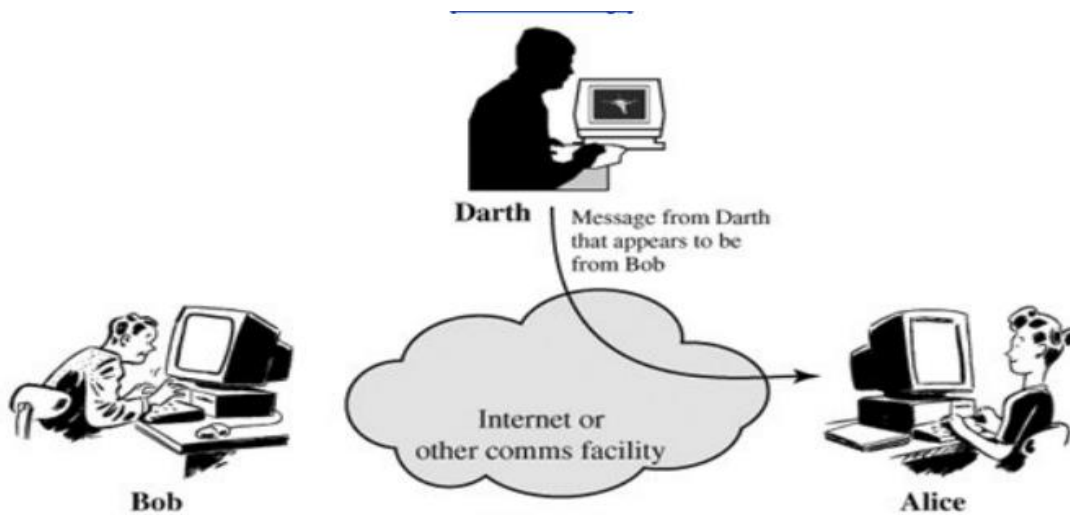
Active Attack:

Active attacks refer to types of attacks that involve the attacker actively disrupting or altering system, network or device activity. Active attacks are typically focused on causing damage or disruption rather than gathering information or intelligence. Here, both the sender and receiver have no clue that their message/ data is modified by some third-party intruder. The message/ data transmitted doesn't remain in its usual form and shows deviation from its usual behavior. This makes active attacks dangerous as there is no information provided of the attack happening in the communication process and the receiver is not aware that the data/ message received is not from the sender. It is further divided into 4 types:

- a) Masquerade
- b) Replay
- c) Modification of Messages
- d) Denial of Service

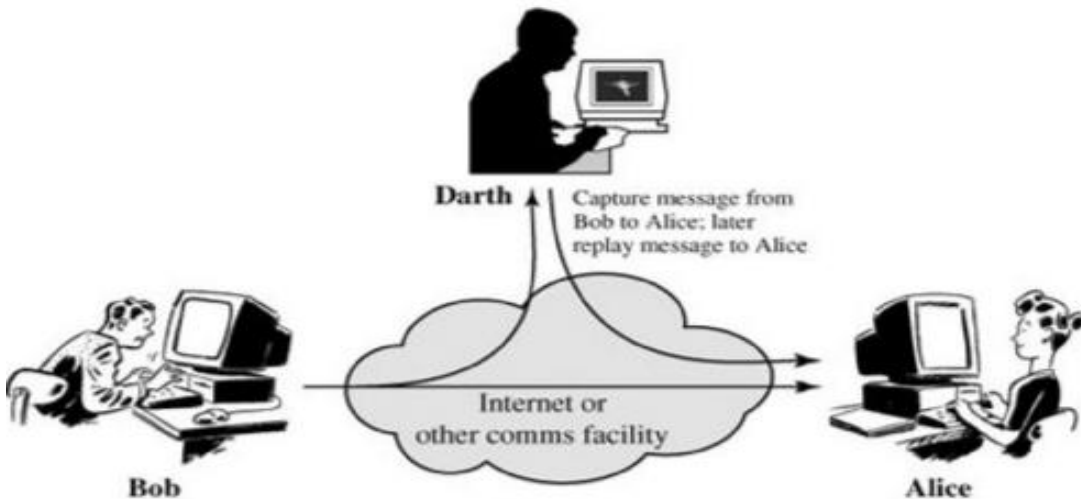
a) Masquerade:

A masquerade takes place when one entity pretends to be a different entity in order to gain unauthorised access to a system. This type of attack can involve the attacker using stolen or forged credentials, or manipulating authentication or authorization controls in some other way.



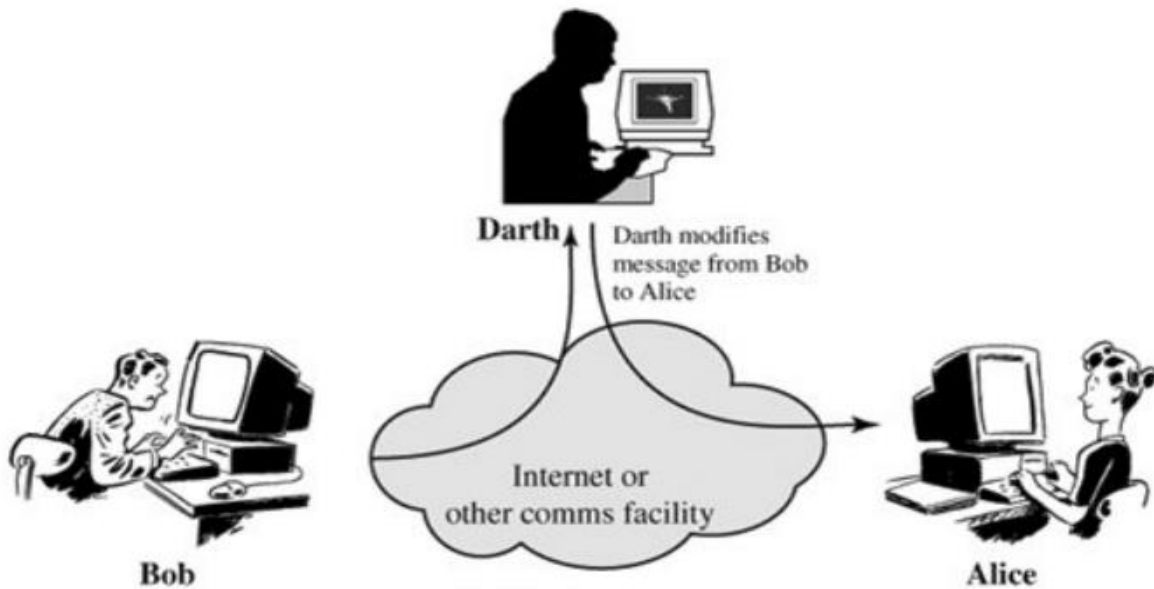
b) Replay:

Replay involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect.



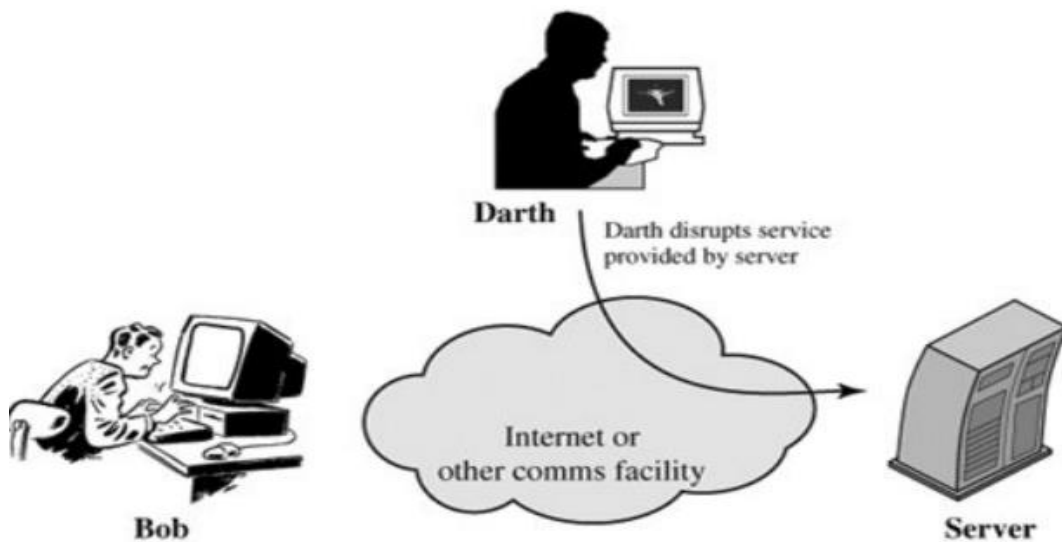
c) Modification of Message:

Modification of messages simply means that some portion of a legitimate message is altered or that messages are delayed or reordered, to produce an unauthorized effect. For example, a message meaning "Allow John Smith to read confidential file accounts" is modified to mean "Allow Fred Brown to read confidential file accounts."



d) Denial of Service:

The denial of service prevents or inhibits the normal use or management of communications facilities. This attack may have a specific target. For example, an entity may suppress all messages directed to a particular destination (e.g., the security audit service). Another form of service denial is the disruption of an entire network, either by disabling the network or by overloading it with messages to degrade performance.



It is quite difficult to prevent active attacks because of the wide variety of potential physical, software and network vulnerabilities. Instead, the goal is to detect active attacks and to recover from any disruption or delays caused by them.

NETWORK SECURITY & MANAGEMENT

Active Attack	Passive Attack
In an active attack, modification in information takes place.	In a passive attack, modification in the information does not take place.
Active Attack is a danger to integrity as well as availability.	Passive Attack is a danger to confidentiality.
In an active attack, attention is on prevention.	In a passive attack, attention is on detection.
Due to active attacks, the execution system is always damaged.	Due to passive attack, there is no harm to the system.
In an active attack, the victim gets informed about the attack.	In a passive attack, the victim does not get informed about the attack.
Active attacks are tough to restrict from entering systems or networks.	A passive attack is easy to prohibit in comparison to an active attack.
The prevention possibility of active attack is high.	The prevention possibility of passive attack is low.
The duration of an active attack is short.	The duration of a passive attack is long.
The purpose of an active attack is to harm the system.	The purpose of a passive attack is to learn the system.

Security Services:

Various security services are explained as follows:

- 1) Confidentiality:** It ensures the protection of data from unauthorized disclosure.
- 2) Authentication:** It is the assurance that the communicating entity is the one that it claims to be.
- 3) Integrity:** It is the assurance that data received are exactly as sent by an authorized entity (i.e., contain no modification, insertion, deletion or replay).
- 4) Non-repudiation:** It protects against denial by one of the entities involved in a communication of having participated in all or part of the communication.
- 5) Access Control:** The prevention of unauthorized use of a resource (i.e., this service controls who can have access to a resource, under what conditions access can occur and what those accessing the resource are allowed to do).
- 6) Availability:** It states that resources/information should be available to authorized parties at all times.

Security Mechanisms:

Security Mechanisms are classified into two types:

- 1) Specific Security Mechanism**
- 2) Pervasive Security Mechanism**

Specific Security Mechanism

Specific Security Mechanisms may be incorporated into the appropriate protocol layer to provide some of the OSI security services.

- a) **Encipherment:** The use of mathematical algorithms to transform data into a form that is not readily intelligible. The transformation and subsequent recovery of the data depend on an algorithm and zero or more encryption keys.
- b) **Digital Signature:** A digital signature is an authentication mechanism that enables the creator of a message to attach a code that acts as a signature.
- c) **Access Control:** A variety of mechanisms that enforce access rights to resources.
- d) **Data Integrity:** A variety of mechanisms used to assure the integrity of a data unit or stream of data units.
- e) **Authentication Exchange:** A mechanism intended to ensure the identity of an entity by means of information exchange.
- f) **Traffic Padding:** The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.
- g) **Routing Control:** Enables selection of particular physically secure routes for certain data and allows routing changes, especially when a breach of security is suspected.
- h) **Notarization:** The use of a trusted third party to assure certain properties of a data exchange.

Pervasive Security Mechanism

Pervasive Security Mechanisms are not specific to any particular OSI security service or protocol layer.

- a) **Trusted Functionality:** That which is perceived to be correct with respect to some criteria (e.g., as established by a security policy).
- b) **Security Label:** The marking bound to a resource (which may be a data unit) that names or designates the security attributes of that resource.
- c) **Event Detection:** Detection of security-relevant events.
- d) **Security Audit Trail:** Data collected and potentially used to facilitate a security audit, which is an independent review and examination of system records and activities.
- e) **Security Recovery:** Deals with requests from mechanisms, such as event handling and management functions, and takes recovery actions.

1.3 TYPES OF COMPUTER AND NETWORK ATTACKS

1) Eavesdropping

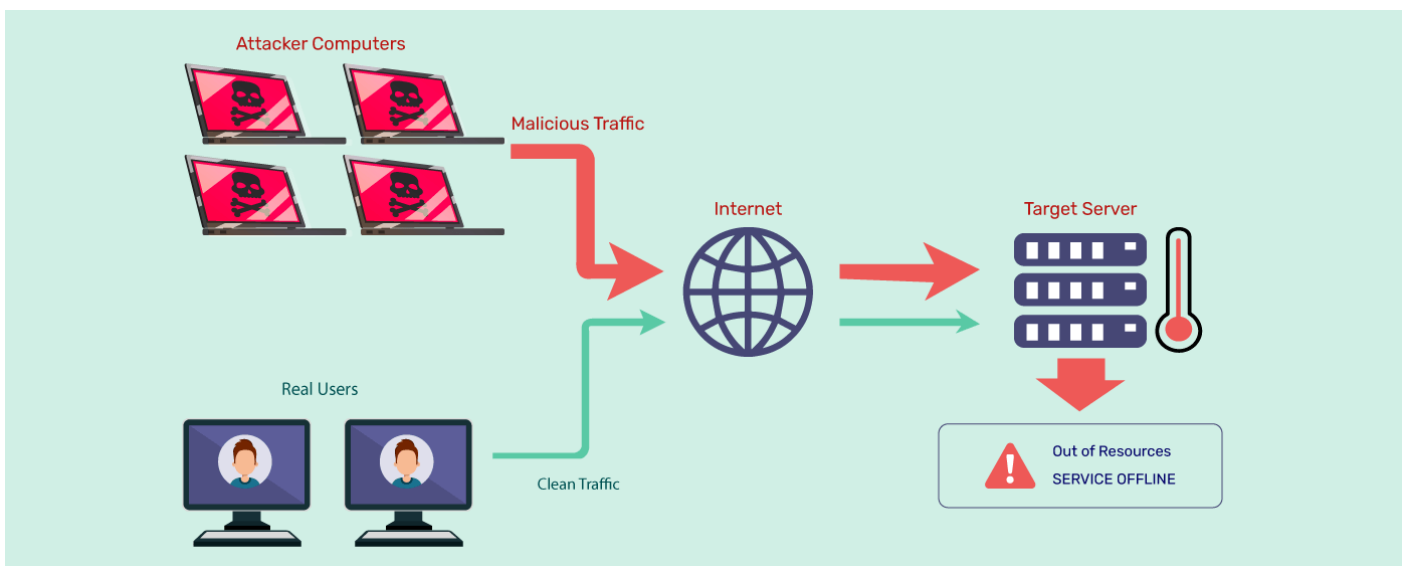
Eavesdropping attacks involve the bad actor intercepting traffic as it is sent through the network. In this way, an attacker can collect usernames, passwords, and other confidential information like credit cards. Eavesdropping can be active or passive. With active eavesdropping, the hacker inserts a piece of software within the network traffic path to collect information that the hacker analyses for useful data. Passive eavesdropping attacks are different in that the hacker “listens in,” or eavesdrops, on the transmissions, looking

for useful data they can steal. Both active and passive eavesdropping are types of MITM attacks. One of the best ways of preventing them is by encrypting your data, which prevents it from being used by a hacker, regardless of whether they use active or passive eavesdropping.

2) Distributed Denial of Service

A distributed denial-of-service (DDoS) attack is a malicious attempt to disrupt the normal traffic of a targeted server, service, or network by overwhelming the target or its surrounding infrastructure with a flood of Internet traffic. DDoS attacks achieve effectiveness by utilizing multiple compromised computer systems as sources of attack traffic. Exploited machines can include computers and other networked resources such as IoT devices. From a high level, a DDoS attack is like an unexpected traffic jam clogging up the highway, preventing regular traffic from arriving at its destination.

DDoS attacks can be segregated by which layer of the Open Systems Interconnection (OSI) model they attack. They are most common in the Network (layer 3), Transport (Layer 4), Presentation (Layer 6) and Application (Layer 7) Layers.



DDoS attacks are carried out with networks of Internet-connected machines. These networks consist of computers and other devices (such as IoT devices) that have been infected with malware, allowing them to be controlled remotely by an attacker. These individual devices are referred to as bots (or zombies), and a group of bots is called a botnet. Once a botnet has been established, the attacker is able to direct an attack by sending remote instructions to each bot. When a victim's server or network is targeted by the botnet, each bot sends requests to the target's IP address, potentially causing the server or network to become overwhelmed, resulting in a denial of service to normal traffic. Because each bot is a legitimate Internet device, separating the attack traffic from normal traffic can be difficult.

3) Malware

Malware is a general term for malicious software. Malware infects a computer and changes how it functions, destroys data, or spies on the user or network traffic as it passes through. Malware can either spread from one device to another or remain in place, only impacting its host device.

Several of the attack methods can involve forms of malware, including MITM attacks, Phishing, Ransomware, SQL injection, Trojan horses, Drive-by attacks, and XSS attacks.

In a malware attack, the software has to be installed on the target device. This requires an action on the part of the user. Therefore, in addition to using firewalls that can detect malware, users should be educated regarding which types of software to avoid, the kinds of links they should verify before clicking, and the emails and attachments they should not engage with.

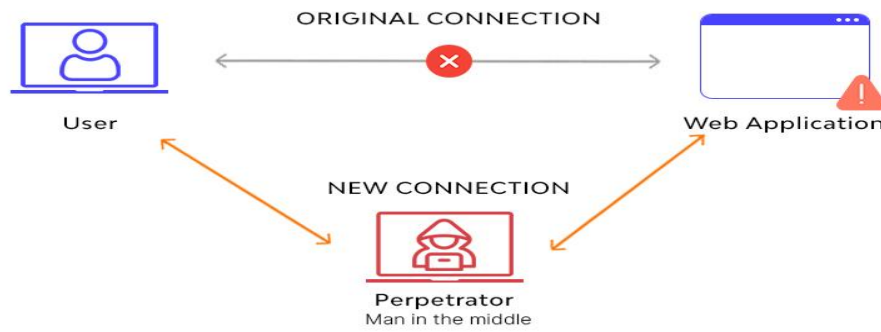
4) Man-in-the-Middle Attack

Man-in-the-middle (MITM) type of cyber-attack refers to breaches in cybersecurity that make it possible for an attacker to eavesdrop on the data sent back and forth between two people, networks, or computers. It is called a “man in the middle” attack because the attacker positions themselves in the “middle” or between the two parties trying to communicate. In effect, the attacker is spying on the interaction between the two parties. In an MITM attack, the two parties involved feel like they are communicating as they normally do. What they do not know is that the person sending the message illicitly modifies or accesses the message before it reaches its destination. Some ways to protect yourself and your organization from MITM attacks are by using strong encryption on access points or by using a virtual private network (VPN).

For example, the attacker can re-route a data exchange. When computers are communicating at low levels of the network layer, the computers might not be able to determine with whom they are exchanging data. If an attacker puts himself between a client and a webpage, a Man-in-the-Middle (MITM) attack occurs. This form of assault comes in many different ways.

For example: To intercept financial login credentials, a fraudulent banking website can be used. Between the user and the real bank webpage, the fake site lies "in the middle."

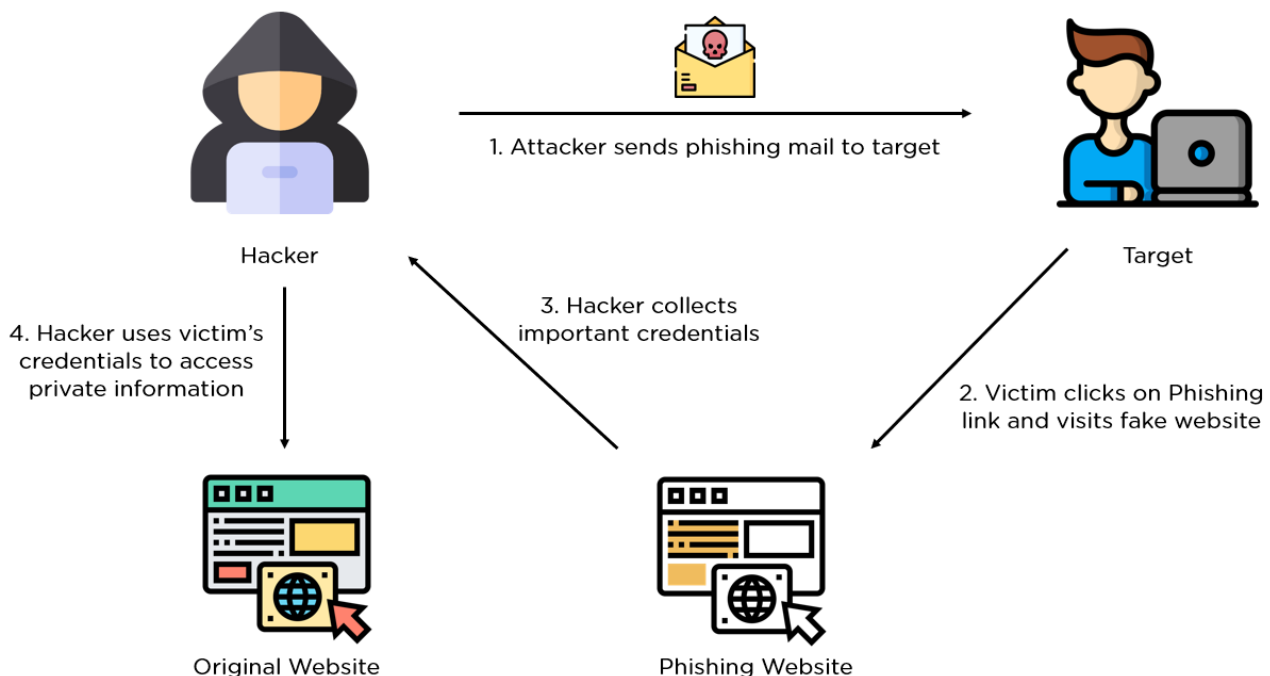
Typically, these attacks are carried out through a two-step process known as data interception and decryption. Data interception consists of an attacker intercepting a data transfer between a client and a server. The attacker tricks the client and the server into believing that they are exchanging information with each other, while the attacker intercepts the data, creates a connection to the real site, and acts as a proxy to read and insert false information into the communication. The decryption phase is where the intercepted data is unencrypted. This essential step enables the attacker to finally decipher and use the data to their advantage



5) Phishing

Phishing attacks are fraudulent emails, text messages, phone calls, or websites that seem to be coming from trusted, legitimate sources in an attempt to grab sensitive information from the target. Successful phishing attacks often lead to identity theft, credit card fraud, ransomware attacks, data breaches and huge financial losses for individuals and corporations.

To execute the attack, the attacker may send a link that brings you to a website that then fools you into downloading malware such as viruses or giving the attacker your private information. In many cases, the target may not realize they have been compromised, which allows the attacker to go after others in the same organization without anyone suspecting malicious activity.



You can prevent phishing attacks from achieving their objectives by thinking carefully about the kinds of emails you open and the links you click on. Pay close attention to email headers, and do not click on anything that looks suspicious. Check the parameters for “Reply-to” and “Return-path.” They need to connect to the same domain presented in the email.

Type	Description
Spear Phishing	Spear-phishing is a type of phishing attack that targets specific individuals or organizations typically through malicious emails. The goal of spear phishing is to steal sensitive information such as login credentials or infect the targets' device with malware.
Whaling	A whaling attack is a type of social engineering attack specifically targeting senior or C-level executive employees with the purpose of stealing money or information, or gaining access to the person's computer in order to execute further cyberattacks.
SMiShing	Smishing is the act of sending fraudulent text messages designed to trick individuals into sharing sensitive data such as passwords, usernames and credit card numbers. A smishing attack may involve cybercriminals pretending to be your bank or a shipping service you use.
Vishing	Vishing, a voice phishing attack, is the fraudulent use of phone calls and voice messages pretending to be from a reputable organization to convince individuals to reveal private information such as bank details and passwords.

6) SQL Injection:

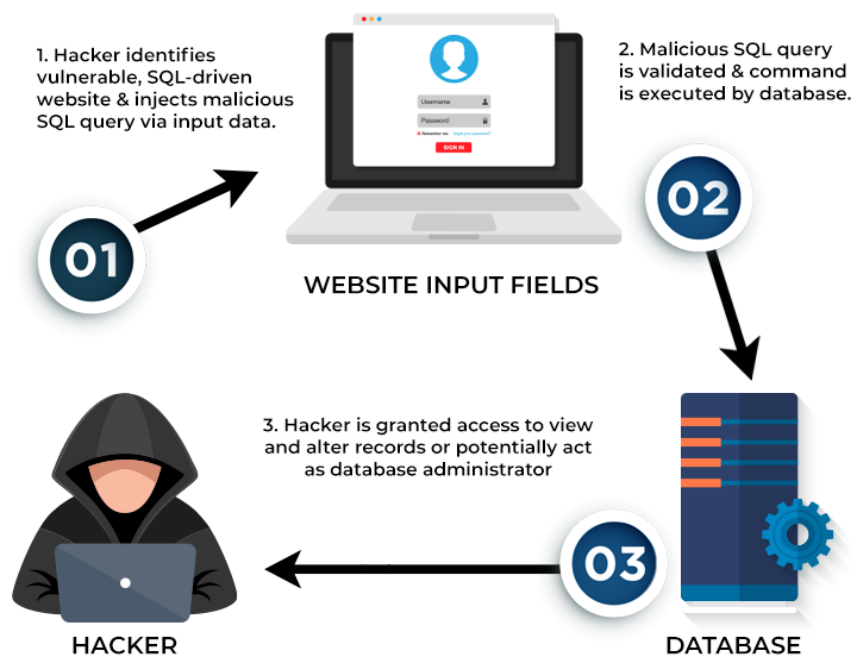
SQL injection is a technique used to extract user data by injecting web page inputs as statements through SQL commands. Malicious users can use these instructions to manipulate the application's web server. SQL injection is a code injection technique that can compromise your database. SQL injection is one of the most common web hacking techniques. SQL injection is the injection of malicious code into SQL statements via web page input.

An SQL injection attack can be done with the following intentions:

- To dump the whole database of a system,
- To modify the content of the databases, or
- To perform different queries that are not allowed by the application.

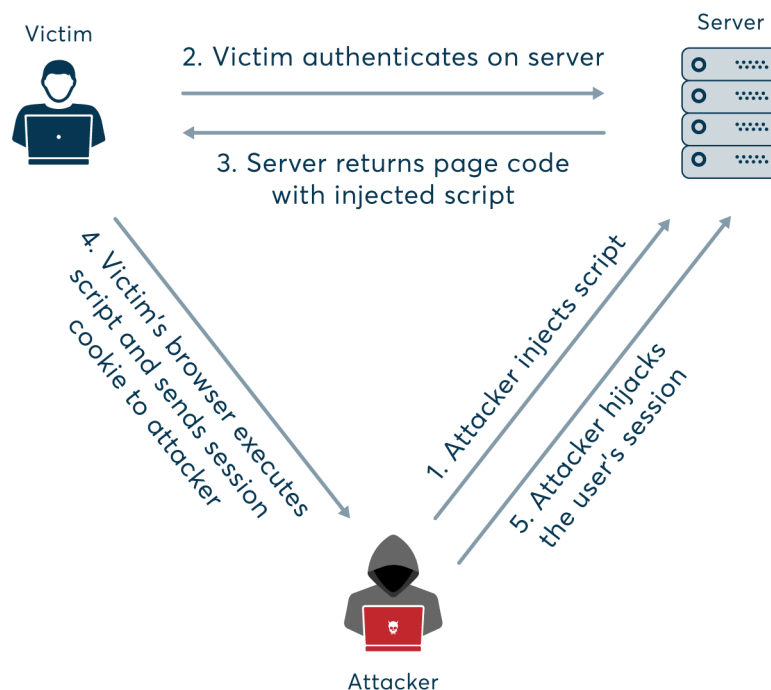
This type of attack works when the applications don't validate the inputs properly, before passing them to an SQL statement. Injections are normally placed in address bars, search fields, or data fields. The easiest way to detect if a web application is vulnerable to an SQL injection attack is to use the " ' " character in a string and see if you get any error.

FUNCTIONING OF AN SQL INJECTION



7) Session Hijacking

Session hijacking is one of multiple types of MITM attacks. The attacker takes over a session between a client and the server. The computer being used in the attack substitutes its Internet Protocol (IP) address for that of the client computer, and the server continues the session without suspecting it is communicating with the attacker instead of the client. This kind of attack is effective because the server uses the client's IP address to verify its identity. If the attacker's IP address is inserted partway through the session, the server may not suspect a breach because it is already engaged in a trusted connection.



To prevent session hijacking, use a VPN to access business-critical servers. This way, all communication is encrypted and an attacker cannot gain access to the secure tunnel created by the VPN.

8) Insider Threat

Sometimes, the most dangerous actors come from within an organization. People within a company's doors pose a special danger because they typically have access to a variety of systems and in some cases, admin privileges that enable them to make critical changes to the system or its security policies.

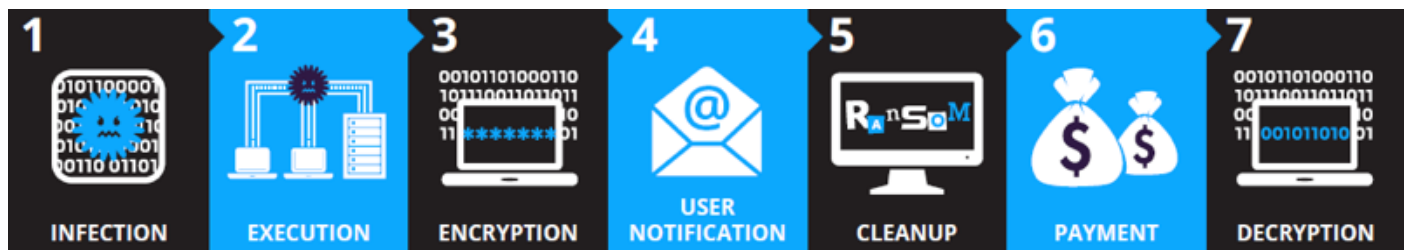
In addition, people within the organization often have an in-depth understanding of its cybersecurity architecture, as well as how the business reacts to threats. This knowledge can be used to gain access to restricted areas, make changes to security settings, or deduce the best possible time to conduct an attack.

One of the best ways to prevent insider threats in organizations is to limit employees' access to sensitive systems to only those who need them to perform their duties. Also, for the select few who need access, use MFA, which will require them to use at least one thing they know in conjunction with a physical item they have to gain access to a sensitive system.

9) Ransomware

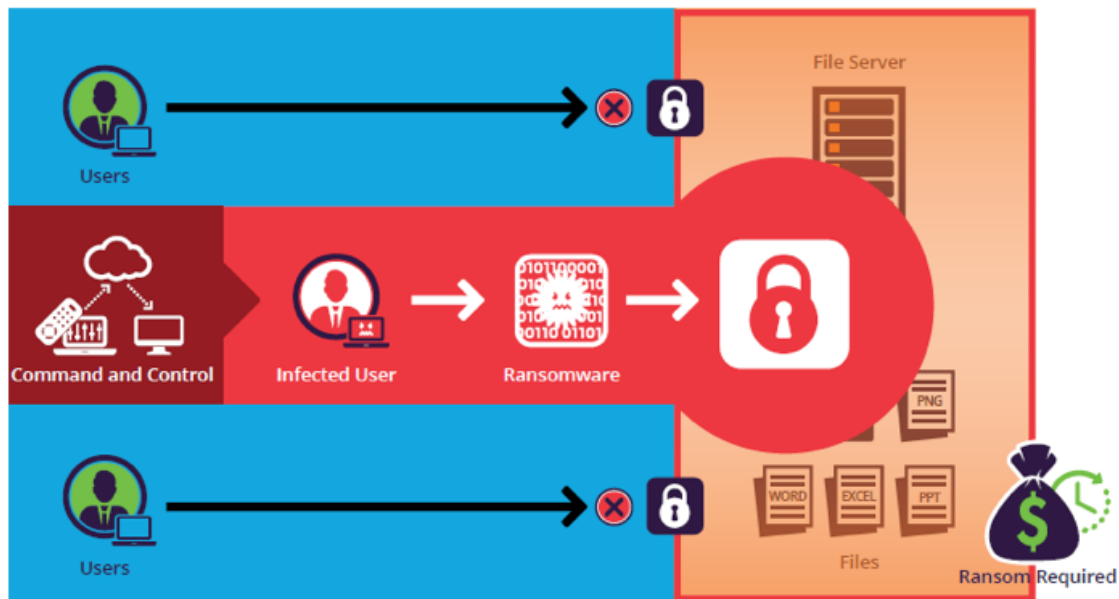
With Ransomware, the victim's system is held hostage until they agree to pay a ransom to the attacker. After the payment has been sent, the attacker then provides instructions regarding how the target can regain control of their computer. The name "ransomware" is appropriate because the malware demands a ransom from the victim.

7 Stages of Ransomware Attack:



Stages	Description
Infection	Ransomware is covertly downloaded and installed on the device.
Execution	Ransomware scans and maps locations for targeted file types, including locally stored files, and mapped and unmapped network-accessible systems. Some ransomware attacks also delete or encrypt any backup files and folders.
Encryption	Ransomware performs a key exchange with the Command and Control Server, using the encryption key to scramble all files discovered during the Execution step. It also locks access to the data.
User Notification	Ransomware adds instruction files detailing the pay-for-decryption process, then uses those files to display a ransom note to the user.

Cleanup	Ransomware usually terminates and deletes itself, leaving only the payment instruction files.
Payment	Victim clicks a link in the payment instructions, which takes the victim to a web page with additional information on how to make the required ransom payment.
Decryption	After the victim pays the ransom, usually via the attacker's Bitcoin address, the victim may receive the decryption key. However, there is no guarantee the decryption key will be delivered as promised.



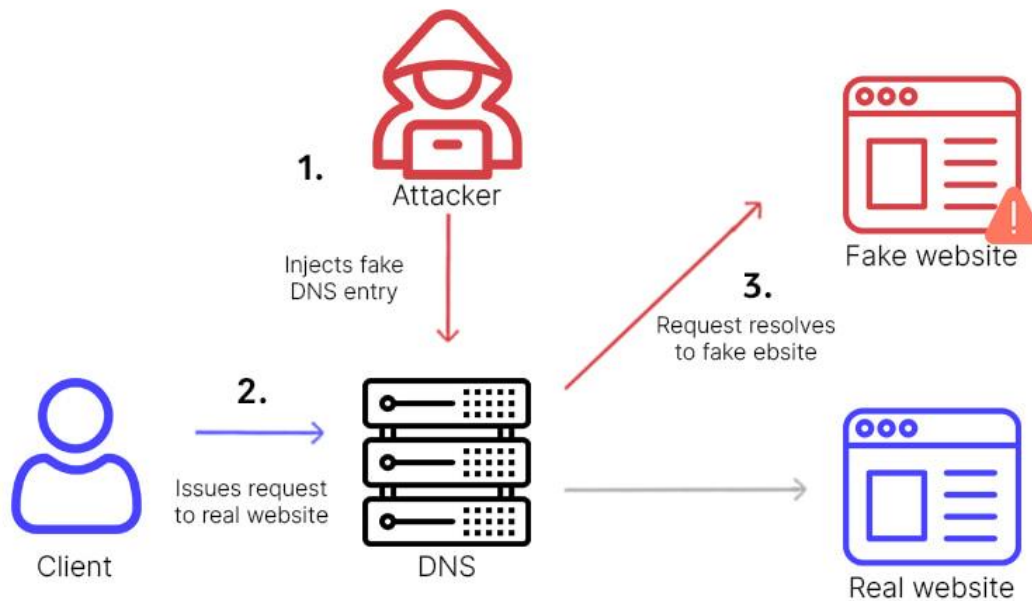
10) DNS Spoofing

DNS Spoofing is a type of computer attack wherein a user is forced to navigate to a fake website disguised to look like a real one, to divert traffic or steal the credentials of the users. Spoofing attacks can go on for a long period without being detected and can cause serious security issues.

In a DNS spoofing attack, the attacker takes advantage of the fact that the user thinks the site they are visiting is legitimate. This gives the attacker the ability to commit crimes in the name of an innocent company, at least from the perspective of the visitor.

Domain Name Server (DNS) resolves the alphabetical domain names like `www.example.com` into respective IP addresses that are used for locating and communicating between nodes on the Internet. DNS spoofing is done by replacing the IP addresses stored in the DNS server with the ones under the control of the attacker. Once it is done, whenever users try to go to a particular website, they get directed to the false websites placed by the attacker in the spoofed DNS server.

To prevent DNS spoofing, make sure your DNS servers are kept up-to-date. Attackers aim to exploit vulnerabilities in DNS servers, and the most recent software versions often contain fixes that close known vulnerabilities.



1.4 TYPES OF CRYPTOGRAPHY

1.4.1 Introduction to Cryptography

Cryptography is a technique of securing information and communications through the use of codes so that only those people for whom the information is intended can understand and process it. Thus, preventing unauthorized access to information. The prefix “crypt” means “hidden” and the suffix “graphy” means “writing”. In Cryptography, the techniques that are used to protect information are obtained from mathematical concepts and a set of rule-based calculations known as algorithms to convert messages in ways that make it hard to decode them. These algorithms are used for cryptographic key generation, digital signing and verification to protect data privacy, web browsing on the internet and to protect confidential transactions such as credit card and debit card transactions.

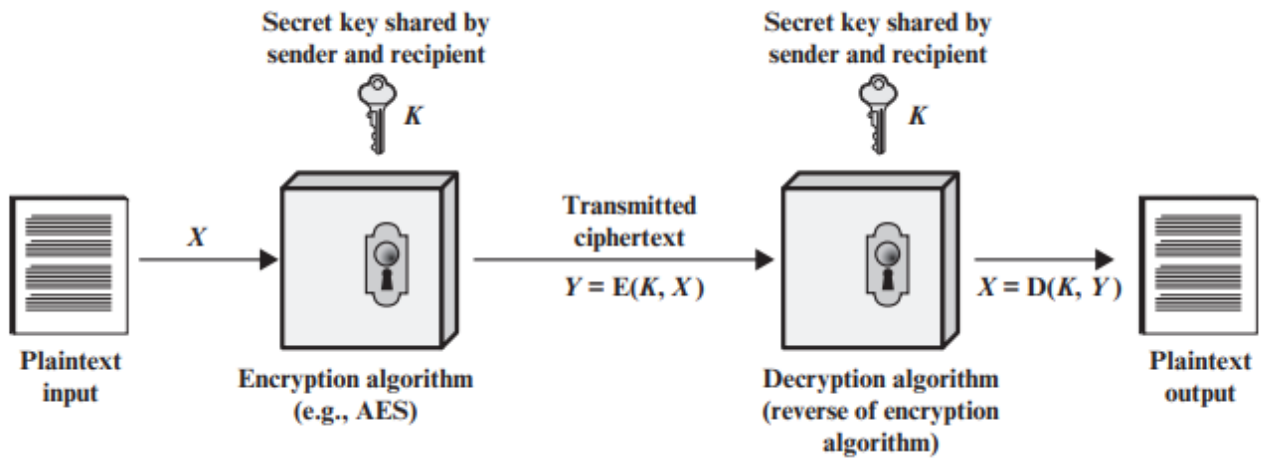
Cryptographic systems are characterized along three independent dimensions:

- 1) The type of operations used for transforming plaintext to ciphertext (Substitution and Transposition).
- 2) The number of keys used (Symmetric Key and Asymmetric Key).
- 3) The way in which the plaintext is processed (Block Cipher and Stream Cipher).

Features of Cryptography:

- 1) Confidentiality
- 2) Integrity
- 3) Non-repudiation
- 4) Authentication

1.4.2 Symmetric Cipher Model



A symmetric cipher scheme consists of 5 main components:

- 1) **Plain Text:** The original message, before being transformed, is called plaintext.
- 2) **Encryption Algorithm:** The process of transforming the plaintext into ciphertext is known as enciphering or encryption. It takes plain text and key value as input to produce cipher text. The encryption algorithm performs various substitutions and transformations on the plaintext.
- 3) **Secret Key:** The secret key is also input to the encryption algorithm. The key is a value independent of the plaintext and the algorithm. The algorithm will produce a different output depending on the specific key being used at the time. The exact substitutions and transformations performed by the algorithm depend on the key.
- 4) **Cipher Text:** The original message after being transformed into a non-readable message is called ciphertext. This is the scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts.
- 5) **Decryption Algorithm:** The process of transforming the cipher text back into plain text is known as deciphering or decryption. It takes cipher text and key value as input to produce plain text back.

1.4.3 Cryptanalysis

Typically, the objective of attacking an encryption system is to recover the key in use rather than simply to recover the plaintext of a single ciphertext. There are two general approaches to attacking a conventional encryption scheme:

- 1) **Cryptanalysis:** Cryptanalytic attacks rely on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext ciphertext pairs. This type of attack exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used.

- 2) **Brute-force attack:** The attacker tries every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained. On average, half of all possible keys must be tried to achieve success.

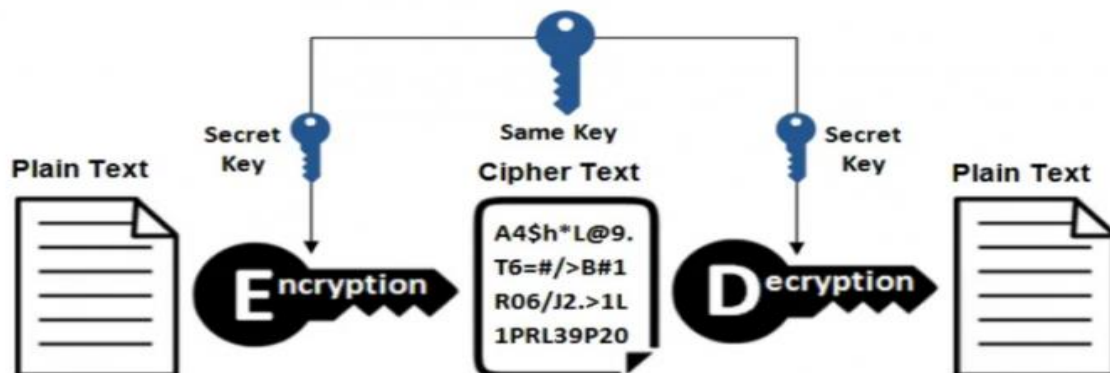
1.4.4 Types of Cryptography

There are three types of Cryptography:

- 1) Symmetric Key Cryptography
- 2) Asymmetric Key Cryptography
- 3) Hash Function

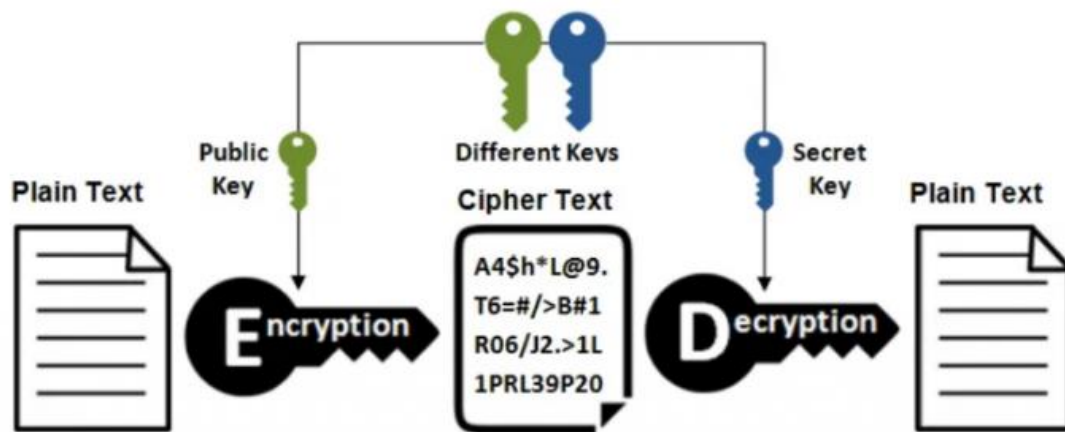
Symmetric Key Cryptography:

Symmetric key cryptography is also known as secret-key cryptography, and in this type of cryptography, you can use only a single key. The sender and the receiver can use that single key to encrypt and decrypt a message. Because there is only one key for encryption and decryption, the symmetric key system has one major disadvantage: the two parties must exchange the key securely. The most popular symmetric key cryptography systems are Data Encryption Standard (DES), Advanced Encryption Standard (AES), Blowfish.



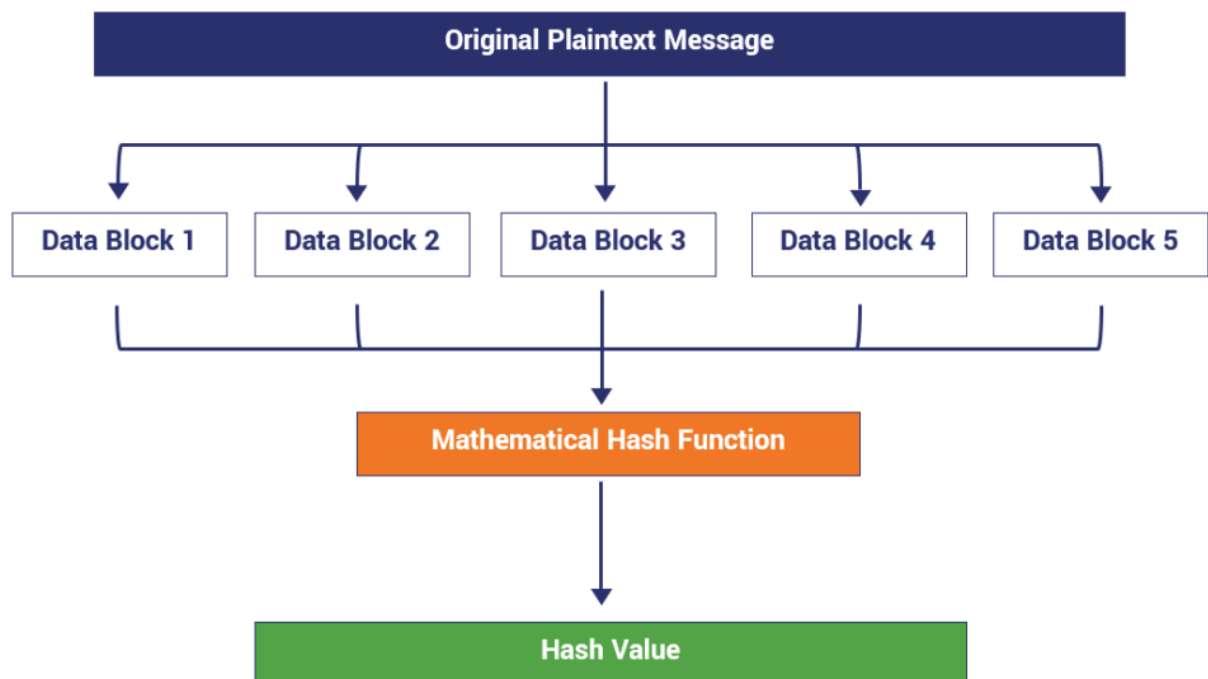
Asymmetric Key Cryptography:

Asymmetric key cryptography is also known as public key cryptography and it employs the use of two keys. This cryptography differs from and is more secure than symmetric key cryptography. In this system, each user encrypts and decrypts using two keys or a pair of keys (private key and public key). Each user keeps the private key secret and the public key is distributed across the network so that anyone can use those public keys to send a message to any other user. You can use any of those keys to encrypt the message and can use the remaining key for decryption. The most popular asymmetric key cryptography system is DSA and RSA.



Hash Function:

It is a type of cryptography in which an algorithm followed by a hash function takes an arbitrary length of the message as input and returns a fixed length of the output. It is also referred to as a mathematical equation because it uses numerical values as input to generate the hash message. This method does not require a key because it operates in a one-way scenario. Each round of hashing operations considers input as an array of the most recent block and generates the last round of activity as output. Commonly used hash algorithms include: Message Digest 5 (MD5), SHA (Secure Hash Algorithm)



UNIT-2

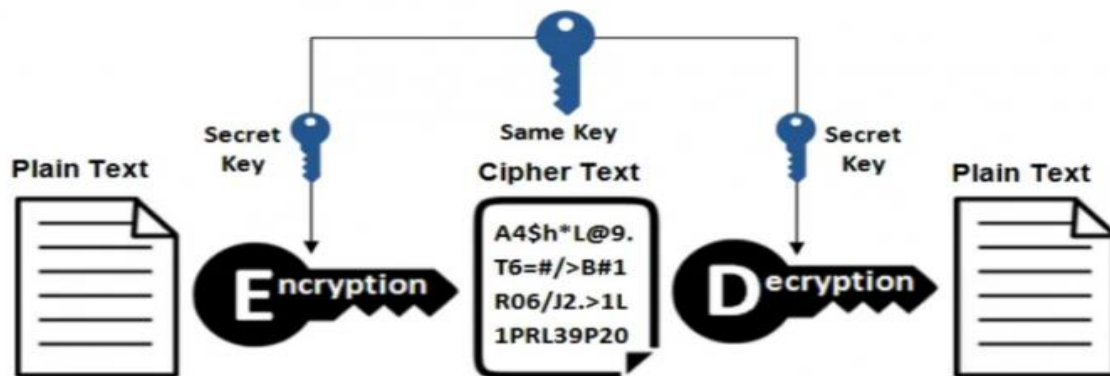
CRYPTOGRAPHY IN NETWORK

2.1 INTRODUCTION TO SYMMETRIC ENCRYPTION & ASYMMETRIC ENCRYPTION

2.1.1 Symmetric Encryption

Symmetric Encryption is a type of encryption where only one key (a secret key) is used to encrypt and decrypt electronic data. The entities communicating via symmetric encryption must exchange the key so that it can be used in the decryption process.

By using symmetric encryption algorithms, data is "scrambled" so that it can't be understood by anyone who does not possess the secret key to decrypt it. Once the intended recipient who possesses the key has the message, the algorithm reverses its action so that the message is returned to its original readable form. The secret key that the sender and recipient both uses could be a specific password/code or it can be a random string of letters or numbers that have been generated by a secure Random Number Generator (RNG).



There are two types of Symmetric Encryption Algorithms:

- 1) **Block Algorithms:** Set lengths of bits are encrypted in blocks of electronic data with the use of a specific secret key. As the data is being encrypted, the system holds the data in its memory as it waits for complete blocks.
- 2) **Stream Algorithms:** Data is encrypted as it streams instead of being retained in the system's memory.

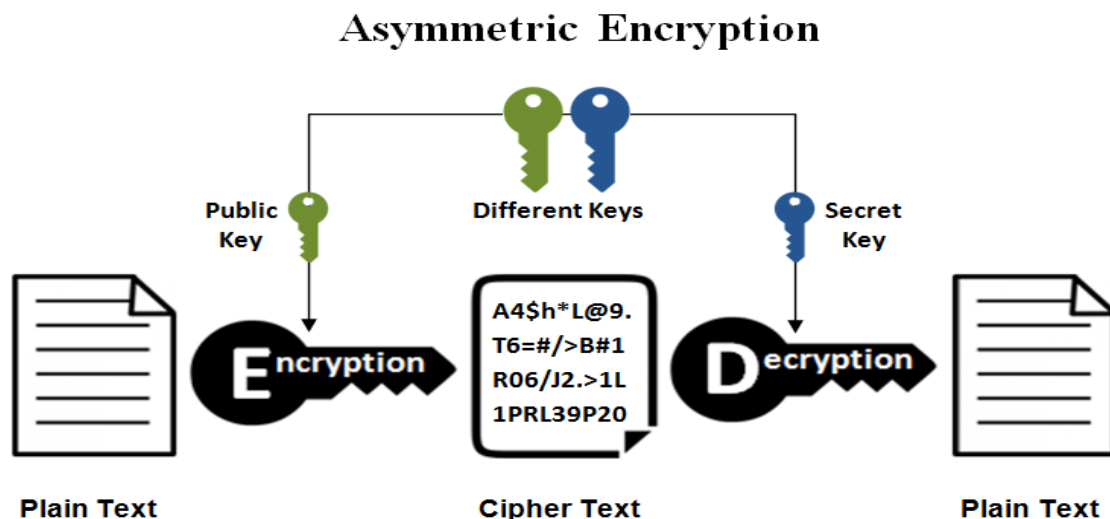
Some examples of symmetric encryption algorithms include:

- 1) AES (Advanced Encryption Standard)
- 2) DES (Data Encryption Standard)
- 3) IDEA (International Data Encryption Algorithm)
- 4) Blowfish (Replacement for DES or IDEA)
- 5) RC4 (Rivest Cipher 4)
- 6) RC5 (Rivest Cipher 5)
- 7) RC6 (Rivest Cipher 6)

AES, DES, IDEA, Blowfish, RC5 and RC6 are block ciphers. RC4 is a stream cipher.

2.1.2 Asymmetric Encryption

Asymmetric encryption, also known as public-key cryptography, is a type of encryption that uses a pair of keys to encrypt and decrypt data. The pair of keys includes a public key, which can be shared with anyone, and a private key, which is kept secret by the owner. In asymmetric encryption, the sender uses the recipient's public key to encrypt the data. The recipient then uses their private key to decrypt the data. This approach allows secure communication between two parties without the need for both parties to have the same secret key. Asymmetric encryption is commonly used in various applications like secure online communication including email encryption, e-commerce, online banking, digital and secure data transfer. Examples of asymmetric encryption algorithms include RSA (Named after persons who invented it - Rivest, Shamir, Adelman), DSA (Digital Signature Algorithm), Diffie-Hellman and Elliptic Curve Cryptography (ECC). Digital Signature which is used to confirm the legitimacy of digital documents and messages is another application of it.



2.1.3 Difference between Symmetric Encryption and Asymmetric Encryption

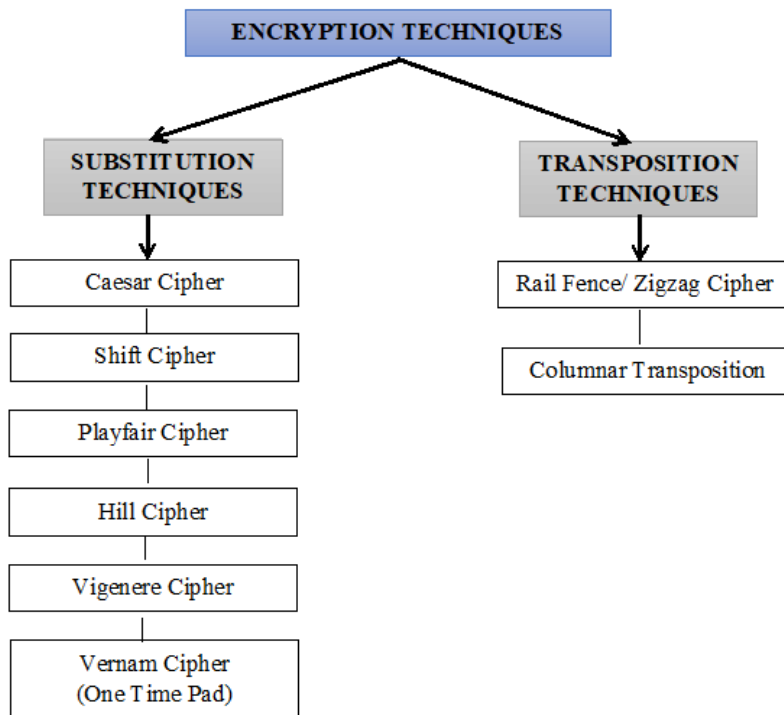
Parameters	Symmetric Encryption	Asymmetric Encryption
Keys used	It uses a single shared key (secret key) to encrypt and decrypt the message.	It uses two different keys for encryption and decryption.
Size	The size of ciphertext in symmetric encryption could be the same or smaller than the plain text.	The size of ciphertext in asymmetric encryption could be the same or larger than the plain text.
Efficiency	It is efficient as this technique is recommended for large amounts of text.	It is inefficient as this technique is used only for short messages.

Speed	The encryption process of symmetric encryption is faster as it uses a single key for encryption and decryption.	The encryption process in asymmetric encryption is slower as it uses two different keys; both keys are related to each other through a complicated mathematical process.
Purpose	Symmetric encryption is mainly used to transmit bulk data.	It is mainly used in smaller transactions. It is used for establishing a secure connection channel before transferring the actual data.
Security	It is less secure as there is a use of a single key for encryption.	It is safer as there are two keys used for encryption and decryption.
Algorithms	The algorithms used in symmetric encryption are 3DES, AES, DES and RC4.	RSA, DSA, Diffie-Hellman, ECC.
Existence	It is an old technique.	It is a new technique.

2.2 SUBSTITUTION TECHNIQUES FOR ENCRYPTION AND DECRYPTION

2.2.1 Introduction

The two basic building blocks of all encryption techniques are substitution and transposition. A substitution technique is one in which the letters of plaintext are replaced by other letters or by numbers or symbols. In simple terms, the plaintext characters are substituted and additional substitute letters, numerals and symbols are implemented in their place. If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns. A character's identity is changed, but its place remains constant in the substitution technique. There are various methods for substitution techniques such as Caesar Cipher, Shift Cipher, Monoalphabetic Cipher, Playfair Cipher, Polyalphabetic Cipher (Vigenere Cipher), One Time Pad (Vernam Cipher), Hill Cipher.



2.2.2 Caesar Cipher Substitution Technique

It is the earliest known use of a substitution cipher and the simplest method. It was invented by Julius Caesar. The Caesar Cipher involves replacing each letter of the alphabet with the letter standing three places further down the alphabet. The encryption can be represented using modular arithmetic by first transforming the letters into numbers.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

The formula of Encryption is:

$$CT = E(K, PT) = (PT + K) \bmod 26$$

The formula of Decryption is:

$$PT = D(K, CT) = (CT - K) \bmod 26$$

Where

PT = Plain Text,

CT = Cipher Text,

K = Key,

E = Encryption,

D = Decryption

In any case, during decryption, if the value becomes negative (-ve), then in that case, 26 will be added to that particular negative value and then decryption will be carried out.

EXAMPLE:

Plain Text: SECURITY, Key: 3

ENCRYPTION:

Plain Text	Encryption CT = (PT+3) Mod 26	Cipher Text
S (18)	$(18 + 3) \bmod 26 = 21 \bmod 26 = 21$	V
E (04)	$(04 + 3) \bmod 26 = 07 \bmod 26 = 07$	H
C (02)	$(02 + 3) \bmod 26 = 05 \bmod 26 = 05$	F
U (20)	$(20 + 3) \bmod 26 = 23 \bmod 26 = 23$	X
R (17)	$(17 + 3) \bmod 26 = 20 \bmod 26 = 20$	U
I (08)	$(08 + 3) \bmod 26 = 11 \bmod 26 = 11$	L
T (19)	$(19 + 3) \bmod 26 = 22 \bmod 26 = 22$	W
Y (24)	$(24 + 3) \bmod 26 = 27 \bmod 26 = 01$	B

Cipher Text: VHFXULWB

DECRYPTION:

Cipher Text	Decryption PT = (CT-3) Mod 26	Plain Text
V (21)	$(21 - 3) \bmod 26 = 18 \bmod 26 = 18$	S
H (07)	$(07 - 3) \bmod 26 = 04 \bmod 26 = 04$	E
F (05)	$(05 - 3) \bmod 26 = 02 \bmod 26 = 02$	C
X (23)	$(23 - 3) \bmod 26 = 20 \bmod 26 = 20$	U
U (20)	$(20 - 3) \bmod 26 = 17 \bmod 26 = 17$	R
L (11)	$(11 - 3) \bmod 26 = 08 \bmod 26 = 08$	I
W (22)	$(22 - 3) \bmod 26 = 19 \bmod 26 = 19$	T
B (01)	$(01 - 3) \bmod 26 = -2 \bmod 26$ $= (-2+26) \bmod 26$ $= 24 \bmod 26 = 24$	Y

Plain Text: SECURITY

2.2.3 Playfair Cipher Technique

The Playfair cipher was invented by Sir Charles Wheatstone, but it was popularized by his friend Lyon Playfair, hence the name "Playfair Cipher." The Playfair cipher is a digraphic substitution cipher, meaning it operates on pairs of letters (digraphs) rather than individual letters, which adds an extra layer of complexity compared to simpler substitution ciphers like the Caesar Cipher. The best-known multiple-letter encryption

cipher is the Playfair. The Playfair algorithm is based on the use of a 5 x 5 matrix of letters constructed using a keyword. The matrix is constructed by filling in the letters of the keyword (minus duplicates) from left to right and from top to bottom, and then filling in the remainder of the matrix with the remaining letters in alphabetic order. The letters I and J count as one letter. Plaintext is encrypted two letters at a time, according to the following rules:

- 1) Firstly, break down plain text in the pair of 2. For example, **playfair** would be **pl ay fa ir**.
- 2) While making a pair from plain text, if the last letter is single then add a filler letter such as x. For example, the **technique** would be **te ch ni qu ex**.
- 3) Repeating plain text letters that are in the same pair are separated with a filler letter such as x. For example, the **balloon** would be treated as **ba lx lo on**.
- 4) If repeating plain text letters that are in a different pair then there is no need to separate it with any other filler letter.
- 5) Two plain text letters that fall in the same row of the matrix are each replaced by the letter to the right, with the first element of the row circularly following the last.
- 6) Two plain text letters that fall in the same column are each replaced by the letter beneath, with the top element of the column circularly following the last.
- 7) Otherwise, each plain text letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plain text letter.
- 8) For decryption, if two cipher text letters that fall in the same row of the matrix are each replaced by the letter to the left, with the last element of the row circularly following the first.
- 9) For decryption, if two ciphertext letters that fall in the same column are each replaced by the letter above, with the bottom element of the column circularly following the first.
- 10) Otherwise, each cipher text letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other cipher text letter.

EXAMPLE:

- 1) **Plain Text: COMPUTER, Key: NETWORK**

N	E	T	W	O
R	K	A	B	C
D	F	G	H	I/J
L	M	P	Q	S
U	V	X	Y	Z

Solution (for encryption):

- First, break plain text into pairs of 2. i.e. CO MP UT ER
- Here CO is in the same column. So, replace it with the below letter in that column. So, CO will be IC.
- Here MP is in the same row. So, replace it with the next letter in that row. So, MP will be PQ.
- Here UT is neither in the same row nor in the same column. Then as per rule 7, UT will be replaced by XN.
- Here ER is neither in the same row nor in the same column. Then as per rule 7, ER will be replaced by NK.
- Therefore, the cipher text will be **ICPQXNNK**.

Solution (for decryption):

- First, break cipher text into a pair of 2. i.e. IC PQ XN NK.
- Here IC is in the same column. So, replace it with the above letter in that column. So, IC will be CO.
- Here PQ is in the same row. So, replace it with the previous letter in that row. So, PQ will be MP.
- Here XN is neither in the same row nor in the same column. Then as per rule 10, XN will be replaced by UT.
- Here NK is neither in the same row nor in the same column. Then as per rule 10, NK will be replaced by ER.

2) Plain Text: INSTRUMENTS

Key: MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Solution (for encryption):

- First, break plain text into pairs of 2. i.e. IN ST RU ME NT S
- Here, the last letter is single. So as per rule no. 2, add filler letter X.
- Therefore, it would be now IN ST RU ME NT SX.
- Here IN is neither in the same row nor in the same column. Then as per rule 7, IN will be replaced by GA.
- Here ST is in the same row. So, replace it with the next letter in that row. So, ST will be TL. (Here, T is the last letter in that particular row. So, it is replaced with the very first letter of that particular row).

- Here RU is neither in the same row nor in the same column. Then as per rule 7, RU will be replaced by MZ.
- Here ME is in the same column. So, replace it with the below letter in that column. So, ME will be CL.
- Here NT is neither in the same row nor in the same column. Then as per rule 7, NT will be replaced by RQ.
- Here SX is in the same column. So, replace it with the below letter in that column. So, SX will be XA. (Here, X is the last letter in that particular column. So, it is replaced with the very first letter of that particular column).
- Therefore, the cipher text will be **GATLMZCLRQXA**.

Solution (for decryption):

- First, break cipher text into a pair of 2. i.e. GA TL MZ CL RQ XA.
- Here GA is neither in the same row nor in the same column. Then as per rule 10, GA will be replaced by IN.
- Here TL is in the same row. So, replace it with the next letter in that row. So, TL will be ST. (Here, L is the first letter in that particular row. So, it is replaced with the very last letter of that particular row).
- Here MZ is neither in the same row nor in the same column. Then as per rule 10, MZ will be replaced by RU.
- Here CL is in the same column. So, replace it with the below letter in that column. So, CL will be ME.
- Here RQ is neither in the same row nor in the same column. Then as per rule 10, RQ will be replaced by NT.
- Here XA is in the same column. So, replace it with the below letter in that column. So, XA will be SX. (Here, A is the first letter in that particular column. So, it is replaced with the very last letter of that particular column).
- Therefore, plain text will be **INSTRUMENTSX**.

2.2.4 Shift Cipher Technique

Shift Cipher Technique is one of the earliest and simplest known substitution techniques. It is similar to the Caesar Cipher Technique. The only difference is that in the Caesar cipher, the key value is fixed i.e. 3 whereas in the shift cipher, the key value ranges from 0 to 25. A given plain text is encrypted into cipher text by shifting each letter of the plain text by n positions.

The encryption/decryption can be represented using modular arithmetic by first transforming the letters into numbers.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

The formula of Encryption is:

$$\mathbf{CT = E (K, PT) = (PT + K) \bmod 26}$$

The formula of Decryption is:

$$\mathbf{PT = D (K, CT) = (CT - K) \bmod 26}$$

In any case, during decryption, if the value becomes negative (-ve), then in that case, 26 will be added to that particular negative value and then decryption will be carried out.

EXAMPLE:

1) Plain Text: HELLO

Key: 6

ENCRYPTION:

Plain Text	Encryption CT = (PT+K) Mod 26	Cipher Text
H (07)	(07 + 06) mod 26 = 13 mod 26 = 13	N
E (04)	(04 + 06) mod 26 = 10 mod 26 = 10	K
L (11)	(11 + 06) mod 26 = 17 mod 26 = 17	R
L (11)	(11 + 06) mod 26 = 17 mod 26 = 17	R
O (14)	(14 + 06) mod 26 = 20 mod 26 = 20	U

Cipher Text: NKRRU

DECRYPTION:

Cipher Text	Decryption PT = (CT-K) Mod 26	Plain Text
N (13)	(13 - 06) mod 26 = 07 mod 26 = 07	H
K (10)	(10 - 06) mod 26 = 04 mod 26 = 04	E
R (17)	(17 - 06) mod 26 = 11 mod 26 = 11	L
R (17)	(17 - 06) mod 26 = 11 mod 26 = 11	L
U (20)	(20 - 06) mod 26 = 14 mod 26 = 14	O

Plain Text: HELLO

2) Plain Text: LAYOUT

Key: 15

ENCRYPTION:

Plain Text	Encryption CT = (PT+K) Mod 26	Cipher Text
L (11)	$(11 + 15) \bmod 26 = 26 \bmod 26 = 00$	A
A (0)	$(00 + 15) \bmod 26 = 15 \bmod 26 = 15$	P
Y (24)	$(24 + 15) \bmod 26 = 39 \bmod 26 = 13$	N
O (14)	$(14 + 15) \bmod 26 = 29 \bmod 26 = 03$	D
U (20)	$(20 + 15) \bmod 26 = 35 \bmod 26 = 09$	J
T (19)	$(19 + 15) \bmod 26 = 34 \bmod 26 = 08$	I

Cipher Text: APNDJI

DECRYPTION:

Cipher Text	Decryption PT = (CT-K) Mod 26	Plain Text
A (0)	$(00 - 15) \bmod 26 = 11 \bmod 26 = 11$	L
P (15)	$(15 - 15) \bmod 26 = 04 \bmod 26 = 00$	A
N (13)	$(13 - 15) \bmod 26 = -2 \bmod 26$ $= (-2+26) \bmod 26$ $= 24 \bmod 26 = 24$	Y
D (03)	$(03 - 15) \bmod 26 = -12 \bmod 26$ $= (-12+26) \bmod 26$ $= 14 \bmod 26 = 14$	O
J (09)	$(09 - 15) \bmod 26 = -6 \bmod 26$ $= (-6+26) \bmod 26$ $= 20 \bmod 26 = 20$	U
I (08)	$(08 - 15) \bmod 26 = 7 \bmod 26$ $= (-7+26) \bmod 26$ $= 19 \bmod 26 = 19$	T

Plain Text: LAYOUT

2.2.5 Vigenere Cipher Technique

Vigenere Cipher is a method of encrypting alphabetic text. It uses a simple form of polyalphabetic substitution. A polyalphabetic cipher is any cipher based on substitution, using multiple substitution alphabets. This algorithm was first described in 1553 by Giovan Battista Bellaso. It uses a Vigenere Table or Vigenere Square for encryption and decryption of the text. The Vigenere table is also called the Tabula Recta. There are two methods to perform the Vigenere cipher.

Method 1:

When the Vigenere table is given, the encryption and decryption are done using the Vigenere table (26 * 26 matrix) in this method.

		Plaintext																									
Key		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
	C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
	D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
	E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
	F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
	G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
	H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
	I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
	J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
	K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
	L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
	M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
	N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
	O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
	W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
	X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
	Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

For generating the key, the given keyword is repeated circularly until it matches the length of the plain text.

EXAMPLE: The plaintext is "CYBERSECURITY", and the key is "BEST".

C	Y	B	E	R	S	E	C	U	R	I	T	Y
B	E	S	T	B	E	S	T	B	E	S	T	B

ENCRYPTION:

The first letter of the plaintext is combined with the first letter of the key. The column of plain text "C" and the row of key "B" intersects the alphabet of "D" in the Vigenere table, so the first letter of ciphertext is "D".

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Similarly, the second letter of the plaintext is combined with the second letter of the key. The column of plain text "Y" and the row of key "E" intersect the alphabet of "C" in the Vigenere table, so the second letter of ciphertext is "C".

This process continues continuously until the plaintext is finished.

Ciphertext = D C T X S W W V V V A M Z

DECRYPTION:

Decryption is done by the row of keys in the Vigenere table. First, select the row of the key letter, find the ciphertext letter's position in that row, and then select the column label of the corresponding ciphertext as the plaintext.

D	C	T	X	S	W	W	V	V	V	A	M	Z
B	E	S	T	B	E	S	T	B	E	S	T	B

For example, in the row of the key is "B" and the ciphertext is "D" and this ciphertext letter appears in the column "C", which means the first plaintext letter is "C".

Next, in the row of the key is "E" and the ciphertext is "C" and this ciphertext letter appears in the column "Y", which means the second plaintext letter is "Y".

This process continues continuously until the ciphertext is finished.

Plaintext = CYBER SECURITY

Method 2:

When the Vigenere table is not given, the encryption and decryption are done by the algebraic formula in this method (convert the letters (A-Z) into the numbers (0-25)).

The formula of Encryption is:

$$C_i = (P_i + K_i) \bmod 26$$

The formula of Decryption is:

$$P_i = (C_i - K_i) \bmod 26$$

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

① Plain Text :- FORMULA
Key :- DATE

→ Solution:-

- While using Vigenere cipher method, length of plain text & key value should be same.

∴ Plain Text = FORMULA
Key = DATE DAT

→ Encryption:-

$$C_i = (P_i + K_i) \bmod 26$$

(i) For $i=0$	$= (14+0) \bmod 26$
$P_0 = F (5)$	$= 14 \bmod 26$
$K_0 = D (3)$	$= 14$
	$= 0$

$$\begin{aligned} C_0 &= (P_0 + K_0) \bmod 26 \\ &= (5+3) \bmod 26 \\ &= 8 \bmod 26 \\ &= 8 \\ &= I \end{aligned}$$

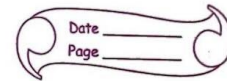
(iii) $P_2 = R (17), K_2 = T (19)$

$$\begin{aligned} \therefore C_2 &= (P_2 + K_2) \bmod 26 \\ &= (17+19) \bmod 26 \\ &= 36 \bmod 26 \\ &= 10 \\ &= K \end{aligned}$$

(ii) For $i=1$

$$\begin{aligned} P_1 &= O (14) \\ K_1 &= A (0) \end{aligned}$$

$$C_1 = (P_1 + K_1) \bmod 26$$



(iv) For $i=3$
 $P_3 = M(12), K = E(4)$

$$\begin{aligned} \therefore C_3 &= (P_3 + K_3) \bmod 26 \\ &= (12 + 4) \bmod 26 \\ &= 16 \bmod 26 \\ &= 16 \\ &= Q \end{aligned}$$

(vii) For $i=6$
 $P_6 = A(0), K_6 = T(19)$

$$\begin{aligned} \therefore C_6 &= (P_6 + K_6) \bmod 26 \\ &= (0 + 19) \bmod 26 \\ &= 19 \bmod 26 \\ &= 19 \\ &= T \end{aligned}$$

(v) For $i=4$
 $P_4 = U(20), K_4 = D(3)$

$$\begin{aligned} \therefore C_4 &= (P_4 + K_4) \bmod 26 \\ &= (20 + 3) \bmod 26 \\ &= 23 \bmod 26 \\ &= 23 \\ &= X \end{aligned}$$

$\therefore$ Cipher Text
 =
 IOKQXLT

$\rightarrow$ Decryption:-

Cipher
 Text = IOKQXLT
 Key = DATE DAT

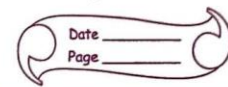
(vi) For $i=5$
 $P_5 = L(11), K_5 = A(0)$

$$\begin{aligned} \therefore C_5 &= (P_5 + K_5) \bmod 26 \\ &= (11 + 0) \bmod 26 \\ &= 11 \bmod 26 \\ &= 11 \\ &= L \end{aligned}$$

$$P_i = (C_i - K_i) \bmod 26$$

(i) For $i=0$
 $C_0 = I(8), K_0 = D(3)$

$$\begin{aligned} \therefore P_0 &= (C_0 - K_0) \bmod 26 \\ &= (8 - 3) \bmod 26 \\ &= 5 \bmod 26 \\ &= 5 \\ &= F \end{aligned}$$



(ii) For $i=1$
 $C_1 = O(14), K_1 = A(0)$

$$\begin{aligned} \therefore P_1 &= (C_1 - K_1) \bmod 26 \\ &= (14 - 0) \bmod 26 \\ &= 14 \bmod 26 \\ &= 14 \\ &= O \end{aligned}$$

(iv) For $i=4$
 $C_4 = X(23), K_4 = D(3)$

$$\begin{aligned} \therefore P_4 &= (C_4 - K_4) \bmod 26 \\ &= (23 - 3) \bmod 26 \\ &= 20 \bmod 26 \\ &= 20 \\ &= U \end{aligned}$$

(iii) For $i=2$
 $C_2 = K(10), K_2 = T(19)$

$$\begin{aligned} \therefore P_2 &= (C_2 - K_2) \bmod 26 \\ &= (10 - 19) \bmod 26 \\ &= -9 \bmod 26 \\ &= (-9 + 26) \bmod 26 \\ &= 17 \bmod 26 \\ &= 17 \\ &= R \end{aligned}$$

(vi) For $i=5$
 $C_5 = L(11), K_5 = A(0)$

$$\begin{aligned} \therefore P_5 &= (C_5 - K_5) \bmod 26 \\ &= (11 - 0) \bmod 26 \\ &= 11 \bmod 26 \\ &= 11 \\ &= L \end{aligned}$$

(iv) For $i=3$
 $C_3 = Q(16), K_3 = E(4)$

$$\begin{aligned} \therefore P_3 &= (C_3 - K_3) \bmod 26 \\ &= (16 - 4) \bmod 26 \\ &= 12 \bmod 26 \\ &= 12 \\ &= M \end{aligned}$$

(vii) For $i=6$
 $C_6 = T(19), K_6 = T(19)$

$$\begin{aligned} \therefore P_6 &= (C_6 - K_6) \bmod 26 \\ &= (19 - 19) \bmod 26 \\ &= 0 \bmod 26 \\ &= 0 \\ &= A \end{aligned}$$

∴ Plain Text = FORMULA

2.2.6 One-Time Pad (Vernam Cipher) Technique

One Time Pad algorithm is the improvement of the Vernam Cipher, that was co-invented by Gilbert S. Vernam (An American Engineer at AT&T Bell Labs) and Joseph Mauborgne(A U.S Army Signal Corp officer). It is the only available algorithm that is unbreakable (completely secure). It is a method of encrypting alphabetic plain text. In this mechanism, we assign a number to each character of the Plain-Text.

Method to take key:

In the Vernam cipher algorithm, we take a key to encrypt the plain text whose length should be equal to the length of the plain text.

ENCRYPTION:

Treat each plaintext character as a number in an increasing sequence from $a = 0$, $b = 1 \dots z = 25$. Do the same for each character of the Key/ OTP. Add each number corresponding to the plain text character to the corresponding character of the key. If the produced cipher text is greater than 25; then subtract 26 from it. Convert each number of the cipher text into corresponding alphabet characters.

EXAMPLE:

Plaintext: COMPUTER

Key: SECURITY

Plain Text	C (02)	O (14)	M (12)	P (15)	U (20)	T (19)	E (04)	R (17)
Key/ OTP	S (18)	E (04)	C (02)	U (20)	R (17)	I (08)	T (19)	Y (24)
PT + Key	20	18	14	35 35-26=09	37 37-26=11	27 27-26=01	23	41 41-26=15
Cipher Text	U (20)	S (18)	O (14)	J (09)	L (11)	B (01)	X (23)	P (15)

Cipher Text: USOJLBXP

DECRYPTION:

Treat each cipher text character as a number in an increasing sequence from $a = 0$, $b = 1 \dots z = 25$. Do the same for each character of the key/ OTP. Subtract each number corresponding to the cipher text character to the corresponding key character number. If the produced cipher text is a negative number; then add 26 to it. Convert each number of the plain text into the corresponding alphabet character.

Cipher Text	U (20)	S (18)	O (14)	J (09)	L (11)	B (01)	X (23)	P (15)
Key/OTP	S (18)	E (04)	C (02)	U (20)	R (17)	I (08)	T (19)	Y (24)
CT - Key	02	14	12	-11 -11+26 = 15	-06 -06+26 = 20	-07 -07+26 =19	04	-09 -09+26 =17
Plain Text	C	O	M	P	U	T	E	R

Plain Text: COMPUTER

2.2.7 Hill Cipher Technique

The Hill Cipher was invented by Lester S. Hill in 1929 and like the other digraphic ciphers, it acts on groups of letters. Unlike the others though it is extendable to work on different-sized blocks of letters. So, technically it is a polygraphic substitution cipher, as it can work on digraphs, trigraphs (3 letter blocks) or theoretically any sized blocks.

The Hill Cipher uses an area of mathematics called linear algebra and in particular, requires the user to have an elementary understanding of matrices. It also makes use of modulo arithmetic. Because of this, the cipher has a significantly more mathematical nature than some of the others. However, it is this nature that allows it to act (relatively) easily on larger blocks of letters.

ENCRYPTION:

To encrypt the text using a hill cipher, we need to perform the following operation.

$$E(K, P) = (K * P) \bmod 26$$

Where K is the key matrix and P is plain text in vector form. Matrix multiplication of K and P generates the encrypted ciphertext.

Step 1: Convert the key using a substitution scheme into a $n * n$ key matrix.

Step 2: Now, we will convert our plain text into vector form. Since the key matrix is $n * n$, the vector must be $n * 1$ for matrix multiplication. (Suppose the key matrix is 2×2 , a vector will be a 2×1 matrix.)

Step 3: Multiply the key matrix with each $n * 1$ plain text vector, and take the modulo of the result by 26.

DECRYPTION:

To decrypt the text using a hill cipher, we need to perform the following operation.

$$D(K, C) = (K^{-1} * C) \bmod 26$$

Where K is the key matrix and C is the ciphertext in vector form. Matrix multiplication of the inverse of key matrix K and ciphertext C generates the decrypted plain text.

Step 1: Calculate the inverse of the key matrix. First, we need to find the determinant of the key matrix (must be between 0-25). Here the Extended Euclidean algorithm is used to get the modulo multiplicative inverse of the key matrix determinant

Step 2: Now, we multiply the $n \times 1$ blocks of ciphertext and the inverse of the key matrix. The resultant block after concatenation is the plain text that we have encrypted.

EXAMPLE:**1) Plain Text: HI**

Key: BEAT

Solution (Encryption):

Convert key into 2×2 matrix and then convert it into numeric form ($A = 0, B = 1 \dots\dots Z = 25$)

$$\begin{bmatrix} B & E \\ A & T \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 19 \end{bmatrix}$$

Convert plain text into a 2×1 matrix and then convert it into numeric form. $P = \begin{bmatrix} H \\ I \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$

$$E = KP \text{ mod } 26$$

$$= \begin{bmatrix} 1 & 4 \\ 0 & 19 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 1 * 7 + 4 * 8 \\ 0 * 7 + 19 * 8 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 7 + 32 \\ 0 + 152 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 39 \\ 152 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 13 \\ 22 \end{bmatrix}$$

$$= \begin{bmatrix} N \\ W \end{bmatrix}$$

Cipher Text = NW

2) Plain Text: CIPHER

Key: HILL

Solution:

Convert key into 2×2 matrix and then convert it into numeric form ($A = 0, B = 1 \dots\dots Z = 25$)

$$K = \begin{bmatrix} H & I \\ L & L \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ 11 & 11 \end{bmatrix}$$

Here, plain text of 2×1 is only possible. So, break the given plain text into 3 parts of a 2×1 matrix and then convert it into numeric form.

$$P_1 = \begin{bmatrix} C \\ I \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} P \\ H \end{bmatrix} = \begin{bmatrix} 15 \\ 7 \end{bmatrix}$$

$$P_3 = \begin{bmatrix} E \\ R \end{bmatrix} = \begin{bmatrix} 4 \\ 17 \end{bmatrix}$$

$$E = E_1 + E_2 + E_3$$

$$= K P_1 \bmod 26 + K P_2 \bmod 26 + K P_3 \bmod 26$$

$$= \begin{bmatrix} 7 & 8 \\ 11 & 11 \end{bmatrix} \begin{bmatrix} 2 \\ 8 \end{bmatrix} \bmod 26 + \begin{bmatrix} 7 & 8 \\ 11 & 11 \end{bmatrix} \begin{bmatrix} 15 \\ 7 \end{bmatrix} \bmod 26 + \begin{bmatrix} 7 & 8 \\ 11 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 17 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 78 \\ 110 \end{bmatrix} \bmod 26 + \begin{bmatrix} 161 \\ 242 \end{bmatrix} \bmod 26 + \begin{bmatrix} 164 \\ 231 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 0 \\ 6 \end{bmatrix} + \begin{bmatrix} 5 \\ 8 \end{bmatrix} + \begin{bmatrix} 8 \\ 23 \end{bmatrix}$$

$$= \begin{bmatrix} A \\ G \end{bmatrix} + \begin{bmatrix} F \\ I \end{bmatrix} + \begin{bmatrix} I \\ X \end{bmatrix}$$

Cipher Text = AGFIIX

2.3 TRANSPOSITION TECHNIQUE: RAIL FENCE CIPHER

2.3.1 Introduction

Transposition Technique rearranges the position of the plain text's characters. In the transposition technique, the position of the character is changed but the character's identity is not changed. Transposition is a type of encryption technique where the positions of the letters in the plaintext message are rearranged to form a ciphertext message. This technique does not alter the letters themselves but rather the order in which they appear. Rail Fence Technique and Columnar Transposition are the most commonly used transposition techniques.

2.3.2 Rail Fence Cipher Technique

The rail fence cipher (also called a zigzag cipher) is a form of transposition cipher. It derives its name from how it is encoded.

ENCRYPTION:

In the rail fence cipher, the plain text is written downwards and diagonally on successive rails of an imaginary fence. When we reach the bottom rail, we traverse upwards moving diagonally, after reaching the top rail, the direction is changed again. Thus, the alphabets of the message are written in a zig-zag manner. After each alphabet has been written, the individual rows are combined to obtain the cipher text.

For example, if the message is "THIS IS SECRET MESSAGE" and the number of rails (key) = 3 then the cipher is prepared as:

T				I				C				M				A		
	H		S		S		E		R		T		E		S		G	
		I				S				E				S				E

Its encryption will be done row-wise. Therefore, the cipher text will be: TICMAHSSERTESGISESE

DECRYPTION:

Let cipher-text = "TICMAHSSERTESGISESE", and Key = 3

Number of columns in matrix = length of cipher-text = 13

Number of rows = key = 3

Hence original matrix will be of 3*19, now marking places with text as '*' or any other symbol (-).

The decryption process for the Rail Fence Cipher involves reconstructing the diagonal grid used to encrypt the message. We start writing the message, but leaving a dash in place of the spaces yet to be occupied. Gradually, you can replace all the dashes with the corresponding letters and read off the plaintext from the table.

We start by making a grid with as many rows as the key is and as many columns as the length of the ciphertext. We then place the first letter in the top left square and dash diagonally downwards where the letters will be. When we get back to the top row, we place the next letter in the ciphertext. Continue like this across the row, and start the next row when you reach the end

Here the ciphertext received is " TICMAHSSERTESGISESE ", encrypted with a key of 3, you start by placing the "T" in the first square. You then dash the diagonal down spaces until you get back to the top row and place the "I" here.

-				-				-				-				-		
	-		-		-		-		-		-		-		-		-	
		-				-				-				-				-

T				I				C				M				A		
	-		-		-		-		-		-		-		-		-	
		-				-				-				-				-

Continuing to fill the rows you get the pattern below

T				I				C				M				A		
	H		S		S		E		R		T		E		S		G	
		-				-				-				-				

T				I				C				M				A		
	H		S		S		E		R		T		E		S		G	
		I				S				E				S				E

Therefore, Plain Text: THIS IS SECRET MESSAGE

2.4 ASYMMETRIC ENCRYPTION: DIGITAL SIGNATURE

A digital signature is an electronic, encrypted stamp of authentication on digital information such as messages. The digital signature confirms the integrity of the message. This signature ensures that the information originated from the signer and was not altered, which proves the identity of the organization that created the digital signature. Any change made to the signed data invalidates the whole signature.

The use of digital signatures is important because they can ensure end-to-end message integrity and can also provide authentication information about the originator of a message.

To be the most effective, the digital signature must be part of the application data so that it is generated at the time the message is created.

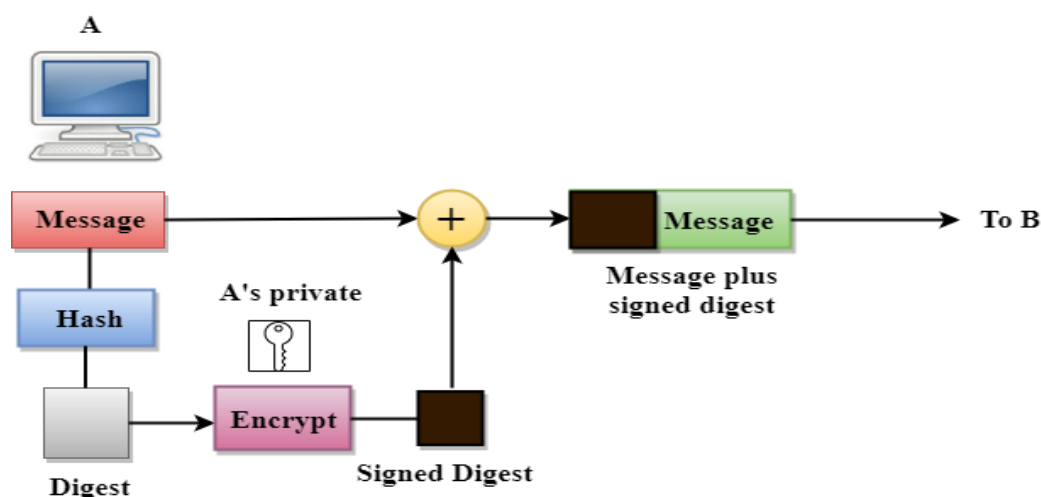
Then, the signature is verified at the time the message is received and processed.

You can choose to sign the entire message or sign parts of the message (even overlapping parts of a message can be signed).

You can choose to sign only parts of a message if a part of the message must be modified before it reaches the consumer.

In this scenario, if the entire message was signed, the whole signature is invalidated if even one part of the message is modified.

A digital signature for an electronic message is created by using a form of cryptography and is equivalent to a personal signature on a written document. A digital certificate owner combines the data to be signed with their private key and then transforms the data with an algorithm. The recipient of the message uses the corresponding certificate public key to decrypt the signature. The public key decryption also verifies the integrity of the signed message and verifies the sender as the source. Only the organization with the private key can create the digital signature. However, anyone who has access to the corresponding public key can verify the digital signature.



The steps of the digital signature process are as follows:

- 1) The sender computes a message digest (with an algorithm such as RSA or SHA1) and then encrypts the digest with their private key, which forms the digital signature. Multiple signatures and signature formats can be attached to a message, each referencing different (or even overlapping) parts of the message.
- 2) The sender transmits the digital signature with the message.
- 3) The receiver decrypts the digital signature with the public key of the sender, thus regenerating the message digest.
- 4) The receiver computes a message digest from the message data that was received and verifies that the two digests are the same. If these digests match, the message is both intact and authentic.

When a content creator digitally signs a message, the signature must meet the following criteria to be valid:

- 1) The certificate that is associated with the digital signature is current (not expired).
- 2) The certificate that is associated with the digital signature is issued to the signing publisher by a reputable certificate authority (CA). The CA signs certificates that it issues. The signature consists of a data string that is encrypted with the private key of the CA. Any user can then verify the signature on the certificate by using the CA public key to decrypt the signature.
- 3) The publisher (the signing organization), is trusted.

A Digital Signature is used to assure:

1) Authenticity:

The identity of the organization that sent the message (the message signer) is confirmed.

2) Integrity:

The message content was not changed or tampered with since it was digitally signed.

3) Nonrepudiation:

The origin of the signed content is verified to all parties so the message signer cannot deny association with the signed content.

UNIT-3

NETWORK SECURITY

3.1 WORKING PRINCIPLES OF FIREWALL

3.1.1 Introduction to Firewall

A Firewall is a hardware or software to prevent a private computer or a network of computers from unauthorized access, it acts as a filter to prevent unauthorized users from accessing private computers and networks. It is a vital component of network security. It is the first line of defense for network security. A firewall has a set of rules which are applied to each packet. The rules decide if a packet can pass or whether it is discarded. It filters network packets and stops malware from entering the user's computer or network by blocking access and preventing the user from being infected. A firewall establishes a barrier between secured internal networks and outside untrusted networks, such as the Internet.

3.1.2 Five Principles of Firewall Design

Firewall design principles are critical to protect your private network and maximize your network security. Here are five principles you can use when establishing your firewall and implementing security policies.

1) Develop a Solid Security Policy

Having a proper security policy is an essential part of designing a firewall. Without it in place, it's a headache to allow users to navigate the company network and restrict intruders. This proper security policy will also help you know the proper protocol if there is a security breach and it is useful for reporting security threats. A properly developed security policy can protect you. A solid security policy includes guidance on proper internet protocol, preventing users from using devices on public networks and recognizing external threats. Simply having a security policy is only the first step. In addition to establishing security policies, one should have frequent training and refreshers for all employees.

2) Use a Simple Design

If you have a complex design, you'll need to find complex solutions anytime a problem arises. A simple design helps alleviate some of the pain you may feel when a problem comes up. Also, complex designs are more likely to suffer from configuration errors that can open paths for external attacks.

3) Choose the Right Device

You need to have the right tools to do the job. If you use the wrong device, you have the wrong tools and are at a disadvantage from the start. Using the right part that fits your design will help you create the best firewall for your network.

4) Build a Layered Defense

Firewalls should have layers to properly protect your network. A multi-layered defense creates a complicated protection system that hackers can't easily break through. Creating layers builds an effective defense and will keep your network safe.

5) Build Protection Against Internal Threats

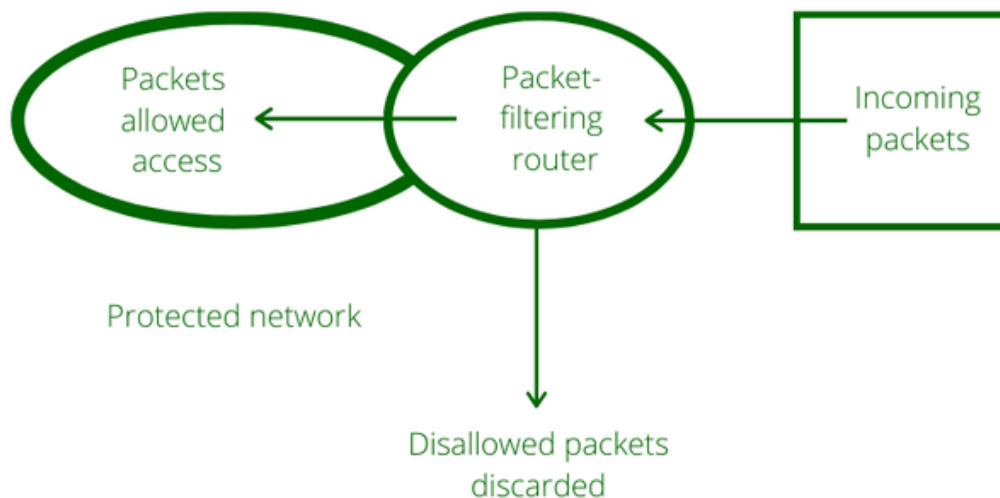
Don't just focus on attacks from external sources. A large percentage of data breaches are the result of internal threats and carelessness. Mistakes made by those internally can open your network to attacks from outside sources. Implementing proper security solutions for your internal network can help prevent this from happening. Something as simple as accessing a web server can expose your network if you aren't protected internally as well as externally.

3.1.3 Types of Firewalls:

There are five main types of firewalls depending upon their operational method:

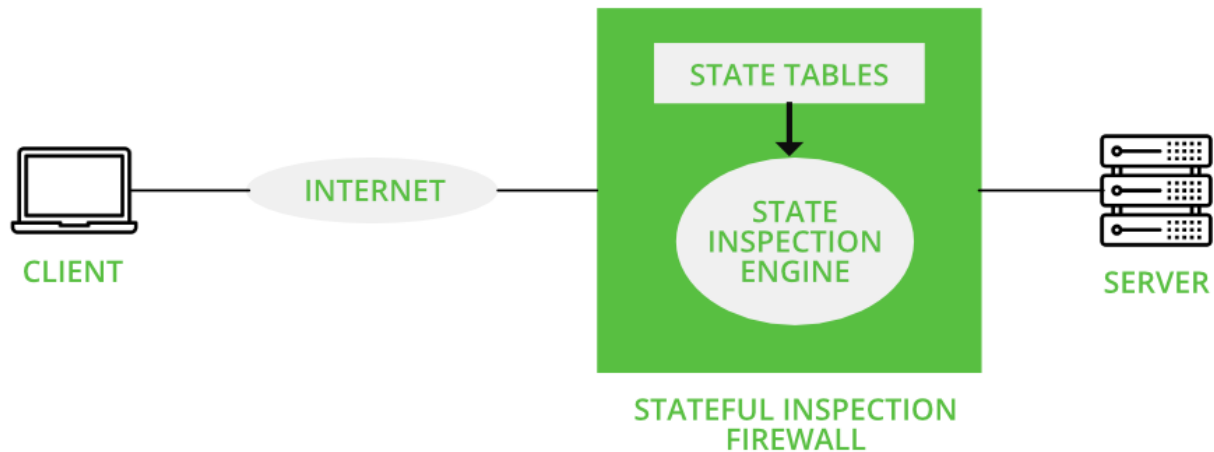
- 1) Stateless or Packet Filtering Firewall
- 2) Stateful Inspection Firewall
- 3) Circuit-Level Gateway
- 4) Application-Level Gateway
- 5) Next-Generation Firewall (NGFW)

1) Stateless or Packet Filtering Firewall:



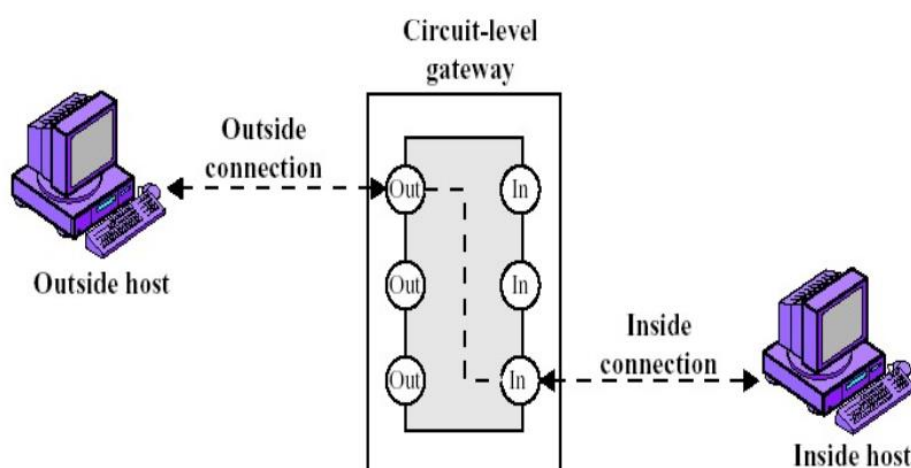
A packet filtering firewall protects the network by analyzing traffic in the transport protocol layer where applications can communicate with each other using specific protocols like Transmission Control Protocol (TCP) and User Datagram Protocol (UDP). The firewall examines the data packets at this layer, looking for malicious code that can infect the network or device. If a data packet is identified as a potential threat, the firewall rejects it. Small businesses that need basic protection from existing cyber threats can benefit from a packet-filtering firewall. Packet-filtering firewalls analyze surface-level details only and do not open the packet to examine the actual data (content payload). They check each one in isolation for destination and IP address, packet type, port number and network protocols but not in context with current traffic streams.

2) Stateful Inspection Firewall:



Stateful inspection firewalls operate at the gateway between systems behind the firewall and resources outside the enterprise network. Stateful inspection firewalls are situated at Layers 3 and 4 of the OSI model. State-aware firewalls examine each packet (stateful inspection) and track and monitor the state of active network connections while analyzing incoming traffic for potential risks. The “state” is the most recent or immediate status of a process or application. Stateful firewalls can detect attempts by unauthorized individuals to access a network, as well as analyze the data within packets to see if they contain malicious code. They are very effective at defending the network against denial of service (DoS) attacks. It is important to monitor the state and context of network communications because this information can be used to identify threats either based on where they are coming from, where they are going, or the content of their data packets. This method offers more security than either packet filtering or circuit monitoring alone but exacts a greater toll on network performance.

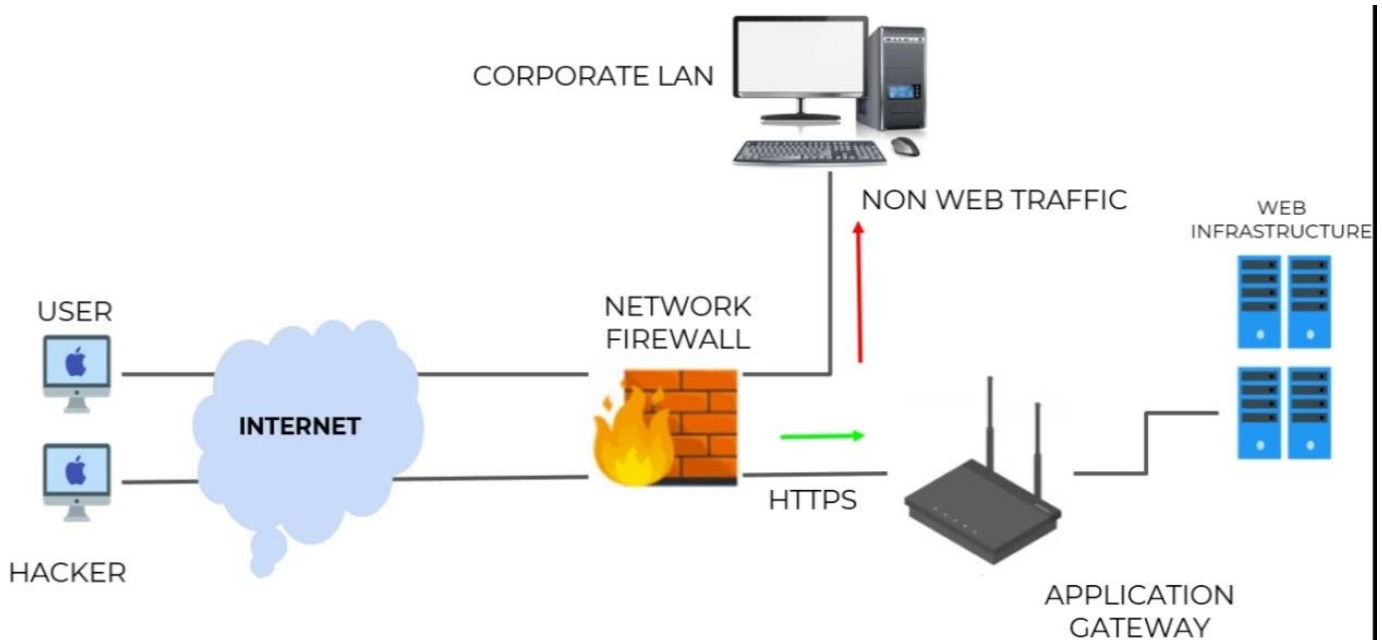
3) Circuit-level Gateway:



Circuit-level gateways operate at the session layer of the OSI model. In the OSI model, a handshake must occur before information can be passed from one cyber entity to another. Circuit-level gateways determine the security of an established connection between the transport layer and the application layer of the TCP/Internet Protocol (TCP/IP) stack by monitoring TCP handshakes between local and remote hosts. While circuit-level gateways have minimal impact on network performance, a data packet containing malware can bypass a

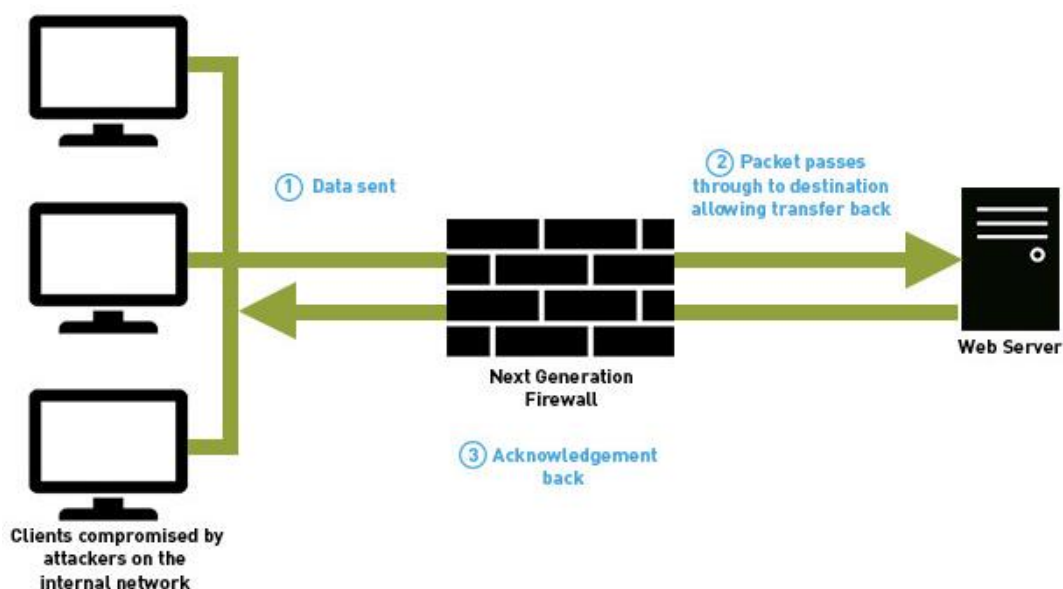
circuit-level gateway easily even if it has a legitimate TCP handshake. This is because circuit-level gateways do not filter the content in data packets. To fill this gap, circuit-level gateways are often paired with another type of firewall that performs content filtering.

4) Application-level Gateway:



It is also referred to as a “proxy firewall” and serves as an intermediate between internal and external systems. An application-level gateway operates at the application layer, the highest of the OSI model. It employs deep packet inspection (DPI) on incoming traffic to check both data packet payloads (content) and headers. This firewall makes sure that only valid data exists at the application level before allowing it to pass through. Application-level gateways follow a set of application-specific policies to determine which communications are allowed to pass to and from an application. They help protect a network by masking clients’ requests before sending them to the host. When network anonymity is required, application-level gateways are often in play. They are ideal for securing web apps from bad actors (malicious intent).

5) Next-Generation Firewall (NGFW):



A Next-Generation Firewall (NGFW) is the only type of firewall that provides the capabilities to protect modern businesses against emerging cyber threats. As malware and threats have become more difficult to detect at the access point, NGFW security has evolved to span the network and monitor behavior and intent. NGFWs provide functions like deep-packet inspection, intrusion prevention (IPS), advanced malware detection, application control and provide overall network visibility through inspection of encrypted traffic. They can be found anywhere from an on-premises network edge to its internal boundaries and can also be employed on public or private cloud networks. NGFWs CPU-intensive capabilities include decryption at a very high-performance level, deep-packet inspection post decryption, detection of malicious URLs, identification of command-and-control activities and download of malware and threat correlation. Due to these advanced security capabilities, NGFWs are critical for heavily regulated industries such as finance or healthcare and are often integrated with other security systems and SIEMs for end-to-end surveillance and reporting.

3.1.4 Characteristics of Firewall

- 1) **Physical Barrier:** A firewall does not allow any external traffic to enter a system or a network without its allowance. A firewall creates a choke point for all the external data trying to enter the system or network and hence can easily block access if needed.
- 2) **Multi-Purpose:** A firewall has many functions other than security purposes. It configures domain names and Internet Protocol (IP) addresses. It also acts as a network address translator. It can act as a meter for internet usage.
- 3) **Flexible Security Policies:** Different local systems or networks need different security policies. A firewall can be modified according to the requirement of the user by changing its security policies.
- 4) **Security Platform:** It provides a platform from which any alert to the issue related to security or fixing issues can be accessed. All the queries related to security can be kept under check from one place in a system or network.
- 5) **Access Handler:** Determines which traffic needs to flow first according to priority or can change for a particular network or system. Specific action requests may be initiated and allowed to flow through the firewall.

3.1.5 Advantages of Firewall

- 1) **Blocks Infected Files:** While surfing the internet we encounter many unknown threats. Any friendly-looking file might have malware in it. The firewall neutralizes this kind of threat by blocking file access to the system.
- 2) **Stop Unwanted Visitors:** A firewall does not allow a cracker to break into the system through a network. A strong firewall detects the threat and then stops the possible loophole that can be used to penetrate through security into the system.

- 3) **Prevents Email spamming:** In this too many emails are sent to the same address leading to the server crashing. A good firewall blocks the spammer source and prevents the server from crashing.
- 4) **Control of Network Access:** By limiting access to specified individuals or groups for particular servers or applications, firewalls can be used to restrict access to particular network resources or services.
- 5) **Monitoring of Network Activity:** Firewalls can be set up to record and keep track of all network activity. This information is essential for identifying and looking into security problems and other kinds of shady behavior.

3.1.6 Disadvantages of Firewall

- 1) **Infected Files:** In the modern world, we come across various kinds of files through emails or the internet. Most of the files are executable under the parameters of an operating system. It becomes impossible for the firewall to keep track of all the files flowing through the system.
- 2) **User Restriction:** Restrictions and rules implemented through a firewall make a network secure but they can make work less effective when it comes to a large organization or a company. Even making a slight change in data can require a permit from a person of higher authority making work slow. The overall productivity drops because of all of this.
- 3) **System Performance:** A software-based firewall consumes a lot of resources of a system. Using the RAM and consuming the power supply leaves very less resources for the rest of the functions or programs. The performance of a system can experience a drop. On the other hand, a hardware firewall does not affect the performance of a system much, because it's very less dependent on the system resources.
- 4) **Complexity:** Setting up and keeping up a firewall can be time-consuming and difficult, especially for bigger networks or companies with a wide variety of users and devices.
- 5) **Cost:** Purchasing many devices or add-on features for a firewall system can be expensive, especially for businesses.

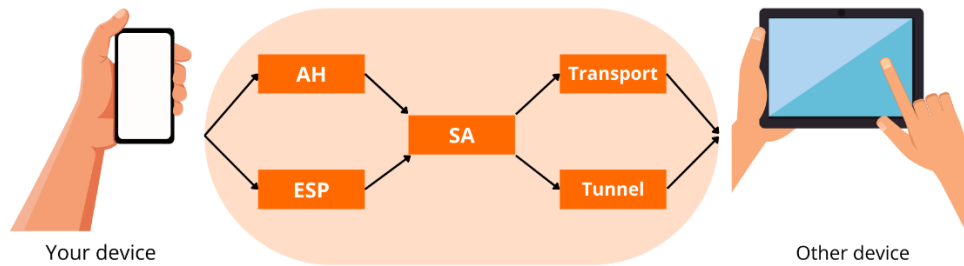
3.2 INTERNET PROTOCOL SECURITY AND ITS USE IN SECURE COMMUNICATION

3.2.1 Introduction

IPsec is a set of protocols to secure internet communication at the network layer. It was developed by the Internet Engineering Task Force (IETF) to provide a secure way to exchange data over the Internet, ensuring that sensitive information is protected from unauthorized access, interception, or modification.

IPsec is the short acronym for Internet Protocol Security. The "IP" stands for Internet Protocol, which is the main routing protocol used on the Internet for sending data to its destination using IP addresses. The "sec" stands for secure, as it provides encryption and authentication to the data transmission process, making it more secure.

IPSEC



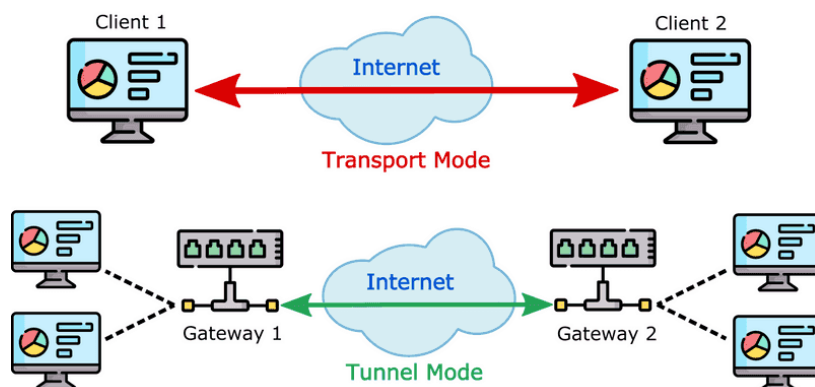
IPsec uses a variety of protocols to establish secure connections and protect data during transmission. IPsec is not one protocol but a suite of protocols. The suite includes the following:

- 1) Authentication Header (AH):** It provides data integrity and authentication and ensures that the transmitted data has not been modified or tampered with. Yet, it does not encrypt data.
- 2) Encapsulating Security Protocol (ESP):** It encrypts both the IP header and the payload of each packet unless transport mode is used, in which case only the payload is encrypted. In addition, ESP adds its header and a trailer to each data packet.
- 3) Security Association (SA):** An SA is a set of security parameters defining how two devices communicate securely. It includes information such as the encryption algorithm, authentication method, and key size. One of the most commonly used SA protocols is the Internet Key Exchange (IKE). IPsec uses port 500 for its IKE (Internet Key Exchange) protocol.

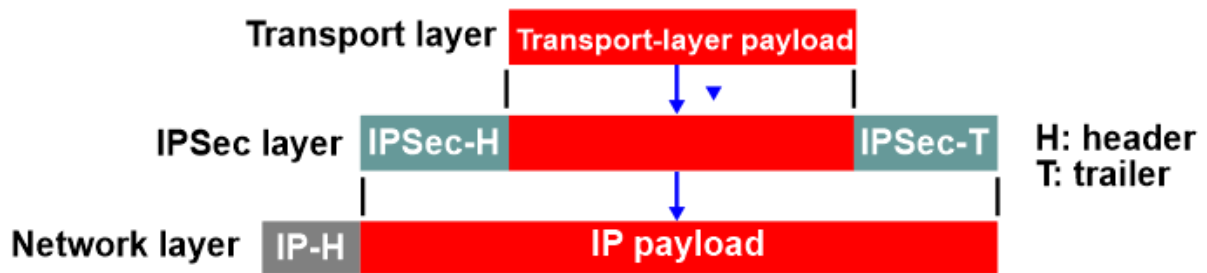
3.2.2 Modes of IPsec

IPsec operates in one of two different modes: Transport Mode or Tunnel Mode

IPsec Modes

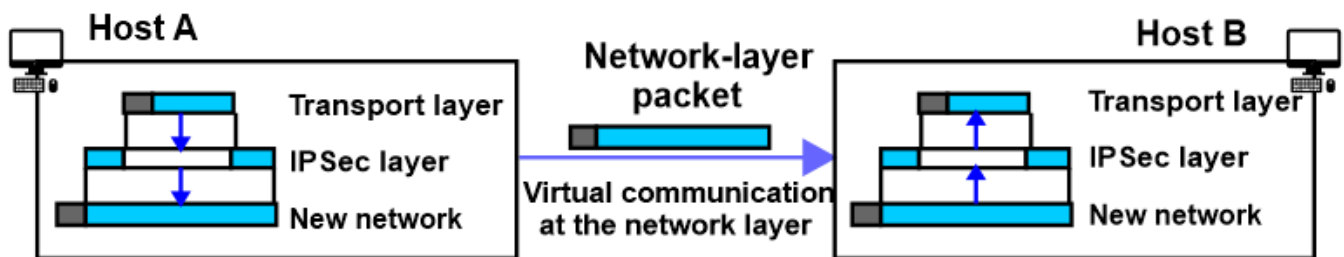


Transport Mode



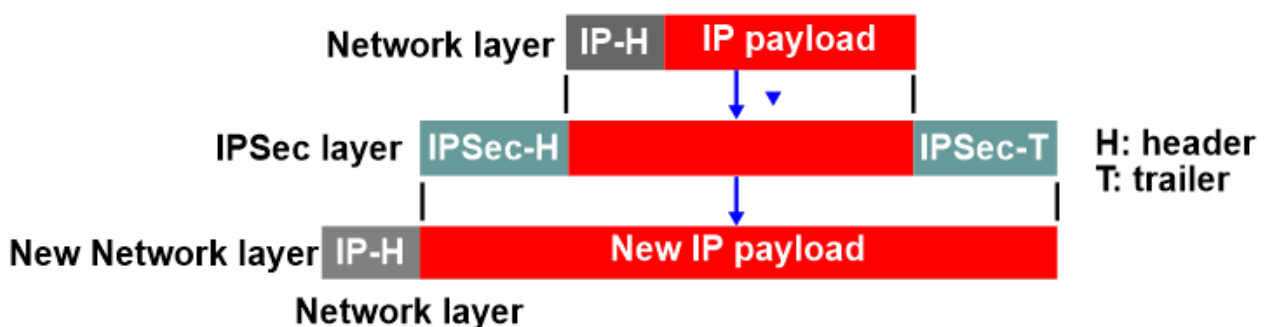
In the transport mode, IPSec protects what is delivered from the transport layer to the network layer. In other words, the transport mode protects the network layer payload, the payload to be encapsulated in the network layer. Note that the transport mode does not protect the IP header. The transport mode does not protect the whole IP packet; it protects only the packet from the transport layer (the IP layer payload). In this mode, the IPSec header and trailer are added to the information coming from the transport layer. The IP header is added later.

Transport mode in action



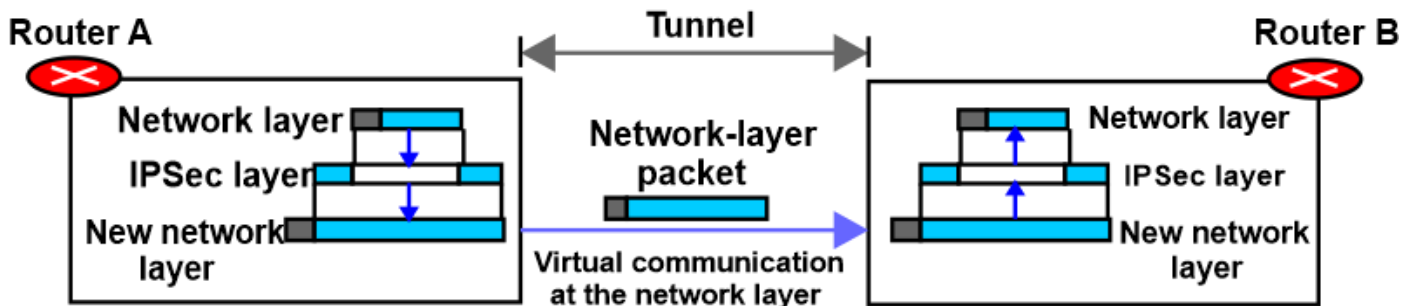
The transport mode is normally used when we need host-to-host (end-to-end) protection of data. The sending host uses IPSec to authenticate and/or encrypt the payload delivered from the transport layer. The receiving host uses IPSec to check the authentication and/or decrypt the IP packet and deliver it to the transport layer. The figure above shows this concept.

Tunnel Mode



In the tunnel mode, IPSec protects the entire IP packet. It takes an IP packet, including the header, applies IPSec security methods to the entire packet, and then adds a new IP header as shown in figure below. The new IP header, as we will see shortly, has different information than the original IF header.

Tunnel mode in action



The tunnel mode is normally used between two routers, between a host and a router, or between a router and a host as shown in figure above. In other words, we use the tunnel mode when either the sender or the receiver is not a host. The entire original packet is protected from intrusion between the sender and the receiver. It's as if the whole packet goes through an imaginary tunnel. IPsec in tunnel mode protects the original IP header.

3.2.3 Features of IPsec

- 1) **Authentication:** IPsec provides authentication of IP packets using digital signatures or shared secrets. This helps ensure that the packets are not tampered with or forged.
- 2) **Confidentiality:** IPsec provides confidentiality by encrypting IP packets, preventing eavesdropping on the network traffic.
- 3) **Integrity:** IPsec provides integrity by ensuring that IP packets have not been modified or corrupted during transmission.
- 4) **Key management:** IPsec provides key management services, including key exchange and key revocation, to ensure that cryptographic keys are securely managed.
- 5) **Tunneling:** IPsec supports tunneling, allowing IP packets to be encapsulated within another protocol, such as GRE (Generic Routing Encapsulation) or L2TP (Layer 2 Tunneling Protocol).
- 6) **Flexibility:** IPsec can be configured to provide security for a wide range of network topologies, including point-to-point, site-to-site, and remote access connections.
- 7) **Interoperability:** IPsec is an open standard protocol, which means that it is supported by a wide range of vendors and can be used in heterogeneous environments.

3.2.4 Advantages of IPsec

- 1) **Strong security:** IPsec provides strong cryptographic security services that help protect sensitive data and ensure network privacy and integrity.
- 2) **Wide compatibility:** IPsec is an open standard protocol that is widely supported by vendors and can be used in heterogeneous environments.

3) Flexibility: IPSec can be configured to provide security for a wide range of network topologies, including point-to-point, site-to-site, and remote access connections.

4) Scalability: IPSec can be used to secure large-scale networks and can be scaled up or down as needed.

5) Improved Network Performance: IPSec can help improve network performance by reducing network congestion and improving network efficiency.

3.2.5 Disadvantages of IPSec

1) Configuration complexity: IPSec can be complex to configure and requires specialized knowledge and skills.

2) Compatibility issues: IPSec can have compatibility issues with some network devices and applications, which can lead to interoperability problems.

3) Performance impact: IPSec can impact network performance due to the overhead of encryption and decryption of IP packets.

4) Key management: IPSec requires effective key management to ensure the security of the cryptographic keys used for encryption and authentication.

5) Limited protection: IPSec only protects IP traffic and other protocols such as ICMP, DNS and routing protocols may still be vulnerable to attacks.

3.2.6 Difference between IPv4 and IPv6

Parameters	IPv4	IPv6
Address length	IPv4 is a 32-bit address.	IPv6 is a 128-bit address.
Fields	IPv4 is a numeric address that consists of 4 fields which are separated by a dot (.).	IPv6 is an alphanumeric address that consists of 8 fields, which are separated by a colon.
Classes	IPv4 has 5 different classes of IP addresses that includes Class A, Class B, Class C, Class D and Class E.	IPv6 does not contain classes of IP addresses.
Number of IP addresses	IPv4 has a limited number of IP addresses.	IPv6 has a large number of IP addresses.
VLSM	It supports VLSM (Virtual Length Subnet Mask). Here, VLSM means that Ipv4 converts IP addresses into a subnet of different sizes.	It does not support VLSM.
Address configuration	It supports manual and DHCP configuration.	It supports manual, DHCP, auto-configuration, and renumbering.

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Address space	It generates 4 billion unique addresses	It generates 340 undecillion unique addresses.
End-to-end connection integrity	In IPv4, end-to-end connection integrity is unachievable.	In the case of IPv6, end-to-end connection integrity is achievable.
Security features	In IPv4, security depends on the application. This IP address is not developed in keeping the security feature in mind.	In IPv6, IPSEC is developed for security purposes.
Address representation	In IPv4, the IP address is represented in decimal.	In IPv6, the representation of the IP address in hexadecimal.
Fragmentation	Fragmentation is done by the senders and the forwarding routers.	Fragmentation is done by the senders only.
Packet flow identification	It does not provide any mechanism for packet flow identification.	It uses flow label field in the header for the packet flow identification.
Checksum field	The checksum field is available in IPv4.	The checksum field is not available in IPv6.
Transmission scheme	IPv4 is broadcasting.	On the other hand, IPv6 is multicasting, which provides efficient network operations.
Encryption and Authentication	It does not provide encryption and authentication.	It provides encryption and authentication.
Number of octets	It consists of 4 octets.	It consists of 8 fields, and each field contains 2 octets. Therefore, the total number of octets in IPv6 is 16.

3.3 VARIOUS TYPES OF IDSs

3.3.1 Introduction

An Intrusion Detection System (IDS) is a system that monitors network traffic for suspicious activity and issues alerts when such activity is discovered. It is a software application that scans a network or a system for harmful activity or policy breaches. Any malicious venture or violation is normally reported either to an administrator or collected centrally using a Security Information and Event Management (SIEM) system. A SIEM system integrates outputs from multiple sources and uses alarm filtering techniques to differentiate malicious activity from false alarms.

Although intrusion detection systems monitor networks for potentially malicious activity, they are also disposed to false alarms. Hence, organizations need to fine-tune their IDS products when they first install them. It means properly setting up the intrusion detection systems to recognize what normal traffic on the network looks like as compared to malicious activity.

IDS and firewall both are related to network security but an IDS differs from a firewall as a firewall looks outwardly for intrusions to stop them from happening. Firewalls restrict access between networks to prevent intrusion and if an attack is from inside the network it doesn't signal. An IDS describes a suspected intrusion once it has happened and then signals an alarm. The most optimal and common position for an IDS is behind the firewall. Although this position varies considering the network.

3.3.2 Detection Methods of IDS

There are mainly 3 detection methods of IDS. These methods define how the IDS analyzes data to identify potential intrusions.

1) Signature-based IDS:

Signature-based IDS detects the attacks based on specific patterns such as the number of bytes or number of 1's or number of 0's in the network traffic. It also detects based on the already known malicious instruction sequence that is used by the malware. The detected patterns in the IDS are known as signatures. Signature-based IDS can easily detect the attacks whose pattern (signature) already exists in the system, but it is quite difficult to detect new malware attacks as their pattern (signature) is not known.

2) Anomaly-based IDS:

Anomaly-based IDS was introduced to detect unknown malware attacks as new malware is developed rapidly. In anomaly-based IDS there is the use of machine learning to create a trustful activity model and anything coming is compared with that model and it is declared suspicious if it is not found in the model. Machine learning-based method has a better- generalized property in comparison to signature-based IDS as these models can be trained according to the applications and hardware configurations.

3) Hybrid IDS:

A hybrid IDS combines both anomaly-based and signature-based detection methods to address the limitations of each approach. A hybrid system leverages signature-based detection for known threats and anomaly-based detection for novel attacks. This enhances the overall effectiveness of intrusion detection.

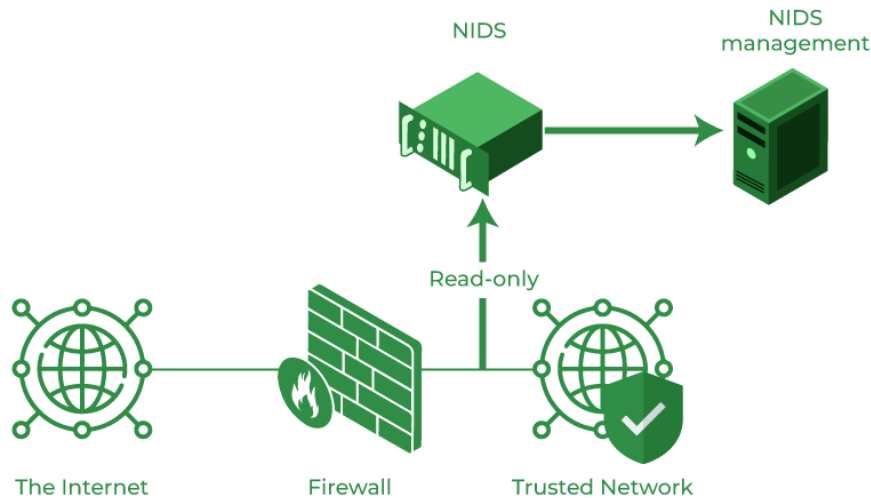
3.3.3 Classification of IDS

Intrusion Detection Systems are classified into 5 types based on deployment and data source:

1) Network Intrusion Detection System (NIDS):

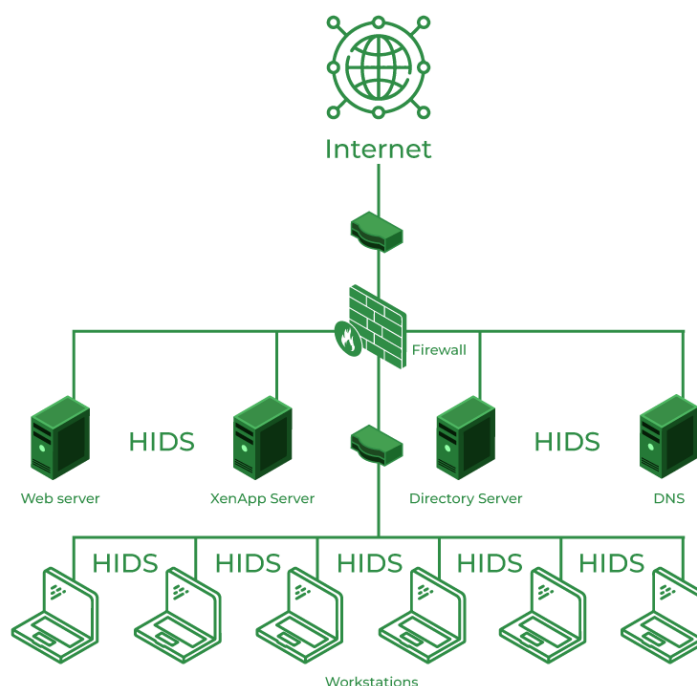
Network intrusion detection systems (NIDS) are set up at a planned point within the network to examine traffic from all devices on the network. It performs an observation of passing traffic on the entire subnet and matches the traffic that is passed on the subnets to the collection of known attacks. Once an attack is identified or

abnormal behavior is observed, the alert can be sent to the administrator. An example of a NIDS is installing it on the subnet where firewalls are located in order to see if someone is trying to crack the firewall.



2) Host Intrusion Detection System (HIDS):

Host Intrusion Detection System (HIDS) run on independent hosts or devices on the network. A HIDS monitors the incoming and outgoing packets from the device only and will alert the administrator if suspicious or malicious activity is detected. It takes a snapshot of existing system files and compares it with the previous snapshot. If the analytical system files were edited or deleted, an alert is sent to the administrator to investigate. An example of HIDS usage can be seen on mission-critical machines, which are not expected to change their layout.



3) Protocol-based Intrusion Detection System (PIDS):

A Protocol-Based Intrusion Detection System (PIDS) comprises a system or agent that would consistently reside at the front end of a server, controlling and interpreting the protocol between a user/device and the

server. It is trying to secure the web server by regularly monitoring the HTTPS protocol stream and accepting the related HTTP protocol. As HTTPS is unencrypted and before instantly entering its web presentation layer then this system would need to reside in this interface, between to use the HTTPS.

4) Application Protocol-based Intrusion Detection System (APIDS):

An Application Protocol-Based Intrusion Detection System (APIDS) is a system or agent that generally resides within a group of servers. It identifies the intrusions by monitoring and interpreting the communication on application-specific protocols. For example, this would monitor the SQL protocol explicitly to the middleware as it transacts with the database in the web server.

5) Hybrid Intrusion Detection System:

Hybrid intrusion detection system is made by the combination of two or more approaches to the intrusion detection system. In the hybrid intrusion detection system, the host agent or system data is combined with network information to develop a complete view of the network system. The hybrid intrusion detection system is more effective in comparison to the other intrusion detection system. Prelude is an example of Hybrid IDS.

3.4 DISTINGUISH HOST-BASED IDS AND NETWORK-BASED IDS

Parameters	HIDS	NIDS
Full Form	Host Intrusion Detection System.	Network Intrusion Detection System.
Type	It doesn't work in real-time.	It operates in real-time.
Concern	HIDS is related to just a single system, as the name suggests it is only concerned with the threats related to the Host system/computer.	NIDS is concerned with the entire network system, NIDS examines the activities and traffic of all the systems in the network.
Installation Point	HIDS can be installed on every computer or server i.e., anything that can serve as a host.	NIDS being concerned with the network is installed at places like routers or servers as these are the main intersection points in the network system.
Execution Process	HIDS operates by taking a snapshot of the current status of the system and comparing it against some already stored	NIDS works in real-time by closely examining the data flow and immediately reporting anything unusual.

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	malicious tagged snapshots stored in the database, this clearly shows that there is a delay in its operation and activities.	
Information about attack	HIDS are more informed about the attacks as they are associated with system files and processes.	As the network is very large making it hard to keep track of the integrating functionalities, they are less informed of the attacks.
Ease of Installation	As it needs to be installed on every host, the installation process can be tiresome.	Few installation points make it easier to install NIDS.
Response Time	Response time is slow.	Response time is fast.

3.5 HIDS AND NIDS COMPONENTS

HIDS Components	Description
Agent Software	Installed on the host; collects data like logs, file changes, or system calls.
File Integrity Checker	Detects unauthorized changes to system files or configurations.
Log Analyzer	Monitors system and application logs for suspicious events.
Policy Database	Contains rules and baselines for what is considered “normal” activity.
Alerting Mechanism	Notifies administrators when anomalies are detected.
Response System	May take predefined actions such as killing a process or alerting the user.
Update/Signature Module	Updates the system with the latest attack signatures or behavioral rules.

NIDS Components	Description
Network Sensors	Capture and monitor network packets at strategic points (e.g., routers, switches).
Packet Sniffer	Analyzes the raw data in packet headers and payloads.
Signature Database	Contains patterns of known threats for comparison.
Protocol Analyzer	Interprets protocol behavior to detect abnormalities.
Traffic Logger	Records traffic data for further analysis.
Alert/Notification System	Sends alerts when suspicious traffic is detected.
Management Console	Provides a user interface to view alerts, logs, and configure rules.

3.6 ADVANTAGES AND DISADVANTAGES OF HIDS, NIDS

Advantages of HIDS

- 1) **Deep visibility:** Monitors system-level activity like file changes, logins, and system calls.
- 2) **Detects insider threats:** Effective at catching internal misuse or malware on a specific machine.
- 3) **Encrypted traffic monitoring:** Can analyze data after decryption since it's running on the host.
- 4) **Detailed logs:** Provides granular information about what happened on a host.

Disadvantages of HIDS

- 1) **High resource usage:** Consumes CPU, RAM, and disk on the host.
- 2) **Harder to manage at scale:** Needs to be installed and updated on every endpoint.
- 3) **Limited network visibility:** Cannot detect threats that don't interact with the monitored host.
- 4) **May be disabled by attackers:** If compromised, the HIDS agent can be tampered with.

Advantages of NIDS

- 1) **Broad network coverage:** Can monitor traffic across multiple systems and subnets.
- 2) **Centralized deployment:** Easy to manage and scale from a few network points.
- 3) **Real-time detection:** Identifies threats like port scans, DDoS, and exploits as they happen.
- 4) **Non-intrusive to hosts:** Doesn't affect the performance of endpoints.

Disadvantages of NIDS

- 1) **Cannot see encrypted data:** Cannot inspect payloads of encrypted traffic (e.g., HTTPS).
- 2) **Limited host visibility:** Doesn't know what happens inside a host (e.g., file tampering).
- 3) **Blind spots:** May miss threats in switched or segmented networks if not properly placed.
- 4) **Higher false positives:** Lacks deep context, which can lead to more alerts for normal network traffic.

UNIT-4

NETWORK ADMINISTRATION PROTOCOLS AND SERVICES

4.1 DIRECTORY SERVICE

Directory Service is defined as a network service that identifies all resources on a network and makes them accessible to users and applications. (Resources include email address, domain name, computer and peripheral devices such as printer). In most networks, you optimize the function of different services by hosting them on different computers. Doing so makes sense. Putting all services on one computer may be a bit complex. Moreover, you can achieve optimal performance, more reliability and higher security by segregating network services in various ways.

Most networks have quite a few services that need to be provided, and often these services run on different servers. Even a relatively simple network now offers the following services:

- File storage and sharing and Printer sharing
- E-mail services
- Web hosting, both for the Internet and an intranet
- Database server services
- Specific application servers
- Internet connectivity
- Fax services
- Domain Name System (DNS) service, Windows Internet Naming Service
- Dynamic Host Configuration Protocol (DHCP) services
- Centralized virus-detection services
- Backup and restore services

Directory services were invented to remove complexities from the network. Basically, directory services work just like a phone book. Instead of using a name to look up an address and phone number in a phone book, you query the directory service for a service name (such as the name of a network folder or a printer), and the directory service tells you where the service is located.

You can also request directory services by property. For instance, if you request the directory service for all items that are “printers,” it can return a complete list, no matter where the printers are located in the organization.

Even better, directory services enable you to browse all the resources on a network easily. We don’t need to make separate user lists on each server. Directory services eliminate this kind of complexity. By sharing itself with all other servers.

To provide the above redundancy directory services usually run on multiple servers in an organization, with each server having an entire copy of the directory service database. To avoid loss of data, it requires making a backup directory server.

There are five types of well-known directory services:

- 1) Novell Directory Services (NDS)
- 2) Microsoft's Windows NT Domains
- 3) Microsoft's Active Directory
- 4) X.500 Directory Access Protocol
- 5) Lightweight Directory Access Protocol

4.2 DIFFERENT DIRECTORIES ACCESS PROTOCOLS

4.2.1 Novell Directory Service (NDS)

Novell eDirectory has been available since 1993, introduced as NDS as part of NetWare 4.x. This product was a real boon and was rapidly implemented in Novell networks, particularly in larger organizations that had many NetWare servers and needed its capabilities. eDirectory is a reliable, robust directory service that has continued to evolve(developed gradually) since its introduction. eDirectory uses a Master/Slave approach to directory servers and also allows partitioning of the tree.

In addition to running on Novell network operating systems, eDirectory is also available for Windows, Solaris, AIX(Advanced Interactive eXecutive) and Linux systems. The product's compatibility with such a variety of systems makes it a good choice for managing all these platforms under a single directory structure.

The NDS tree is managed from a client computer logged into the network with administrative privileges. You can either use a graphical tool designed to manage the tree called NWAdmin or a text-based tool called NET ADMIN. Both allow full management of the tree, although the graphical product is much easier to use.

The NDS tree contains a number of different object types. The standard directory service types are countries, organizations and organizational units. The system also has objects to represent NetWare security groups, NetWare servers and NetWare server volumes.

4.2.2 Windows NT Domains

The Windows NT (New Technology) domain model breaks an organization into chunks called domains, all of which are part of an organization. The domains are usually organized geographically, which helps minimize domain-to-domain communication requirements across WAN links, although you're free to organize domains as you wish.

Each domain is controlled by a primary domain controller (PDC), which might have one or more backup domain controllers (BDCs) to kick in if the PDC fails.

All changes within the domain are made to the PDC, which then replicates those changes to any BDCs. BDCs are read-only, except for valid updates received from the PDC. In case of a PDC failure, BDCs automatically continue authenticating users. To make administrative changes to a domain that suffers PDC failure, any of the BDCs can be promoted to PDC. Once the PDC is ready to come back online, the promoted BDC can be demoted back to BDC status.

Windows NT domains can be organized into one of four domain models:

- 1) **Single domain:** In this model, only one domain contains all network resources.
- 2) **Master domain:** The master model usually puts users in the top-level domain and then places network resources, such as shared folders or printers, in lower-level domains (called resource domains). In this model, the resource domains trust the master domain.
- 3) **Multiple master domain:** This is a slight variation on the master domain model, in which users might exist in multiple master domains, all of which trust one another, and in which resources are located in resource domains, all of which trust all the master domains.
- 4) **Complete trust:** This variation of the single-domain model spreads users and resources across all domains, which all trust each other.

You can choose an appropriate domain model depending on the physical layout of the network, the number of users to be served and other factors. (If you're planning a domain model, you should review the white papers on Microsoft's website for details on planning large domains, because the process can be complex.)

Explicit trust relationships must be maintained between domains using the master or multiple master domain models and must be managed on each domain separately. Maintaining these relationships is one of the biggest difficulties in the Windows NT domain structure approach, at least for larger organizations. If you have 100 domains, you must manage the 99 possible trust relationships for each domain, for a total of 9,900 trust relationships. For smaller numbers of domains (for example, less than 10 domains), management of the trust relationship is less of a problem, although it can still cause difficulties.

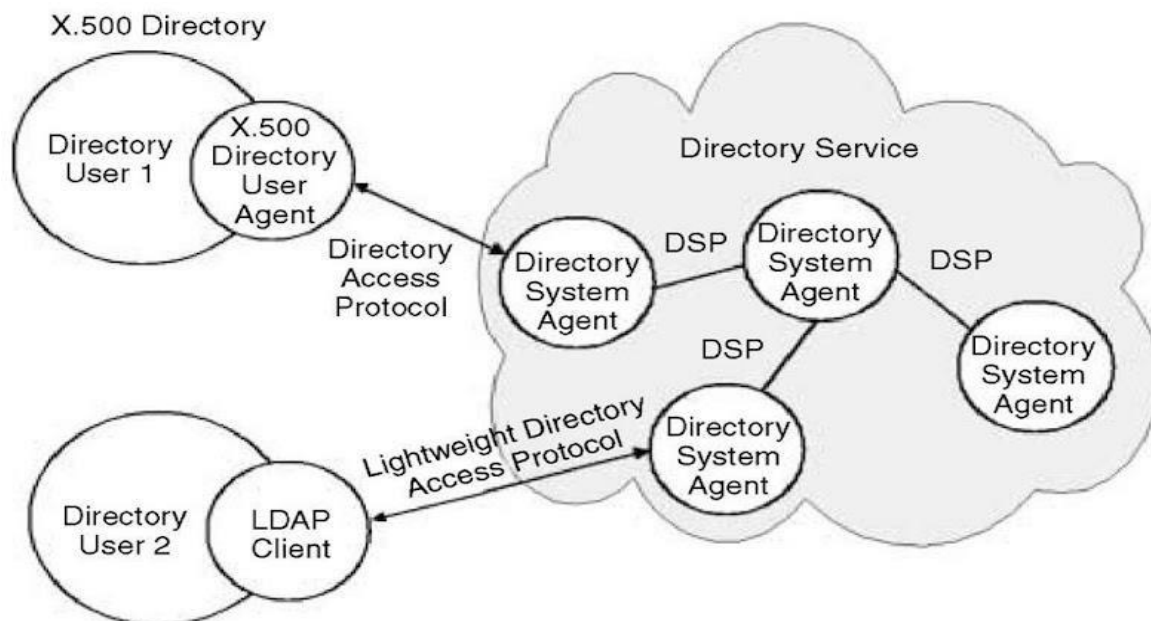
4.2.3 X.500 Directory

The X.500 standard was developed jointly by the International Telecommunications Union (ITU) and the International Standards Organization (ISO). Its purpose was to provide an international standard for directory systems. The primary concept of X.500 is that there is a single Directory Information Tree (DIT), a hierarchical organization of entries that are distributed across one or more servers, called Directory System Agents (DSA). The X.500 protocol architecture consists of a Client-Server communicating via the Open Systems Interconnection (OSI) networking model. The Client is called the Directory Service Agent (DUA) and the Server is called the Directory System Agent (DSA).

X.500 is a directory service used in the same way as a conventional name service, but it is primarily used to satisfy descriptive queries and is designed to discover the names and attributes of other users or system resources.

Users may have a variety of requirements for searching and browsing in a directory of network users, organizations and system resources to obtain information about the entities that the directory contains. The uses for such a service are likely to be quite diverse. They range from inquiries that are directly analogous to the use of telephone directories, such as a simple "white pages" access to obtain a user's electronic mail address or a "yellow pages" query aimed, for example, at obtaining the names and telephone numbers of garages

specializing in the repair of a particular make of car, to the use of the directory to access personal details such as job roles, dietary habits or even photographic images of the individuals.



The above figure shows the model for X.500.

- In the X.500 directory architecture, the client queries and receives responses from one or more servers in the servers Directory Service with the Directory Access Protocol (DAP) controlling the communication between the client and the server.
- The Directory client, called the Directory User Agent (DUA), supports users in searching or browsing through one or more directory databases and in retrieving the requested directory information. The DUA can be implemented in all kinds of user interfaces through dedicated DUA clients, Web-server gateways, e-mail applications or middleware. DUAs are currently available for virtually all types of workstations.
- Directory information is stored in a Directory System Agent (DSA), a hierarchical database designed to provide fast and efficient search and retrieval.
- The Directory System Protocol (DSP) controls the interaction between two or more DSAs. This is done in a way that allows users to access information in the Directory without knowing its exact location.
- The Directory Access Protocol (DAP) is used for controlling communication between a DUA and DSA.

4.2.4 LDAP

LDAP stands for Lightweight Directory Access Protocol.

LDAP was developed to solve the problem associated with x.500. LDAP contains 90% of the functionality of the X.500, by this it is a sub-set of X.500.

LDAP runs over TCP/IP and uses a client/server model.

The LDAP standard describes not only the layout and fields within an LDAP directory but also the methods to be used when a person logs in to a server that uses LDAP or queries or updates the LDAP directory information on an LDAP server.

An LDAP tree starts with a root, which then contains entries. Each entry can have one or more attributes. Each of these attributes has both a type and values associated with it. One example is the CommonName entry (CN), which contains at least two attributes: FirstName and Surname. All attributes in LDAP use the text string data type. Entries are broken up into a tree and managed geographically and then within each organization.

One nice feature of LDAP is an organization can build a global directory structure using a feature called referral, where LDAP directory queries that are managed by a different LDAP server are transparently routed to that server. Because each LDAP server knows its parent LDAP server and its child servers, any user anywhere in the network can access the entire LDAP tree. In fact, the user won't even know he or she is running on different servers in different locales.

The following are 4 basic models that describe LDAP:

- 1) Information Model: It defines the structure of the data stored in the directory.
- 2) Naming Model: It describes how to reference and organize the data.
- 3) Functional Model: It describes how to work with the data.
- 4) Security Model: It defines how to keep the data in the directory secure.

4.3 ACTIVE DIRECTORY

Active Directory was developed by Microsoft. Active Directory is essentially a database of network resources (known as objects) and information about each of these objects. Active Directory is fully compatible with LDAP and also with the Domain Name System (DNS) used on the Internet.

Active Directory uses a peer approach to domain controllers; all domain controllers are full participants at all times. This arrangement is called a Multimaster because there are many “master” domain controllers but no backup controllers.

Active Directory is built on a structure that allows “trees of trees,” which is called a forest. Each tree has its domain and its domain controllers.

Within a domain, separate organizational units are allowed to make administration easier and more logical. Trees are then aggregated into a larger forest structure. According to Microsoft, Active Directory can handle millions of objects through this approach.

Active Directory does not require the management of trust relationships, except when connected to Windows NT 4. x servers that are not using Active Directory. Otherwise, all domains within a tree have automatic trust relationships.

Active Directory uses two types of domain controller:

- 1) Primary Domain Controller
- 2) Backup Domain Controller

Active Directory uses LDAP protocol to access objects. Each domain controller in a domain is capable of accepting requests for changes to the domain database and replicating that information with the other DCs in the domain. The first domain that is created is referred to as the "root domain" and is at the top of the directory tree. After combining this kind of tree becomes one general tree called a domain forest. All subsequent domains will live under the root domain and are referred to as child domains. The child domain names must be unique as shown in the below figure

There are three Directory Components :

- 1) Object: Objects in the database can include printers, users, servers, clients, shares, services, etc. and are the most basic component of the directory. There are two basic types of objects in an active directory:
 - a) Container Object: It is simply an object that stores other objects. These objects essentially function as the branches of the tree.\
 - b) Leaf Object: It stands alone and can't store other objects.
- 2) Attributes: An attribute describes an object. For example, passwords and names are attributes of user objects. Different objects will have a different set of attributes that define them, however, different objects may also share attributes. For example, a printer and Windows Vista computer may both have an IP address as an attribute.
- 3) Schema: A schema defines the list of attributes that describe a given type of object. For example, let's say that all printer objects are defined by name, PDL type and speed attributes. This list of attributes comprises the schema for the object class "printers". The schema is customizable, meaning that the attributes that define an object class can be modified.

4.4 VPN AND ITS PROTOCOLS

4.4.1 Introduction

A VPN is a network that uses a public telecommunication infrastructure, such as the Internet, to provide remote offices or individual users with secure access to their organization's network.

In other words, A VPN is a network technology that creates a secure network connection over a public network such as the Internet or a private network owned by a service provider.

4.4.2 Working of VPN

A VPN works by routing a device's internet connection through a private service rather than the user's regular internet service provider (ISP). The VPN acts as an intermediary between the user getting online and connecting to the internet by hiding their IP address.

Using a VPN creates a private, encrypted tunnel through which a user's device can access the internet while hiding their personal information, location, and other data. All network traffic is sent through a secure connection via the VPN. This means that any data transmitted to the internet is redirected to the VPN rather than from the user's computer.

When the user connects to the web using their VPN, their computer submits information to websites through the encrypted connection created by the VPN. The VPN then forwards that request and sends a response from the requested website back to the connection.

4.4.3 Working of VPN in Practice

Streaming services like Amazon Prime Video, Hulu, and Netflix offer different content to users located in different countries. Using a VPN enables a streaming customer to access the content intended for people living in different countries regardless of their actual location.

It can also enable a user to access a streaming subscription they have in their home country while traveling. For example, a user on holiday in another country could use a VPN to set their location to the U.S. and stream their favorite sports team's live game.

4.4.4 VPN Protocols

VPN protocols work in various ways, but they usually perform two basic functions: authentication and encryption. Authentication ensures your device is communicating with a trusted VPN server and encryption makes the communication itself unreadable to outsiders.

Different encryption standards and authentication methods result in differing levels of speed and security for VPN users. VPN protocols also have differing rules on how to handle potential errors, which affects stability and reliability.

1) Point-to-Point Tunneling Protocol (PPTP)

PPTP is one of the oldest protocols still active on the internet. Created by Microsoft, it uses the Transmission Control Protocol (TCP) control channel and Generic Routing Encapsulation (GRE) tunneling protocol. It relies on the Point-to-Point Protocol (PPP), which is a Layer 2 communications protocol directly between two routers, to implement security functionalities. This protocol has only the capability to encrypt data with 128-bit so it ensures low security. PPTP is fast and simple to deploy but only really applicable to people using older Windows operating systems. It also has several well-known security issues, so any VPN that only uses PPTP should be avoided.

2) Layer Two Tunneling Protocol (L2TP)

Layer Two Tunneling Protocol (**L2TP**) is an extension of the Point-to-Point Tunneling Protocol (PPTP) used by an Internet service provider (ISP) to enable the operation of a virtual private network (VPN) over the Internet. L2TP allows multiprotocol traffic to be encrypted and then sent over any medium that supports point-to-point datagram delivery, such as IP or Asynchronous Transfer Mode (ATM). L2TP is a combination of PPTP and Layer 2 Forwarding (L2F), a technology developed by Cisco Systems, Inc. L2TP represents the best features of PPTP and L2F. Unlike PPTP, the Microsoft implementation of L2TP does not use MPPE (Microsoft Point-to-Point Encryption) to encrypt PPP datagrams. L2TP relies on Internet Protocol security (IPsec) in Transport Mode for encryption services. The combination of L2TP and IPsec is known as L2TP/IPsec. Both L2TP and IPsec must be supported by both the VPN client and the VPN server. L2TP is installed with the TCP/IP protocol. L2TP is available across famous operating systems, including Android, Windows, macOS and iOS.

3) Secure Socket Tunneling Protocol (SSTP)

SSTP is a VPN tunnel created by Microsoft and is a much more secure option. It transports PPP traffic through the secure sockets layer/transport layer security (SSL/TLS) channel, which provides encryption, key negotiation, and traffic integrity checking. As such, only the two parties that transmit the data are able to decode it. Using this over the TCP port 443 ensures that SSTP can travel through most firewalls and proxy servers. SSTP is well-suited for Windows devices. While it supports other platforms—macOS, Linux or mobile devices—may be more limited compared to other VPN protocols.

4) Internet Key Exchange Version 2 (IKEv2)

IKEv2 handles request and response actions to ensure traffic is secure and authenticated, usually using IPsec. It establishes the security attributes of the device and server, then authenticates them, and agrees which encryption methods to use. It supports 256-bit encryption and allows the use of popular ciphers such as Advanced Encryption Standard (AES), Camellia and ChaCha20. IKEv2 is mostly used to secure mobile devices, in which it is particularly effective. The protocol is extensively supported on a wide range of platforms, including MacOS, Windows, Linux, iOS and Android.

5) OpenVPN

OpenVPN is an open-source and highly configurable protocol that is known for its security and versatility. There are two types of OpenVPN protocols: Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) for communication. OpenVPN is widely used due to its ability to provide a balance between speed and security. It uses end-to-end AES 256-bit encryption, intrusion detection and prevention systems to safeguard your data and SSL/TLS encryption. A kill switch feature prevents Domain Name Server (DNS)

attacks and leakage, in case the VPN loses connection. It is available on MacOS, Microsoft Windows, MacOS, Android, Linux and iOS.

6) Wireguard

Launched in 2015, WireGuard is a communication protocol that was developed by Jason A. Donenfeld. Unlike older protocols, it is designed to be lightweight and efficient which makes it one of the fastest protocols.

WireGuard emphasizes simplicity in its design and implementation. Not only is it easier to set up and maintain, but it also supports multiple types of primitives and encryption. WireGuard has gathered attention for its potential to become a next-generation VPN protocol that offers both speed and security. The protocol is available on multiple operating systems such as Windows, Linux, macOS, Android and iOS.

4.5 DHCP ARCHITECTURE, RARP AND BOOTP

4.5.1 DHCP

Introduction

DHCP (Dynamic Host Configuration Protocol) is a network administration protocol that is responsible for the task of assigning an IP address to your system and network device. DHCP works on the Application layer of the TCP/IP Protocol. The DHCP network model is based on the client-server architecture, where the connection is established when the client device sends a request message to the server device to provide the system with an IP address.

DHCP Architecture

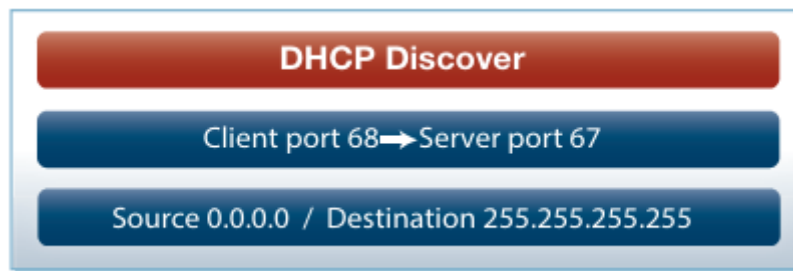
The DHCP architecture is made up of DHCP clients, DHCP servers and DHCP relay agents. The client interacts with servers using DHCP messages in a DHCP conversation to obtain and renew IP address leases.

Here is a brief description of the DHCP components:

- 1) **DHCP Server:** It automatically provides network information (IP address, subnet mask, gateway address) on lease. Once the duration has expired, that network information can be assigned to another machine. It also maintains the data storage which stores the available IP addresses.
- 2) **DHCP Client:** A DHCP client is any IP device connected on the network that has been configured to act as a host requesting configuration parameters such as an IP address from a DHCP server.
- 3) **DHCP Relay Agent:** DHCP relay agents pass DHCP messages between servers and clients where the DHCP server does not reside on the same IP subnet as its clients. Other components include the IP address pool, subnet, lease and DHCP communications protocol.

The following diagram shows the changing port numbers and the source and destination addresses used during the DHCP transaction. UDP port 68 is reserved for DHCP clients and UDP port 67 is reserved for DHCP servers.

Step 1: DHCP DISCOVER



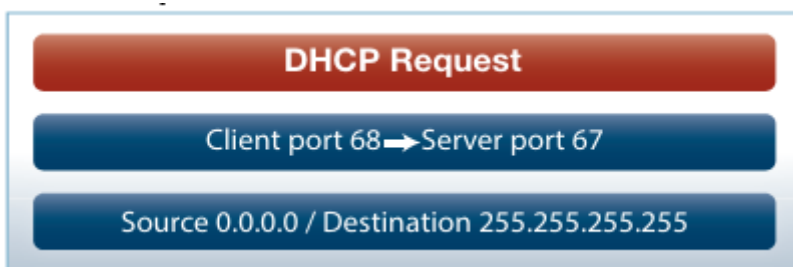
Sent by the client looking for the IP address. The source IP is 0.0.0.0 because the client doesn't have an IP address. The destination is 255.255.255.255, which is the broadcast address, as the client doesn't know where the DHCP server is located, so it broadcasts to all devices on the network.

Step 2: DHCP OFFER



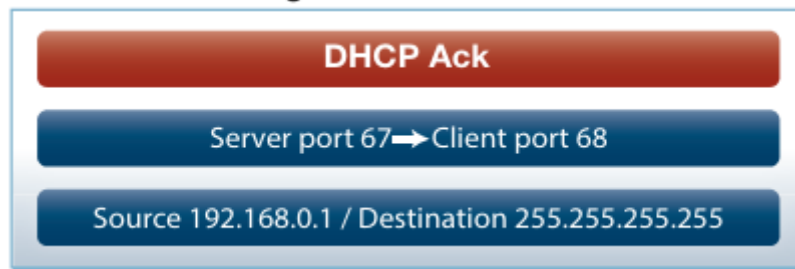
Sent by the DHCP server offering an IP address to the client. The source address is the DHCP server address. The DHCP server doesn't know the client address yet, so it broadcasts the offer to all devices on the network.

Step 3: DHCP REQUEST

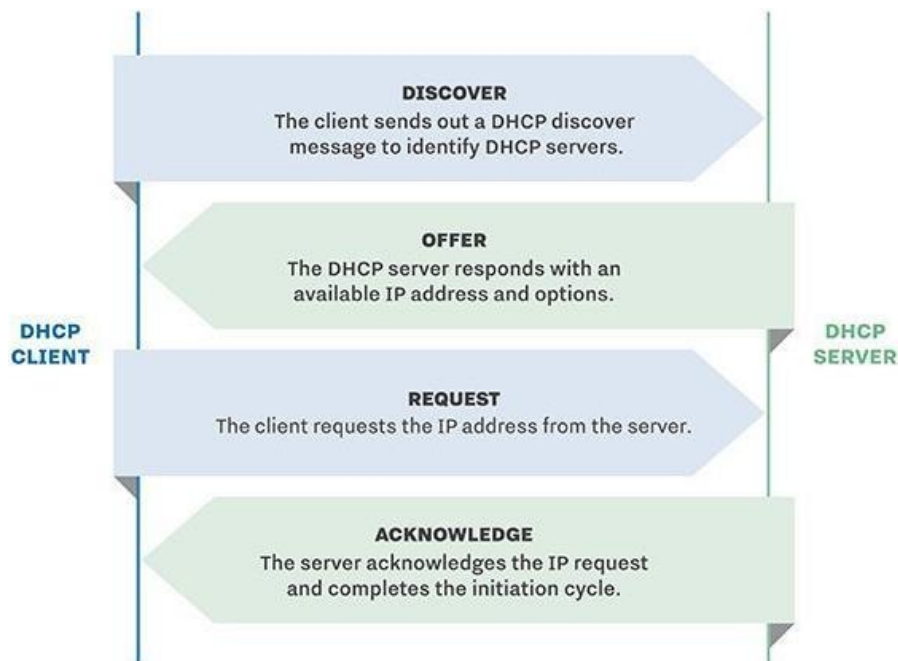


Sent by the client to the DHCP server to say "I will take that IP address, thanks." The client IP address is still 0.0.0.0 and it is again broadcast to all so that any other servers on the network that may have offered an IP address will know to stop communicating with the client for now.

Step 4: DHCP ACKNOWLEDGEMENT

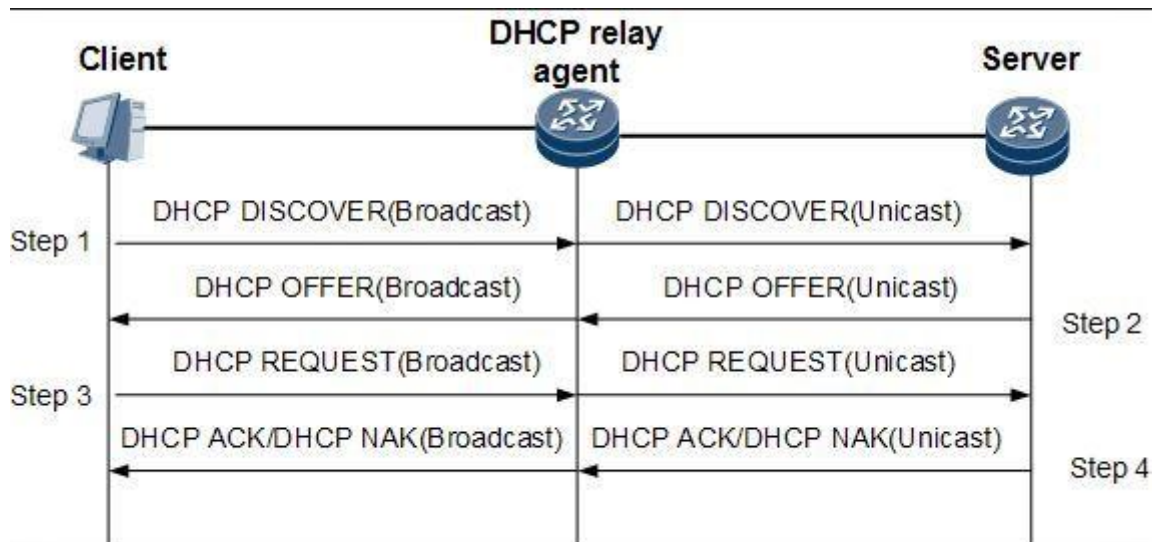


Sent by the DHCP server to the client. It confirms the IP address and other details such as subnet mask, default gateway and lease time with the client. The source address is the DHCP server and the destination is still the broadcast address.



DHCP Client, Server and Relay Agent Model

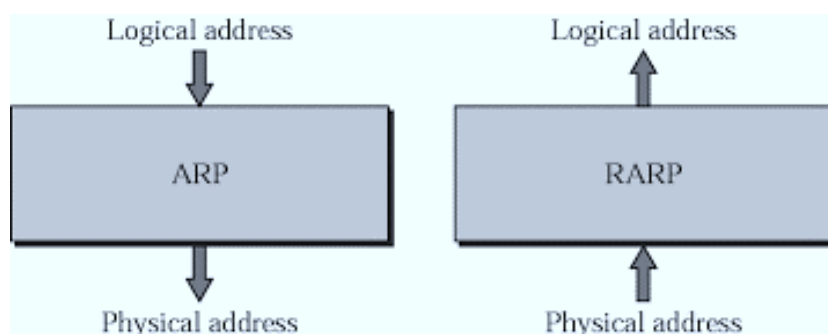
The DHCP relay agent is located between a DHCP client and a DHCP server and forwards DHCP messages between servers and clients as follows:



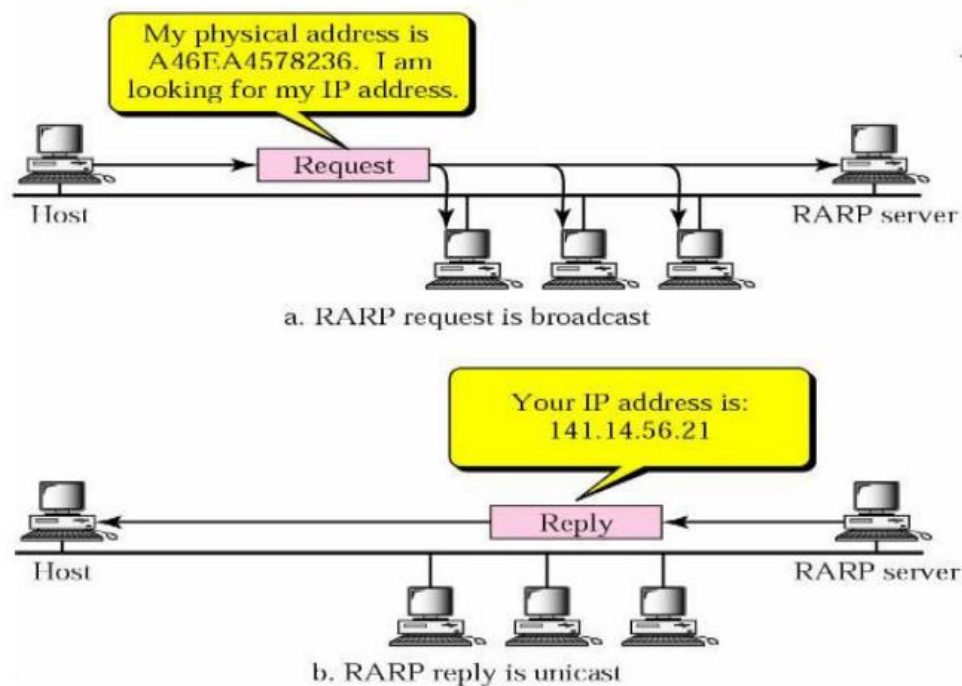
- 1) The DHCP client sends a discover packet to find a DHCP server in the network from which to obtain configuration parameters for the subscriber (or DHCP client), including an IP address.
- 2) The DHCP relay agent receives the discovered packet and forwards copies to each of the two DHCP servers. The DHCP relay agent then creates an entry in its internal client table to keep track of the client's state.
- 3) In response to receiving the discover packet, each DHCP server sends an offer packet to the client. The DHCP relay agent receives the offer packets and forwards them to the DHCP client.
- 4) On receipt of the offer packets, the DHCP client selects the DHCP server from which to obtain configuration information. Typically, the client selects the server that offers the longest lease time on the IP address.
- 5) The DHCP client sends a request packet that specifies the DHCP server from which to obtain configuration information.
- 6) The DHCP relay agent receives the request packet and forwards copies to each of the two DHCP servers.
- 7) The DHCP server requested by the client sends an acknowledgment (ACK) packet that contains the client's configuration parameters.
- 8) The DHCP relay agent receives the ACK packet and forwards it to the client.
- 9) The DHCP client receives the ACK packet and stores the configuration information.
- 10) If configured to do so, the DHCP relay agent installs a host route and Address Resolution Protocol (ARP) entry for this client.
- 11) After establishing the initial lease on the IP address, the DHCP client and the DHCP server use unicast transmission to negotiate lease renewal or release. The DHCP relay agent "snoops" on all of the packets unicast between the client and the server that pass through the router (or switch) to determine when the lease for this client has expired or been released. This process is referred to as lease shadowing or passive snooping.

4.5.2 RARP

The Reverse Address Resolution Protocol (RARP) is a networking protocol that is used to map a physical (MAC) address to an Internet Protocol (IP) address. It is the reverse of the more commonly used Address Resolution Protocol (ARP), which maps an IP address to a MAC address.



RARP was developed in the early days of computer networking as a way to provide IP addresses to diskless workstations or other devices that could not store their own IP addresses. RARP is specified in RFC 903 and operates at the data link layer of the OSI model.



With RARP, the device would broadcast its MAC address and request an IP address and a RARP server on the network would respond with the corresponding IP address.

While RARP was widely used in the past, it has largely been replaced by newer protocols such as DHCP (Dynamic Host Configuration Protocol), which provides more flexibility and functionality in assigning IP addresses dynamically. However, RARP is still used in some specialized applications, such as booting embedded systems and configuring network devices with pre-assigned IP addresses.

4.5.3 BOOTP

Bootstrap Protocol (BOOTP) is a basic protocol that automatically provides each participant in a network connection with a unique IP address for identification and authentication as soon as it connects to the network. This helps the server to speed up data transfers and connection requests.

BOOTP uses a unique IP address algorithm to provide each system on the network with a completely different IP address in a fraction of a second. This shortens the connection time between the server and the client. It starts the process of downloading and updating the source code even with very little information.

BOOTP uses a combination of TFTP (Trivial File Transfer Protocol) and UDP (User Datagram Protocol) to request and receive requests from various network-connected participants and to handle their responses.

In a BOOTP connection, the server and client just need an IP address and a gateway address to establish a successful connection. Typically, in a BOOTP network, the server and client share the same LAN, and the routers used in the network must support BOOTP bridging.

A great example of a network with a TCP / IP configuration is the Bootstrap Protocol network. Whenever a computer on the network asks for a specific request to the server, BOOTP uses its unique IP address to quickly resolve them.

4.6 INTRODUCTION TO DNS AND ITS OBJECTIVES

4.6.1 Introduction to DNS

The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, such as gmail.com or youtube.com. Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources.

Each device connected to the Internet has a unique IP address which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4), or more complex newer alphanumeric IP addresses such as 2400:cb00:2048:1::c629:d7a2 (in IPv6).

How Does DNS Work?

In a usual DNS query, the URL typed in by the user has to go through four servers for the IP address to be provided. The four servers work with each other to get the correct IP address to the client, and they include:

- 1) **DNS Recursor:** The DNS recursor, which is also referred to as a DNS resolver, receives the query from the DNS client. Then it communicates with other DNS servers to find the right IP address. After the resolver retrieves the request from the client, the resolver acts like a client itself. As it does this, it makes queries that get sent to the other three DNS servers: root nameservers, top-level domain (TLD) nameservers, and authoritative nameservers.
- 2) **Root Name Servers:** The root nameserver is designated for the internet's DNS root zone. Its job is to answer requests sent to it for records in the root zone. It answers requests by sending back a list of the authoritative nameservers that go with the correct TLD.
- 3) **TLD Name Servers:** A TLD (Top Level Domain) nameserver keeps the IP address of the second-level domain contained within the TLD name. It then releases the website's IP address and sends the query to the domain's nameserver.
- 4) **Authoritative Name Servers:** An authoritative nameserver is what gives you the real answer to your DNS query. There are two types of authoritative nameservers: a master server or primary nameserver and a slave server or secondary nameserver. The master server keeps the original copies of the zone records, while the slave server is an exact copy of the master server. It shares the DNS server load and acts as a backup if the master server fails.

What is a Top Level Domain?

Top level domains (TLDs) are more commonly known as domain extensions. A top-level domain is everything that follows the last dot in a domain name. For instance, .com is the TLD in the domain name "domainwheel.com".

The second-level domain is the actual website name, and the third-level domain (more often referred to as a subdomain) is reserved for specific sections of your website. Note that all websites have a top-level domain and a second-level domain, but most websites don't have any third-level domains.

Example:

URL: <https://www.example.com>

In this URL:

- 1) **Top-Level Domain (TLD):** ".com" is the top-level domain. It represents the commercial domain category and is one of the most commonly used TLDs on the internet.
- 2) **Second-Level Domain (SLD):** "example" is the second-level domain. It typically represents the name of the organization, business, or entity that owns the domain. In this case, "example" is used as a placeholder or demonstration domain.
- 3) **Subdomain:** "www" is the subdomain. It is an optional prefix added to the domain name to create specific web addresses or to organize content within a domain. In this example, "www" is commonly used to denote the World Wide Web portion of the website.

Together, these components form a complete URL that specifies the protocol (<https://>), subdomain (www), second-level domain (example), and top-level domain (.com).

Top-Level Domain Types

There are five official types of TLDs:

- 1) **Generic Top-level Domains (gTLD):** Domains that can be used by anyone, without restrictions.
- 2) **Country Code Top-level Domains (ccTLD):** Domains that signal that your website operates in a certain country. Many of these domains require you to prove that you live and/or do business in the appropriate country.
- 3) **Sponsored Top-level Domains (sTLD):** These domains are overseen by specific authority companies/organizations and typically have specific requirements for registration.
- 4) **Infrastructure Top-Level Domain (ARPA):** This domain extension is used by the Internet Engineering Taskforce.
- 5) **Test Top-Level Domains (tTLD):** These TLDs are intended for documentation and website testing.

The first two types are the most common on the internet.

Generic top-level domains are some of the oldest types of extensions, and there are eight of them:

Original top-level domain	The original meaning of the domain ending
.com	Open domain for commercial web offers
.org	Open TLD for non-profit organizations
.net	Open address for Internet service providers
.int	Strictly limited extension for internationally operating companies, organizations and programs
.edu	Domain intended for trade schools and universities
.gov	Domain for government institutions
.mil	TLD is available only to departments, services, and agencies of the U.S. Department of Defense

Some of the most popular ccTLDs are:

Country-specific top-level domain	Domain ending meaning
.ch	Switzerland
.cn	China
.de	Germany
.eg	Egypt
.es	Spain
.fr	France
.it	Italy
.in	India

4.6.2 Objectives of DNS

The Domain Name System (DNS) serves several key objectives in computer networking and the internet:

1) Mapping Domain Names to IP Addresses: The primary function of DNS is to translate human-readable domain names (like "example.com") into machine-readable IP addresses (like "192.0.2.1"). This mapping allows users to access websites, send emails, and perform other network activities using familiar domain names rather than numerical IP addresses.

- 2) Hierarchical Structure:** DNS organizes domain names in a hierarchical structure, which helps in efficient and scalable management of the naming system. This hierarchy includes top-level domains (TLDs), such as .com, .org, .net, and country-code TLDs like .uk, .fr, etc.
- 3) Distribution of Authority:** DNS implements a distributed database system, where different parts of the domain namespace are managed by various organizations and servers. This decentralization ensures reliability, scalability, and fault tolerance of the DNS system.
- 4) Load Distribution and Load Balancing:** DNS can be used to distribute incoming network traffic across multiple servers, thereby optimizing resource usage and improving performance. This is achieved through techniques like round-robin DNS and geographic load balancing.
- 5) Alias and Redirection:** DNS supports aliasing and redirection, allowing multiple domain names to point to the same IP address or to different IP addresses based on various criteria such as geographic location or server availability.
- 6) Caching:** DNS servers cache DNS records to reduce the time and resources required to resolve domain names. Cached records can be reused for subsequent requests, improving the overall efficiency of DNS resolution and reducing network latency.
- 7) Security:** DNS plays a crucial role in security by supporting mechanisms like DNSSEC (Domain Name System Security Extensions), which provides data integrity and authentication of DNS data to prevent DNS spoofing and other malicious activities.
- 8) Scalability and Growth:** DNS is designed to accommodate the growth of the internet and the increasing number of domain names and network devices. Its distributed architecture and hierarchical structure make it highly scalable and adaptable to changes in network topology and size.

UNIT-5

NETWORK PLANNING AND IMPLEMENTATION

5.1 NETWORK NEEDS

A computer network is a collection of interconnected computers, servers, and other devices that can communicate with each other, either wirelessly or through physical connections like cables or fiber optics. These interconnected devices can share resources such as files, printers and internet connections, and they can also facilitate communication through email, messaging, and other means. Computer networks can vary in size and complexity, ranging from small local networks within a home or office to large-scale global networks like the Internet.

Networks serve various purposes and fulfill multiple needs for individuals, businesses and organizations. Some of the key needs of a network include:

- 1) Communication:** Networks enable communication between individuals, devices and systems regardless of their physical location. This includes email, instant messaging, voice and video calls and data sharing.
- 2) Resource Sharing:** Networks allow for the sharing of hardware resources such as printers, scanners and storage devices, as well as software resources like applications and databases.
- 3) Data Transfer and Access:** Networks facilitate the transfer of data between devices and provide access to shared data repositories, allowing users to retrieve, store and manipulate information.
- 4) Collaboration:** Networks support collaboration among users by enabling them to work together on projects, share documents and coordinate tasks in real-time, regardless of their geographical location.
- 5) Remote Access:** Networks enable remote access to resources and services, allowing users to connect to their organization's network from external locations securely.
- 6) Internet Access:** Networks provide connectivity to the Internet, allowing users to access a vast array of online resources, services and information.
- 7) Centralized Management:** Networks allow for centralized management of resources, user accounts, security policies and network configurations, streamlining administration and maintenance tasks.
- 8) Backup and Disaster Recovery:** Networks facilitate backup and replication of data to remote locations, ensuring data integrity and enabling disaster recovery in case of hardware failures, natural disasters or cyberattacks.
- 9) Scalability:** Networks can be scaled to accommodate the growth of an organization by adding additional devices, expanding infrastructure capacity, and adapting to changing requirements.
- 10) Security:** Networks provide security mechanisms such as firewalls, encryption, access controls and intrusion detection systems to protect against unauthorized access, data breaches and cyber threats.

5.2 INSTALL AND CONFIGURE WINDOWS SERVER

Windows Server 2012 R2 Hardware Requirements:

As with previous Windows versions, your hardware must meet certain requirements for Windows Server 2012 R2 to function properly. First of all, Windows Server 2012 R2 requires a 64-bit processor.

Component	Minimum Requirement	Microsoft Recommended
Processor	1.4 GHz	2 GHz or faster
Memory	512 MB RAM	2 GB RAM or greater
Available Disk Space	32 GB	40 GB or greater
Optical Drive	DVD-ROM drive	DVD-ROM drive
Display	Super VGA (800x600) monitor	XGA (1024x768) monitor

In addition, you must have the usual I/O peripherals, including a keyboard and mouse or compatible pointing device and a wired or wireless network interface card (NIC). If you can connect to a network location on which you have copied the contents of the Windows Server 2012 R2 DVD-ROM, you are not required to have a DVD-ROM drive on your computer.

Microsoft recommends that you also perform the following actions before installing Windows Server 2012 R2:

- 1) Disconnect uninterruptible power supply (UPS) devices
- 2) Back up data
- 3) Disable antivirus software
- 4) Provide mass storage drivers if needed
- 5) Note that Windows Firewall is on by default
- 6) Prepare your Active Directory environment for Windows Server 2012 R2

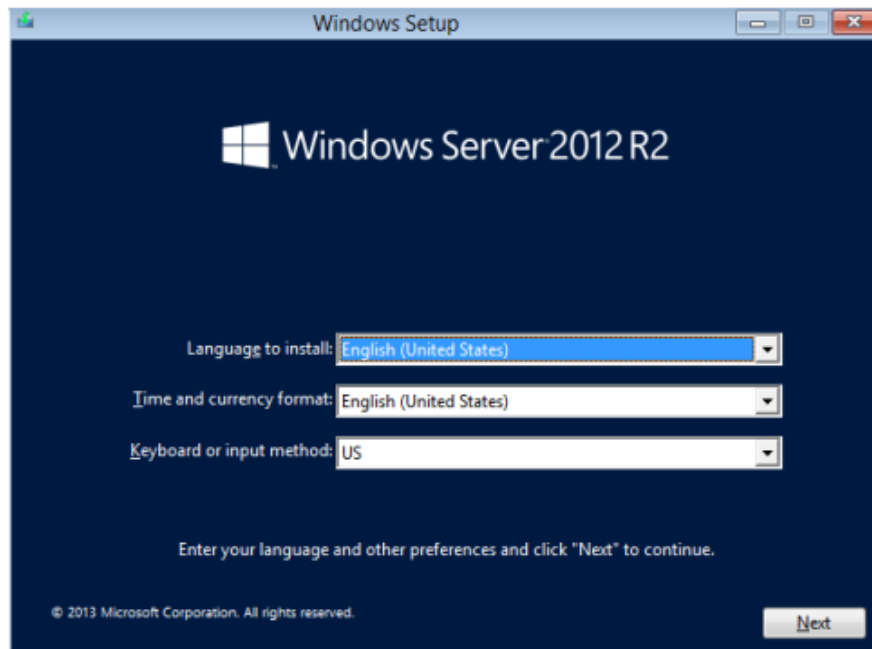
Steps to install Windows Server 2012:

1. Insert the Windows Server 2012 R2 DVD-ROM and turn on your computer. You should see a message informing you that Windows is copying temporary files; if not, you should access the BIOS setup program included with your computer and modify the boot sequence so that the computer boots from the DVD.

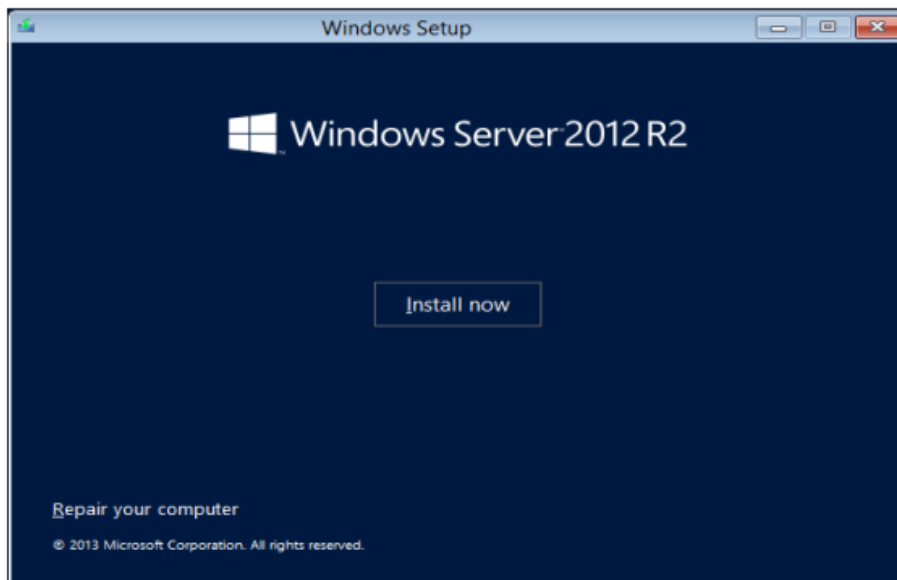
2. Once booted, you should see Windows loading the necessary setup files.



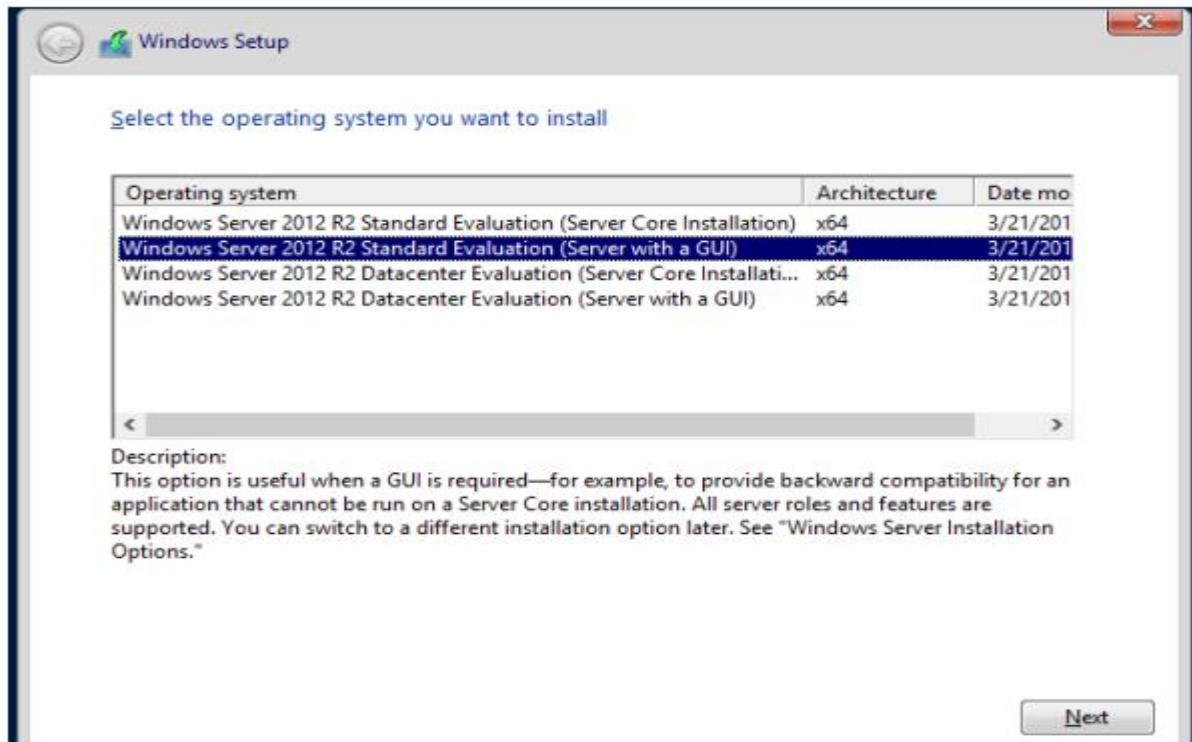
3. In the Windows Setup dialog, set the language, time and currency format, keyboard and input method then press Next.



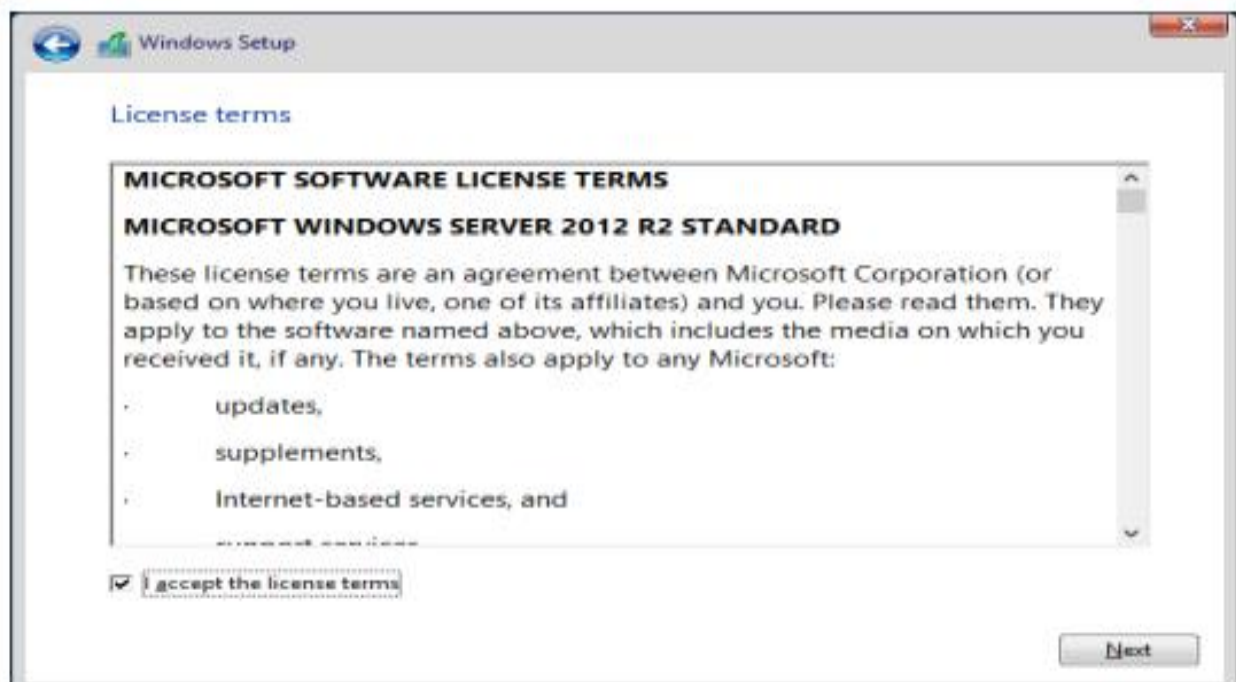
4. Click on Install now



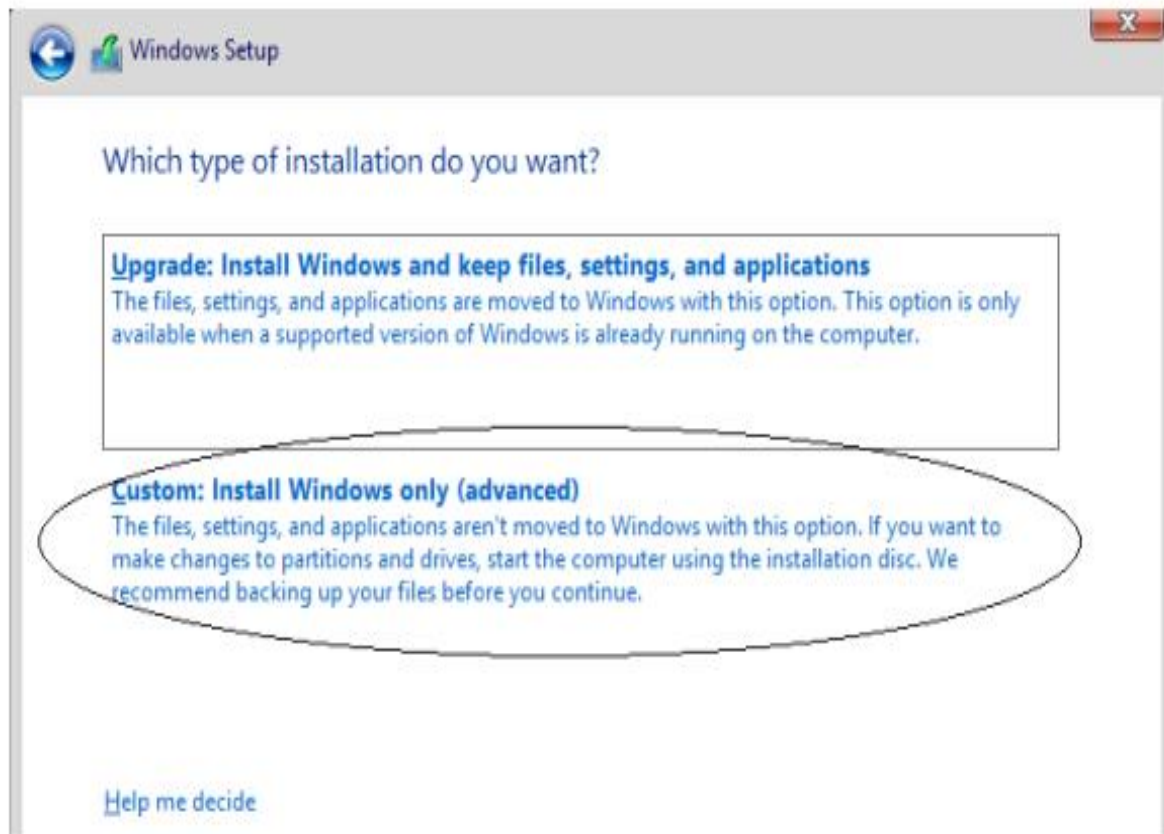
5. Select the operating system that you wish to install. Regardless if you install the Server 2012 R2 Datacenter or the 2012 R2 Standard version, you want the installing package that includes the GUI. Otherwise, you end up with just the Server 2012 Core and no GUI.



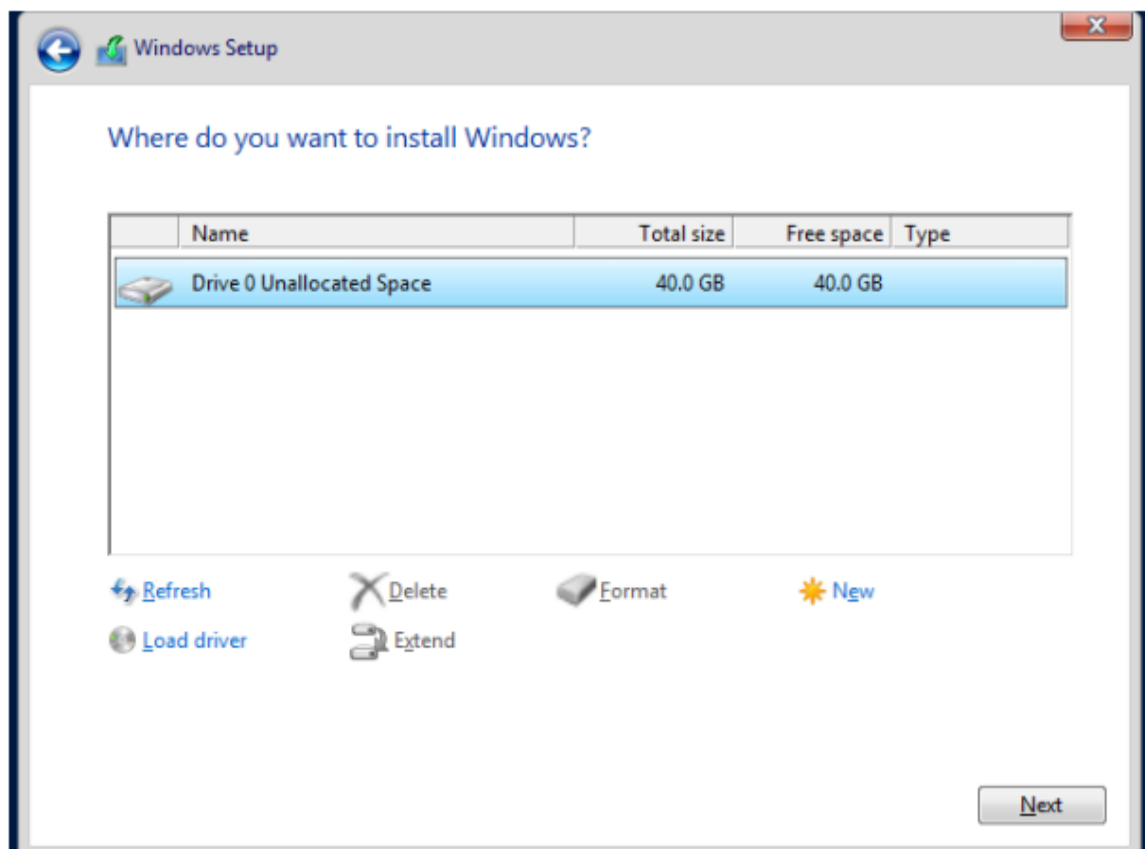
6. Accept the license agreement.



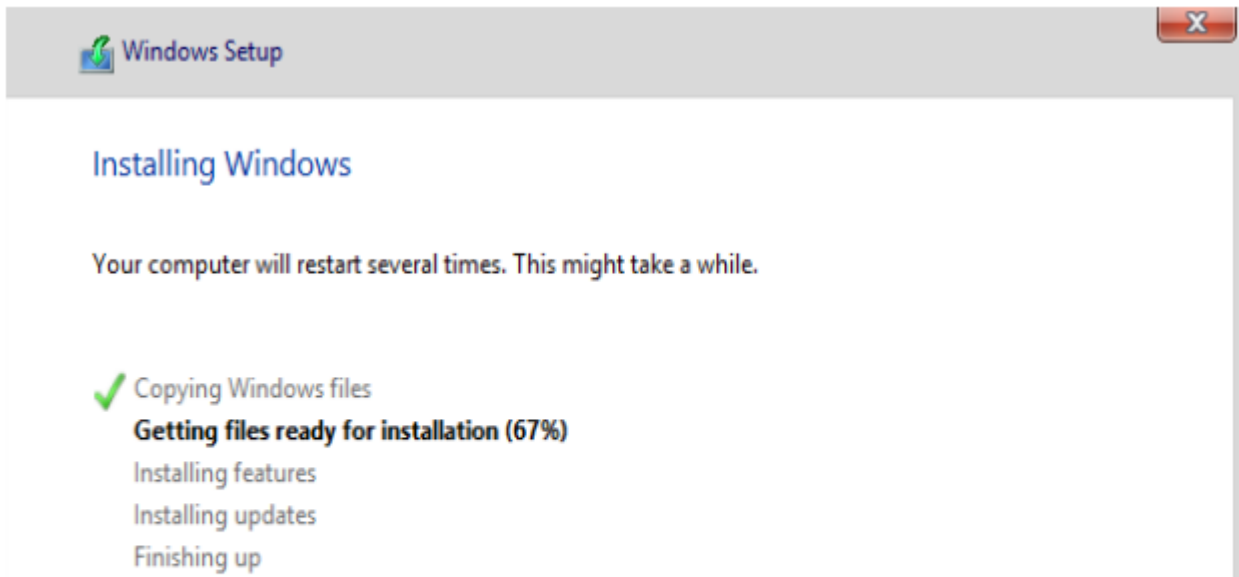
7. Select Custom Install Windows only (advanced).



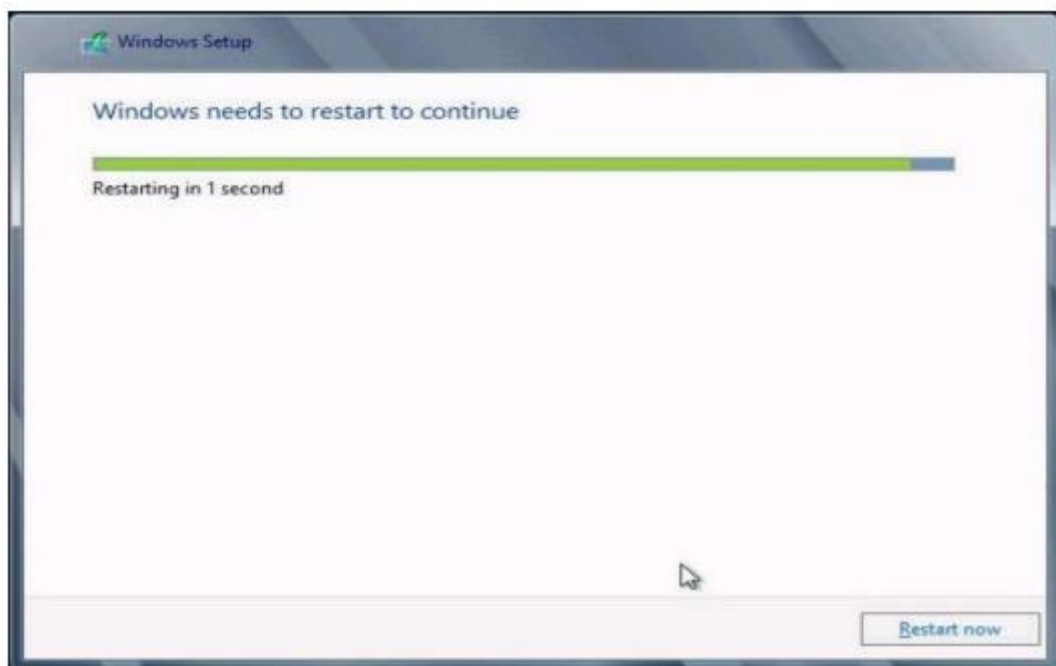
8. Click Next.



9. Server 2012 R2 begins the file copy process.



10. The server will restart automatically and installation will continue further.



11. Before Server 2012 will allow you to log in, you must configure the password for the administrator account.
12. Click Finish. Welcome to Windows Server 2012 R2!

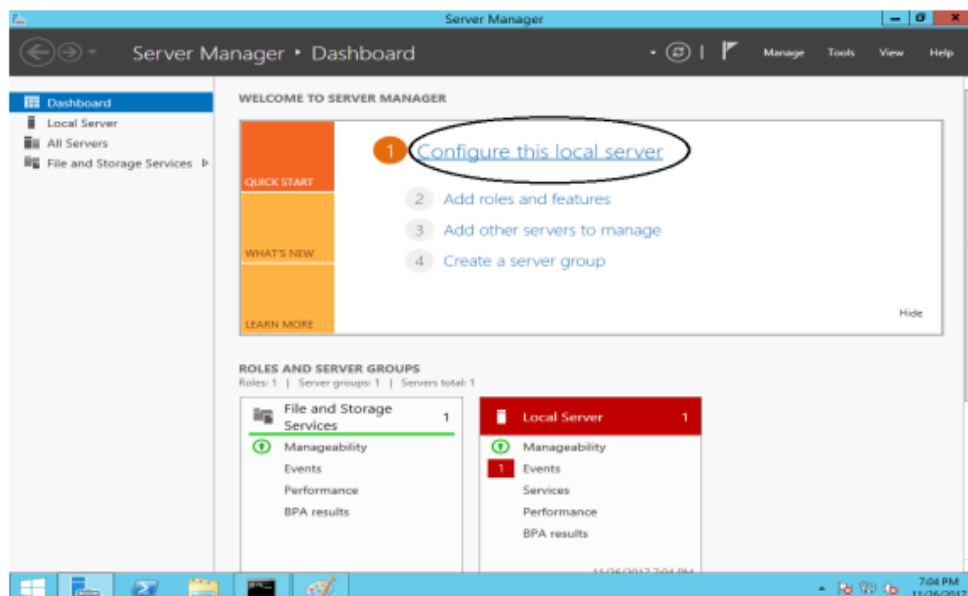


The steps to Change the Server Name are:

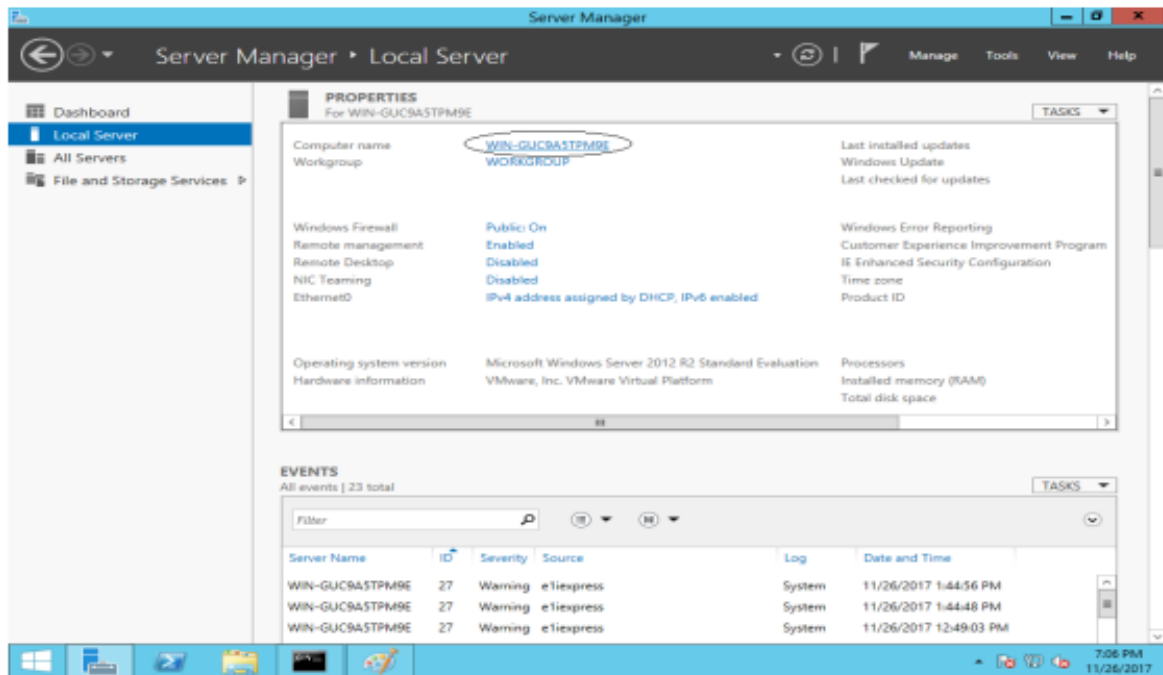
1. Open Server manager



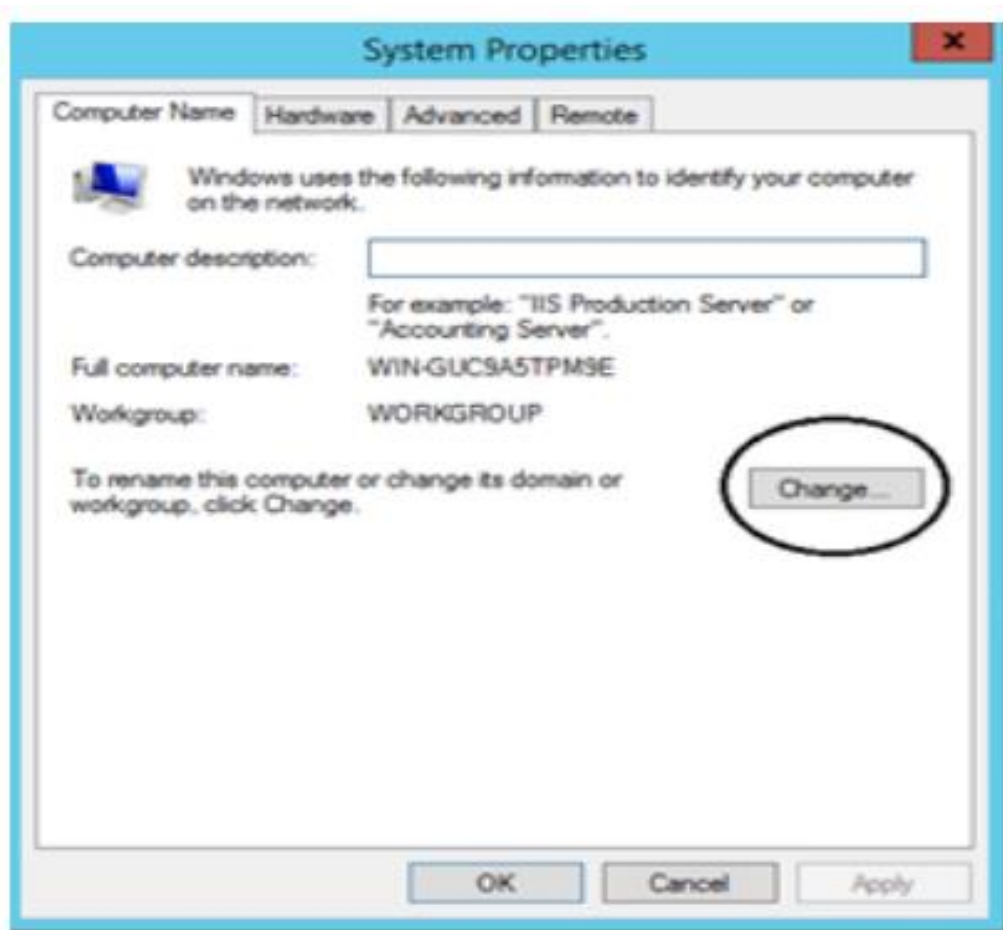
2. Click on Configure this local server



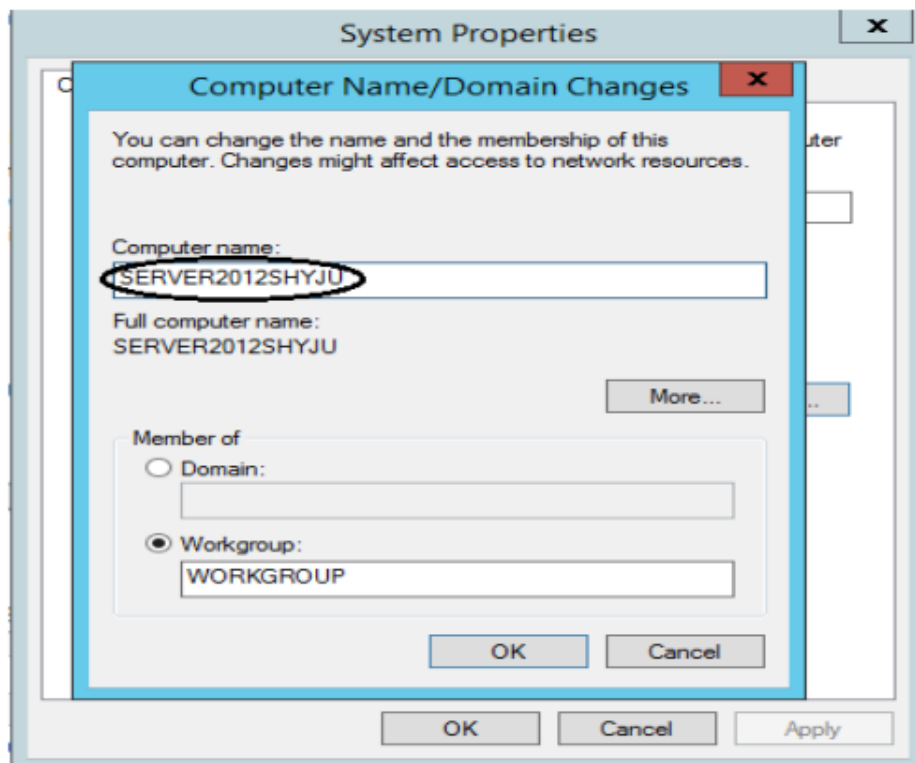
- Click on default Computer name which was given while server 2012 was installed.



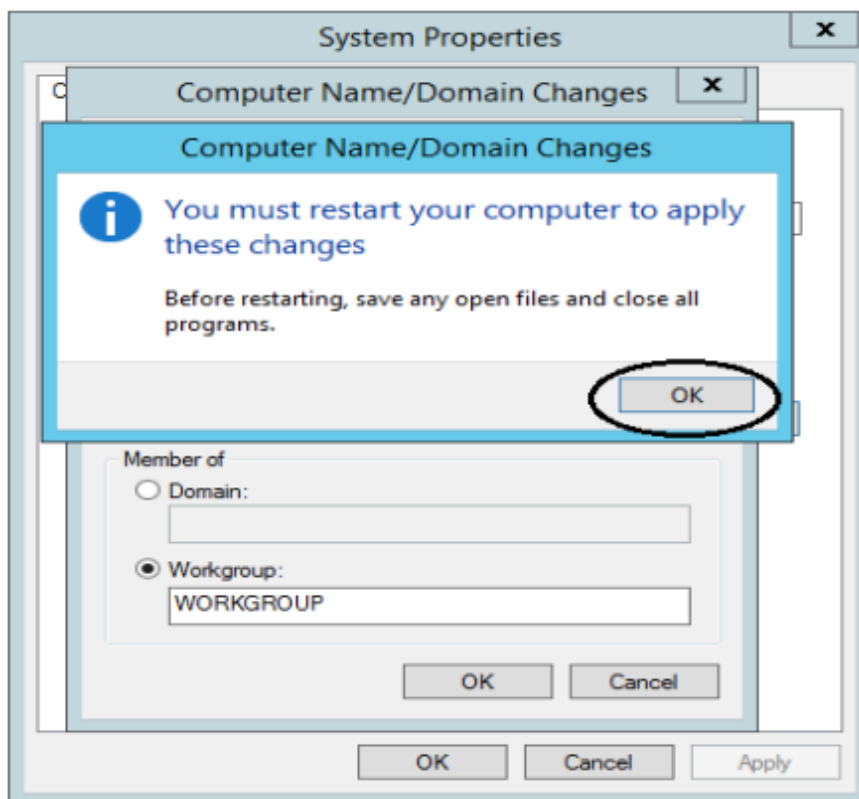
- Click Change in System Properties window



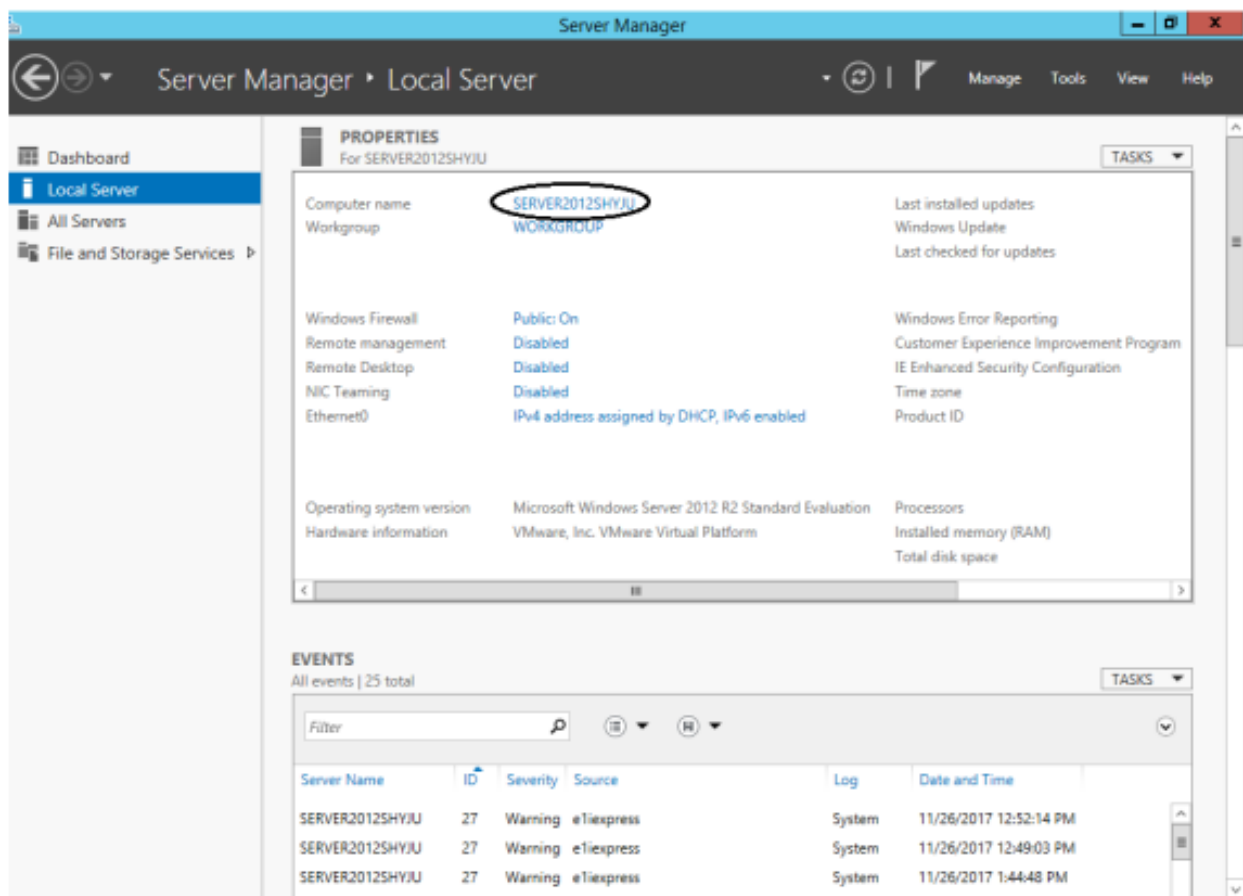
5. Give a new Computer name which is easy to remember.



6. Click OK and Restart your computer.



7. After Restarting your computer server name will be changed



5.3 STEPS TO CREATE DOMAIN CONTROLLER

Set a static IP address

To start, you'll want to ensure your domain controller has a static IP address (that isn't in a DHCP range either) to ensure it cannot change automatically.

Install the Role

From the Server Manager click Manage > Add Roles and Features this will open up the Add Roles and Features Wizard.

Once open, click next until you reach the Server Roles section.

Once there click the check box next to Active Directory Domain Services.

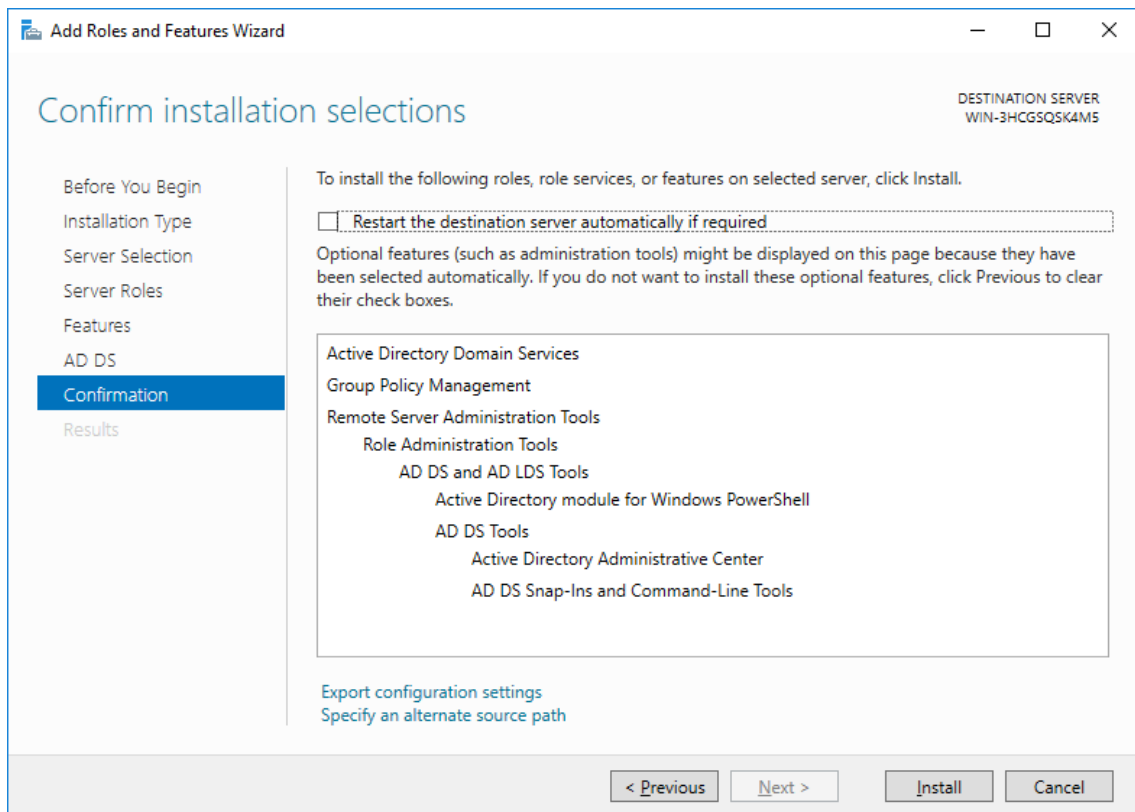
Another window should open asking if you want to install Active Directory Administrative Center and the AD DS Snap-in and Command Line Tools.

We recommend you accept this and click the Add Features button with the Include Management tools (if applicable) box checked as well.

That will be the only role to include this time so you can click next and you can also click next through the features section as well.

Once you reach the confirmation section it will display the list of roles and features to be installed; it should match the image below.

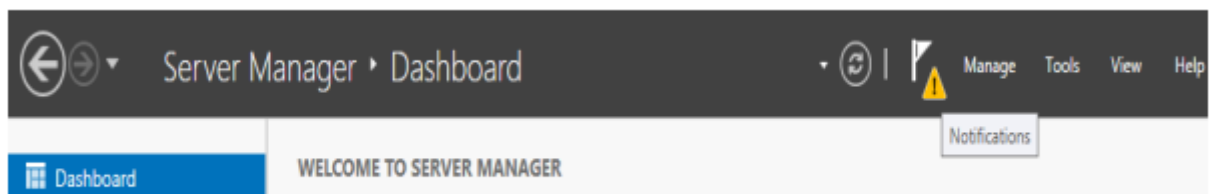
Click install and now we wait!



Promoting Server 2012 R2 to a Domain Controller

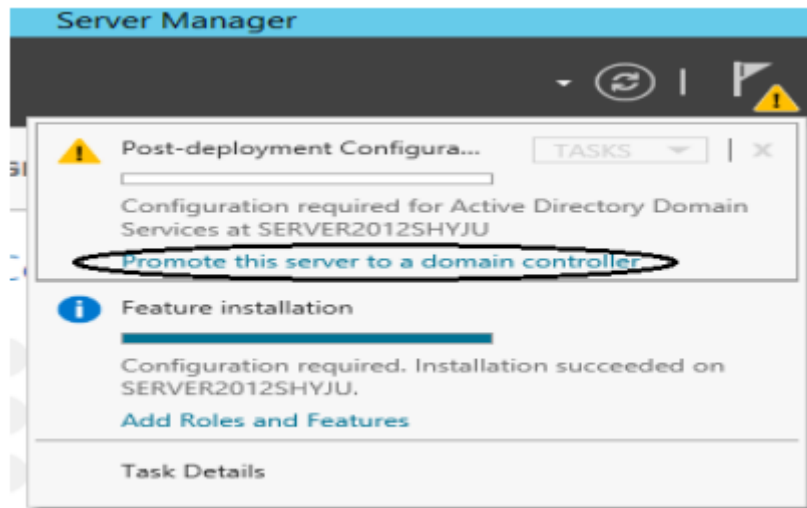
Once the Active Directory Domain Services role has been installed, we can promote our server to a Domain Controller by following the below steps.

1. When we return to Server Manager, there is an information message waiting for us. This is the second part of the role installation that converts the server to a domain controller.

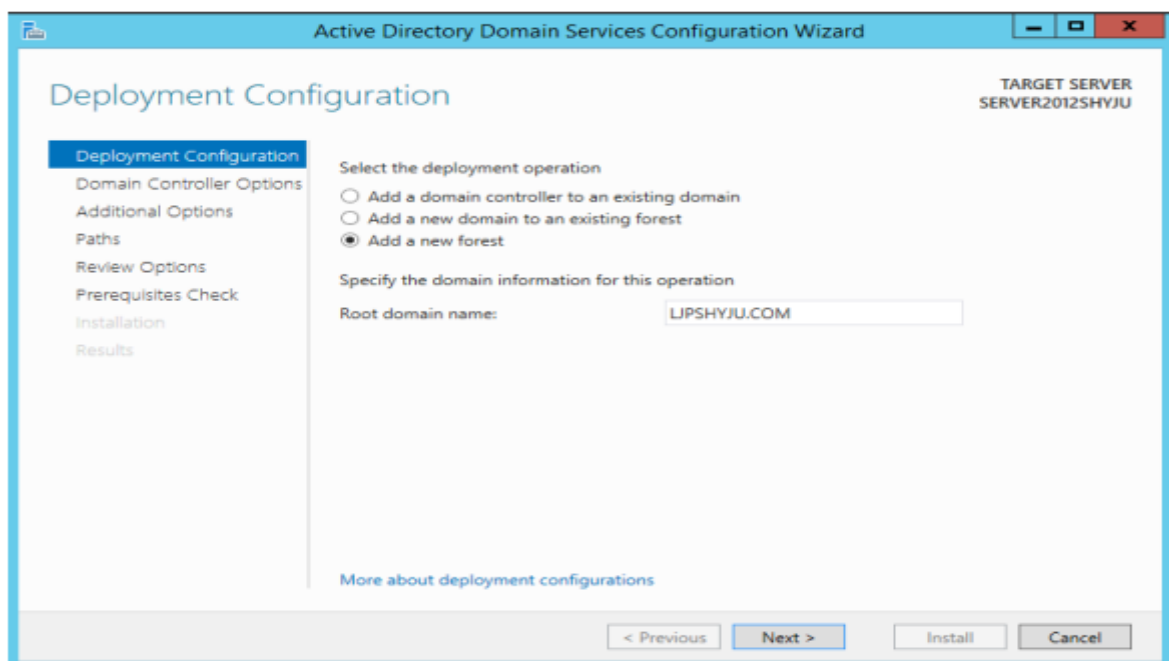


2. Click on Notification you will see the option "Promote this server to a domain controller." Click on this link for role promotion.

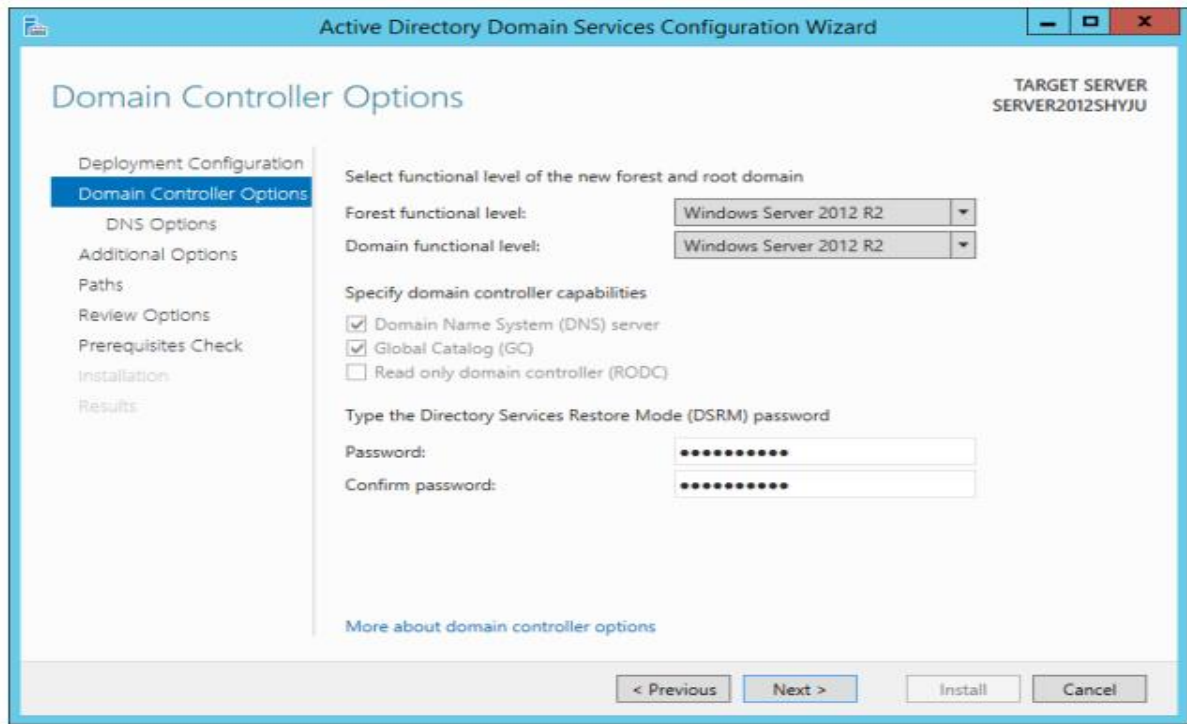
If we click on the message, "Promote this server to a domain controller" the second wizard will start.



3. Click on the “Add a new forest” radio button and enter the name of the new root domain. Click Next

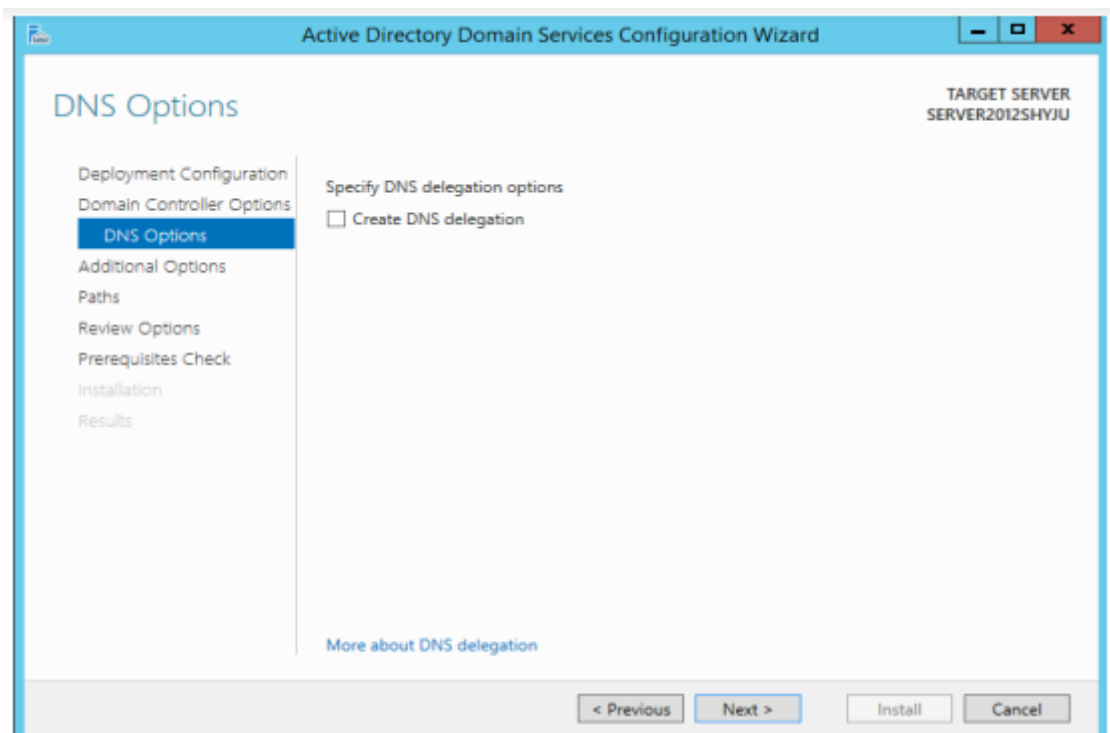


4. If we did not have DNS installed, the option to install DNS would be provided. Check the box to install DNS as it is a requirement for Active Directory. If it is grayed out, you already have it installed and running. Leave the function for Server 2012 R2 as is. If we had server 2008 running on the network, we downgrade the function to accommodate the need for a backward capability. Type the Directory Services Restore Mode (DSRM) password in the Domain Controller Options window.



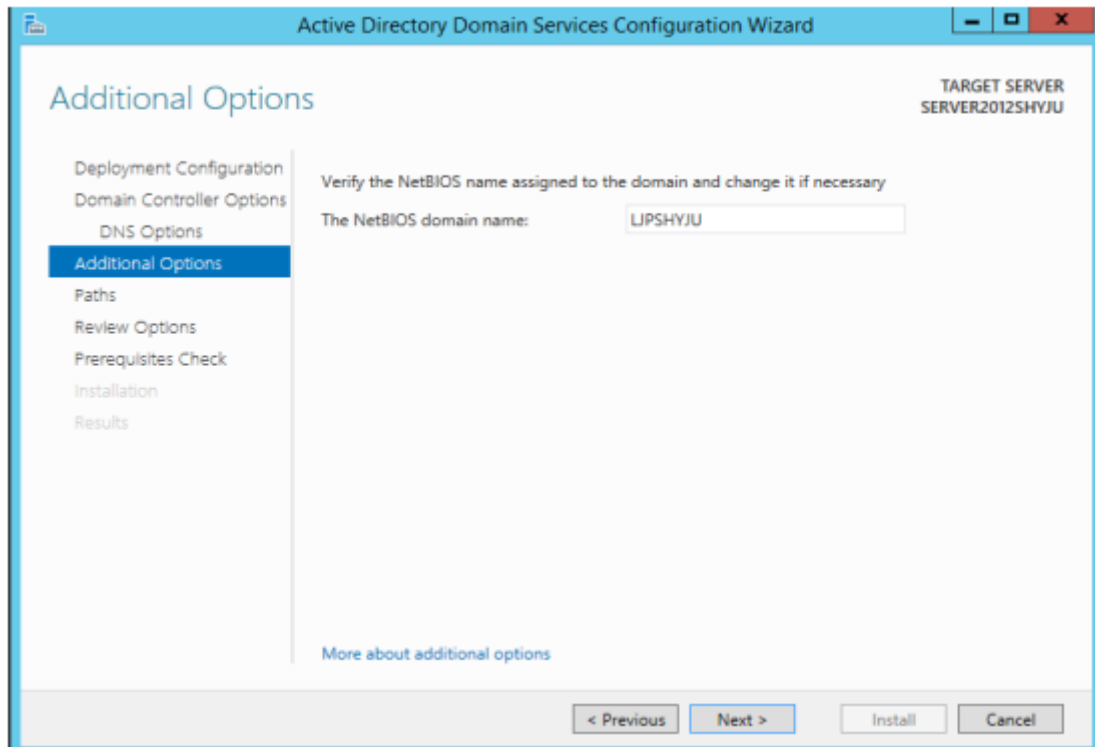
The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo, the text 'Active Directory Domain Services Configuration Wizard', and standard window controls. The main window has a blue header with the title 'Domain Controller Options' and a 'TARGET SERVER' label with the value 'SERVER2012SHYJU'. On the left is a navigation pane with links: 'Deployment Configuration', 'Domain Controller Options' (highlighted), 'DNS Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area is titled 'Select functional level of the new forest and root domain'. It contains two dropdown menus: 'Forest functional level:' and 'Domain functional level:', both set to 'Windows Server 2012 R2'. Below these is a section 'Specify domain controller capabilities' with three checkboxes: 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked). Further down is a section 'Type the Directory Services Restore Mode (DSRM) password' with two password fields labeled 'Password:' and 'Confirm password:', both containing masked characters. At the bottom of the main area is a link 'More about domain controller options'. The bottom of the window features a grey bar with four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

5. Uncheck the box DNS delegation if authoritative zone is available. Click next.

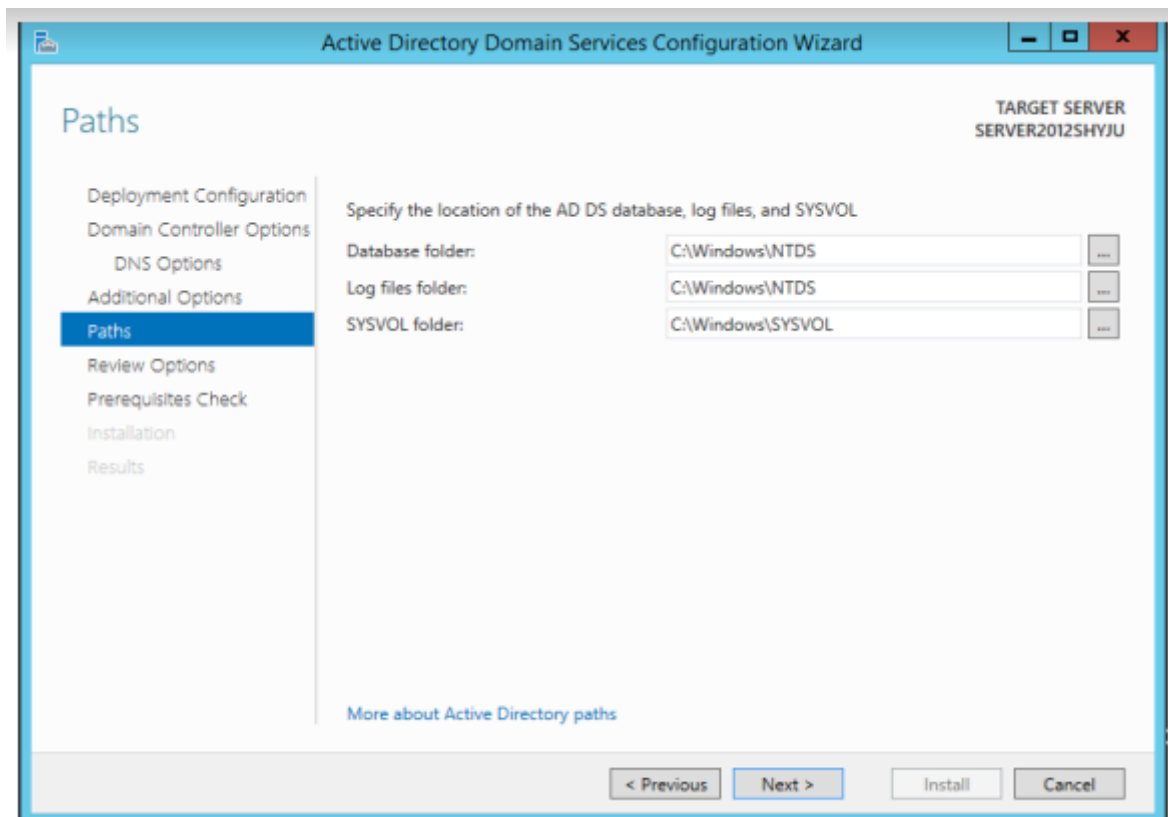


The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window at the 'DNS Options' step. The title bar and window controls are the same as the previous screenshot. The main window has a blue header with the title 'DNS Options' and the same 'TARGET SERVER' label 'SERVER2012SHYJU'. The navigation pane on the left is the same, but 'DNS Options' is now highlighted. The main content area is titled 'Specify DNS delegation options' and contains a single checkbox labeled 'Create DNS delegation', which is currently unchecked. At the bottom of the main area is a link 'More about DNS delegation'. The bottom of the window features a grey bar with four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

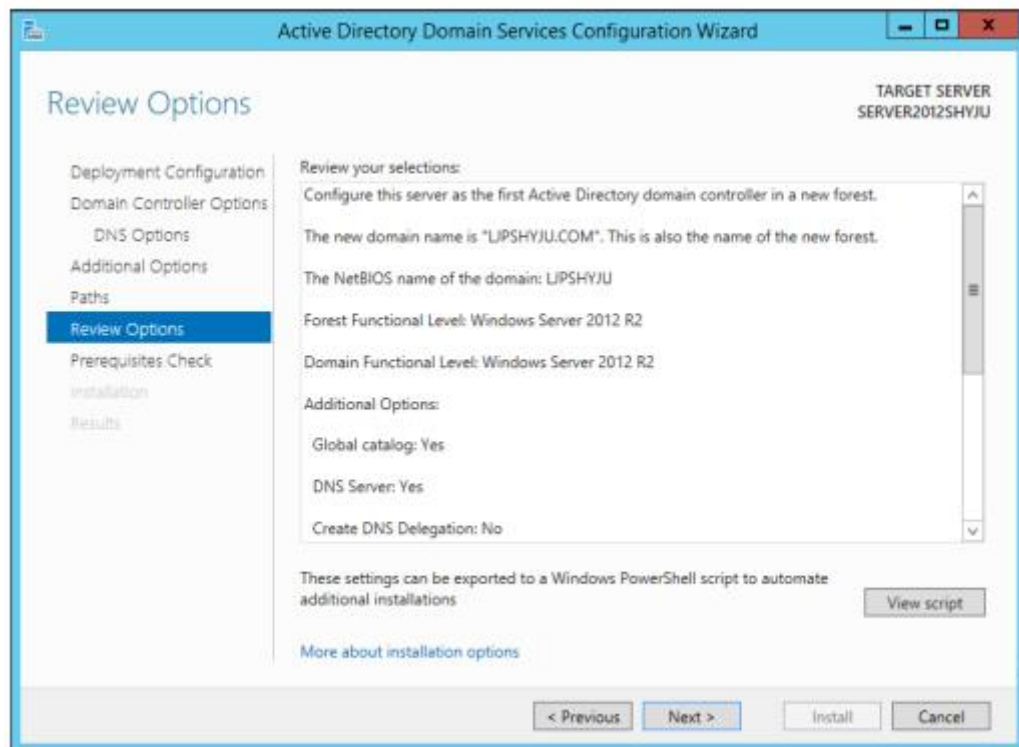
6. Setup will find the NetBIOS domain name. Once found, click on Next.



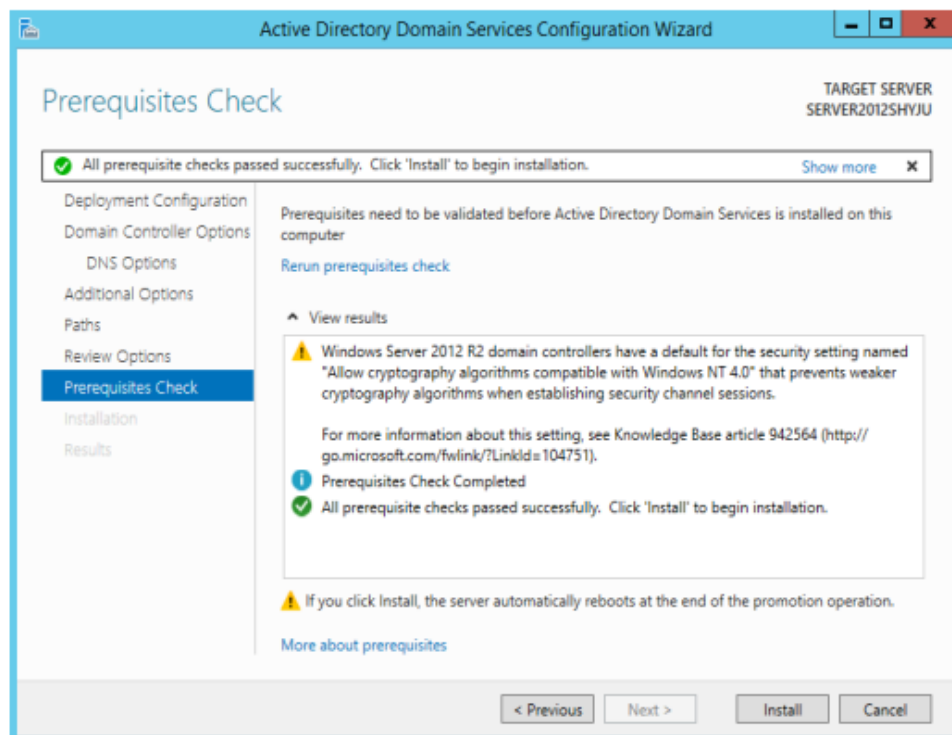
7. Confirm the location of the database, log files and SYSVOL folders.



- Review the summary and click Next to promote the server to Domain Controller.

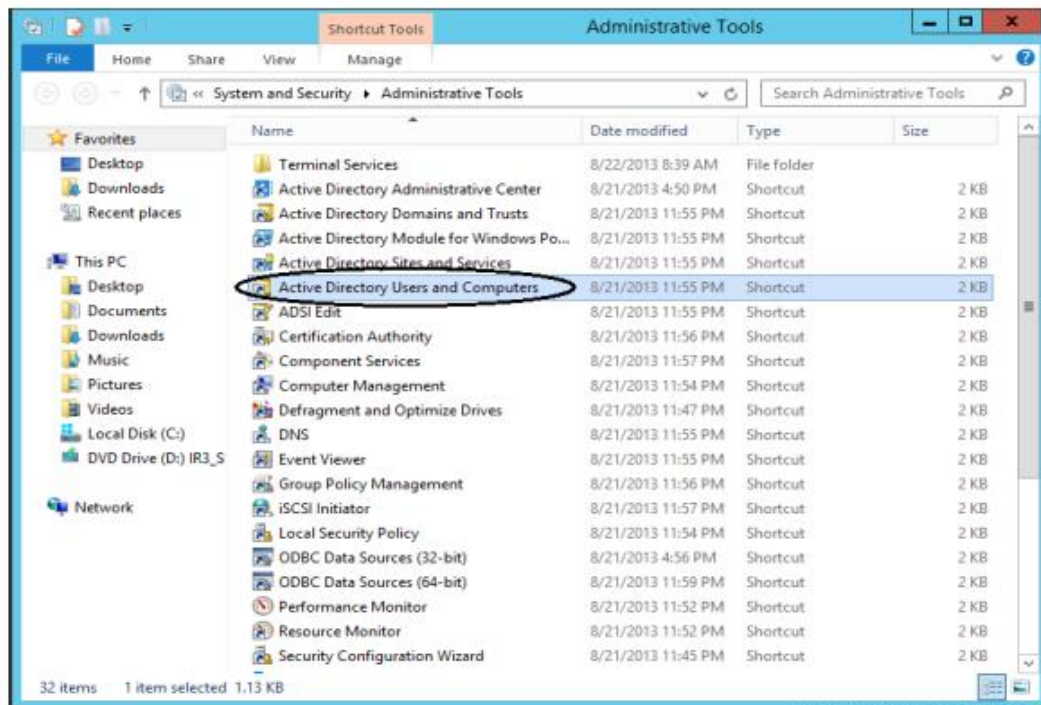


- Once prerequisites have passed successfully, click on Install.



- The system may restart once the installation is completed. Be patient! When the machine restarts and comes back up, you will have an Active Directory Domain controller. To verify if Active Directory tools have been installed, click on the start button, and the Administrative Tools tile. Check out your

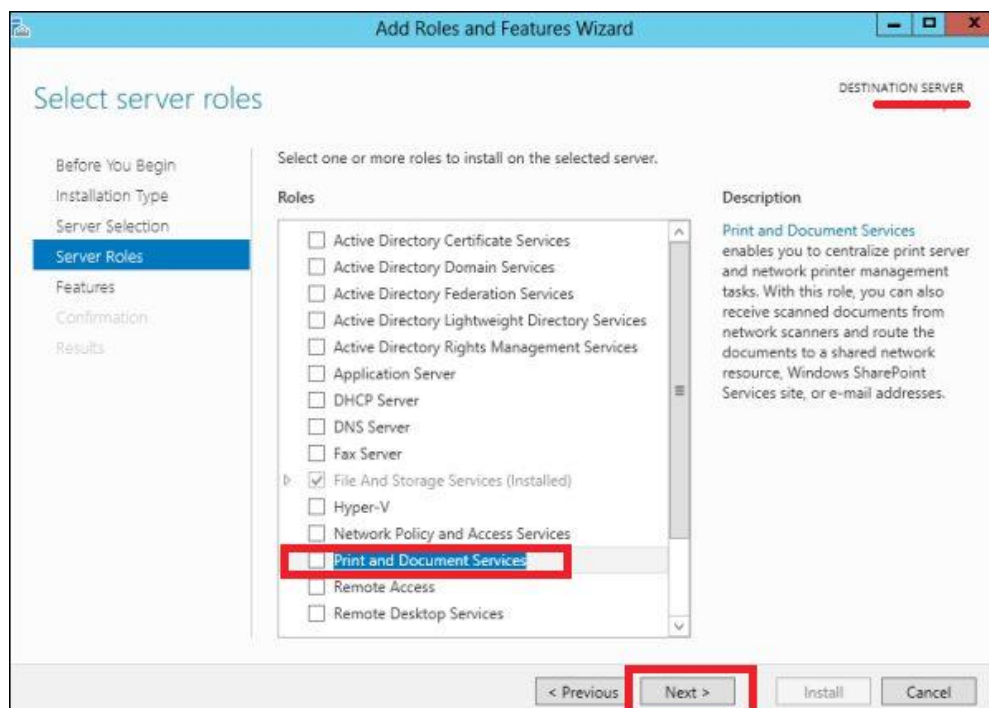
Active Directory tools! The main snap-in for managing users and computers is the Active Directory Users and Computers snap-in.



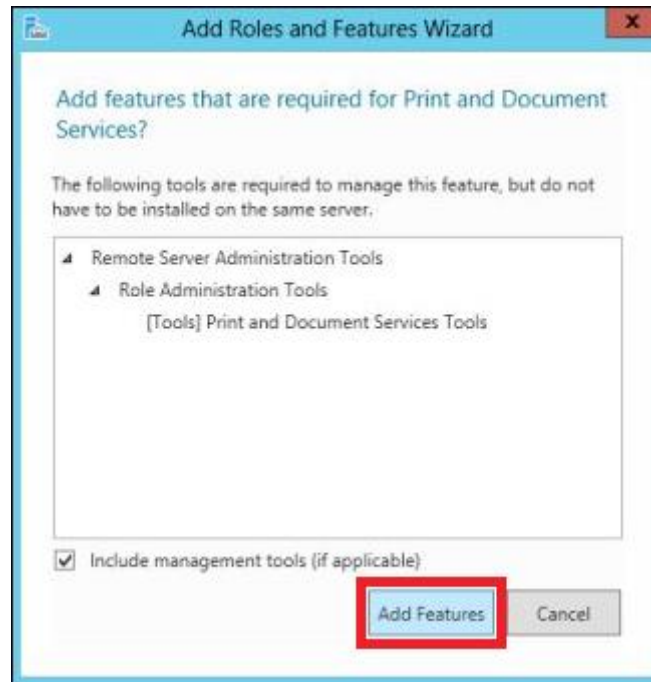
5.3 ADDING FILE SERVER AND PRINTER SERVER

Printer Server

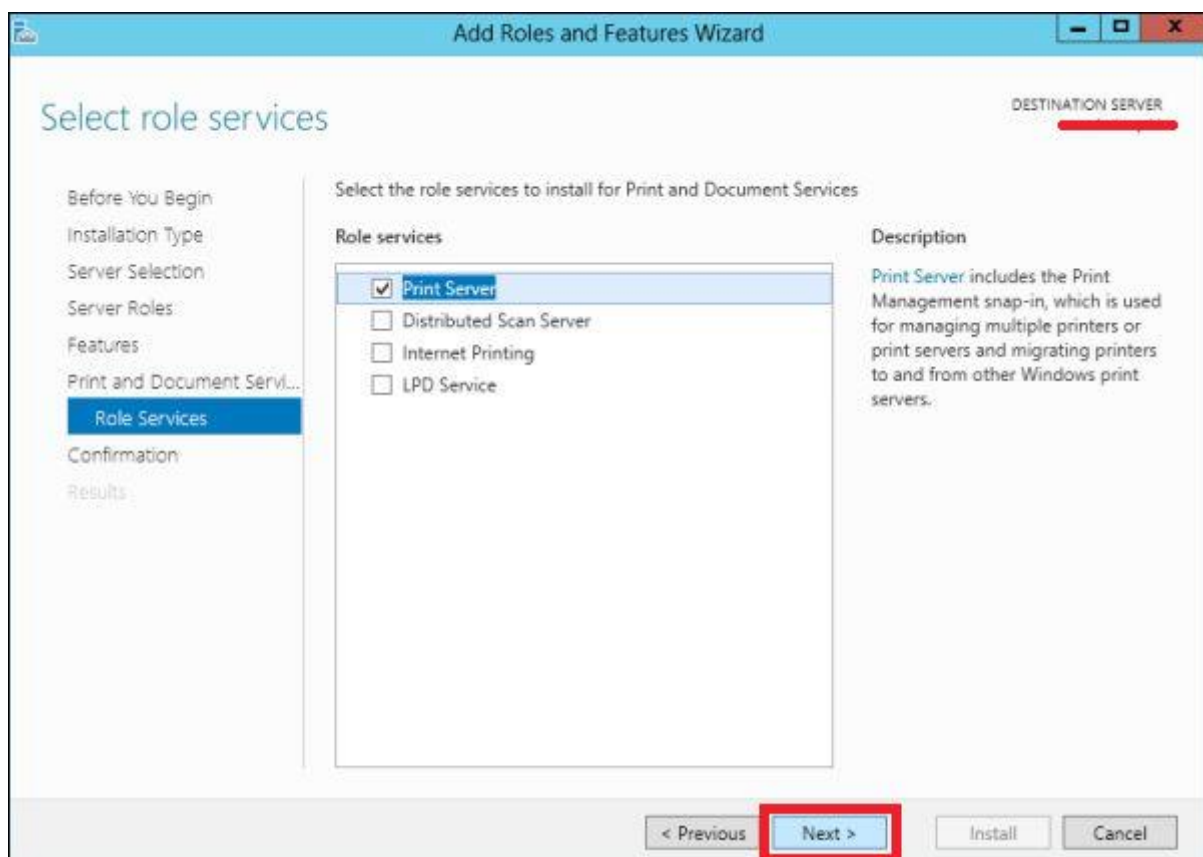
1. Go to Server Manager → Manage → Add Roles and Features → Next → Check on the Role-based or feature-based installation box and → Check on Select a server from the server pool and then → Next. Once it is done then, at the list of the roles find Print and Document Services → A popup window will open.



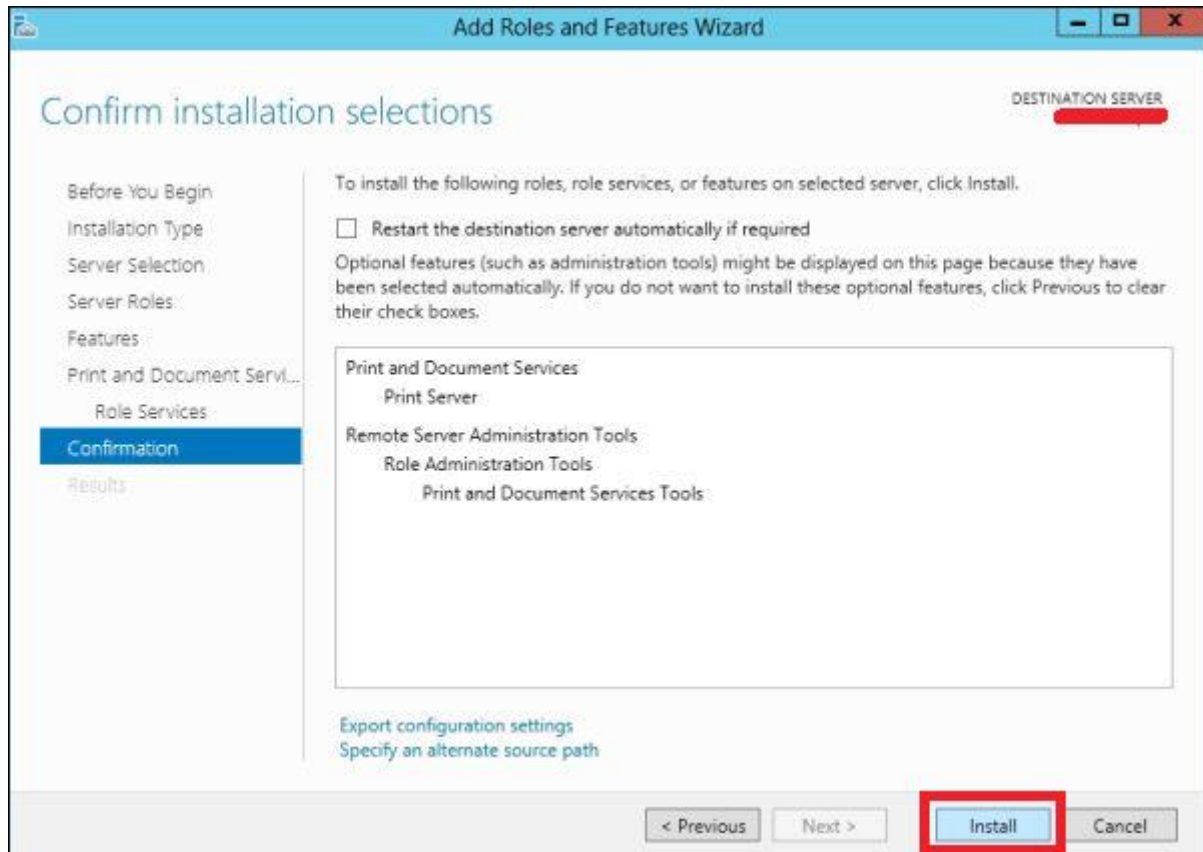
2. Click on Add features and then → Next → Next → Next.



3. Check on the Print Server box and → Next.

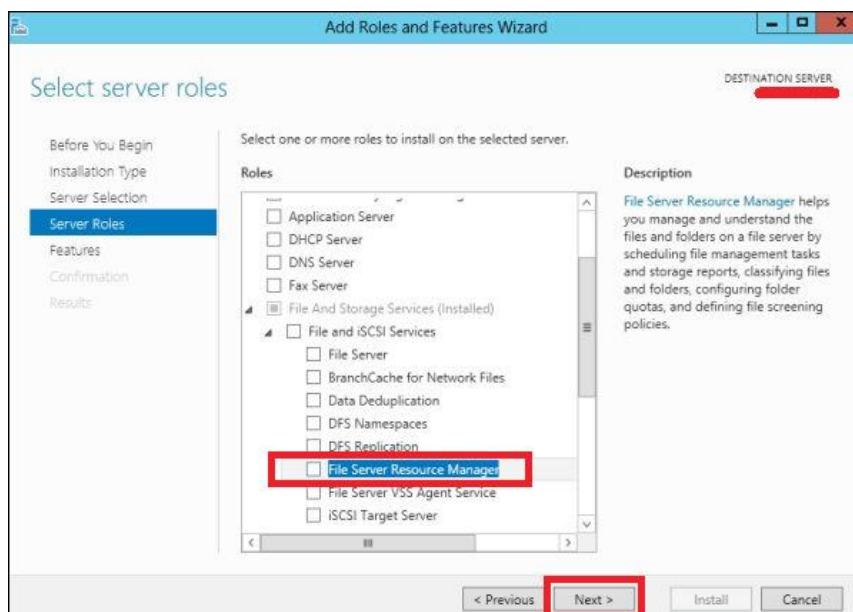


4. Click Install.

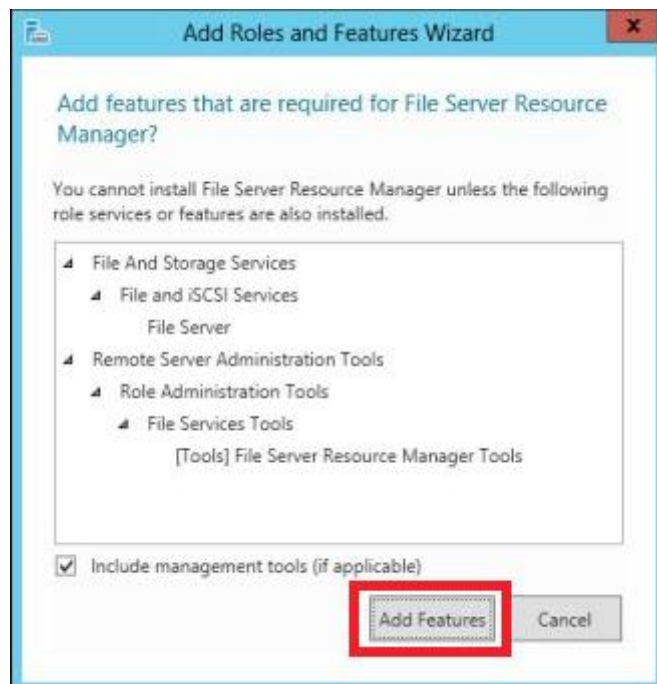


File Server

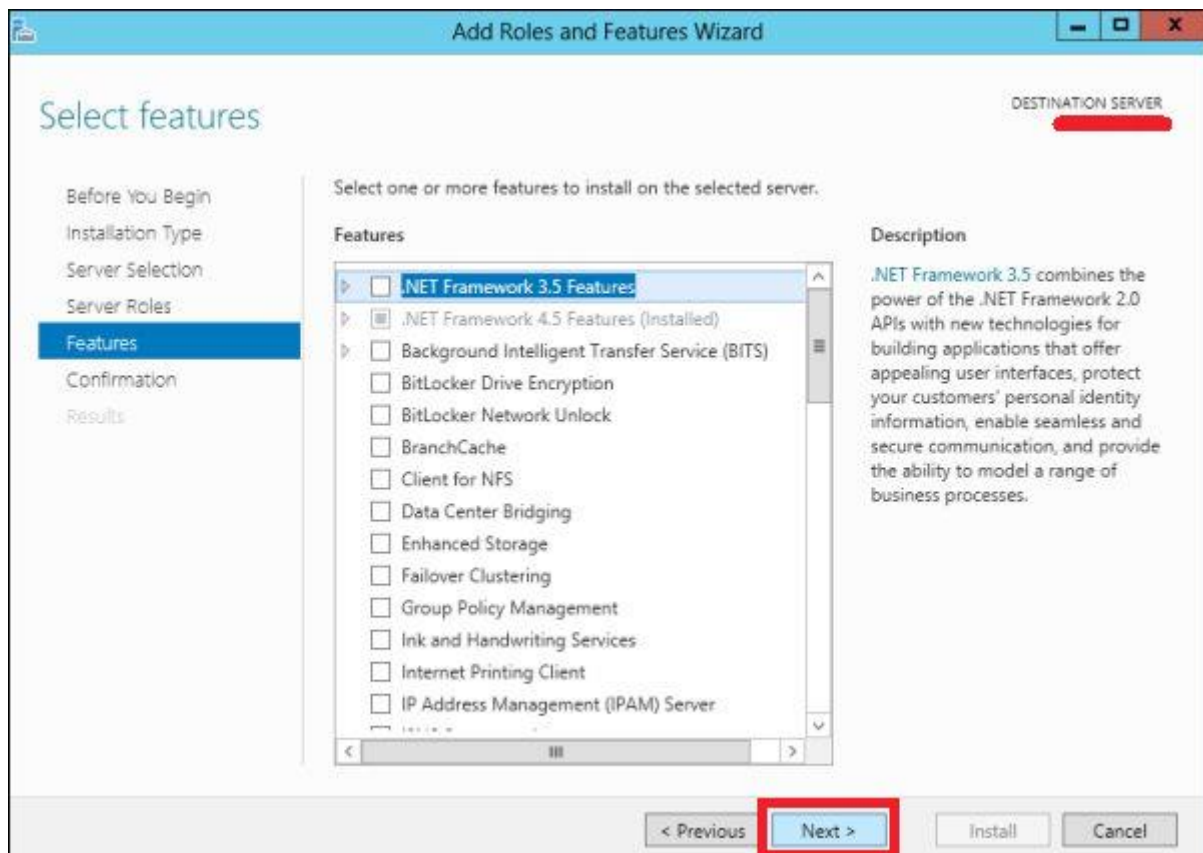
1. Go to Server Manager → Manage → Add Roles and Features → Next → Check on the Role-based or feature-based installation box → Check on the Select a server from the server pool box → Next. Then, at the list of the roles find File and Storage Services and expand it. After this, expand the Files and iSCSI Services → Check on the File Server Resource Manager box and then a pop-up window will open.



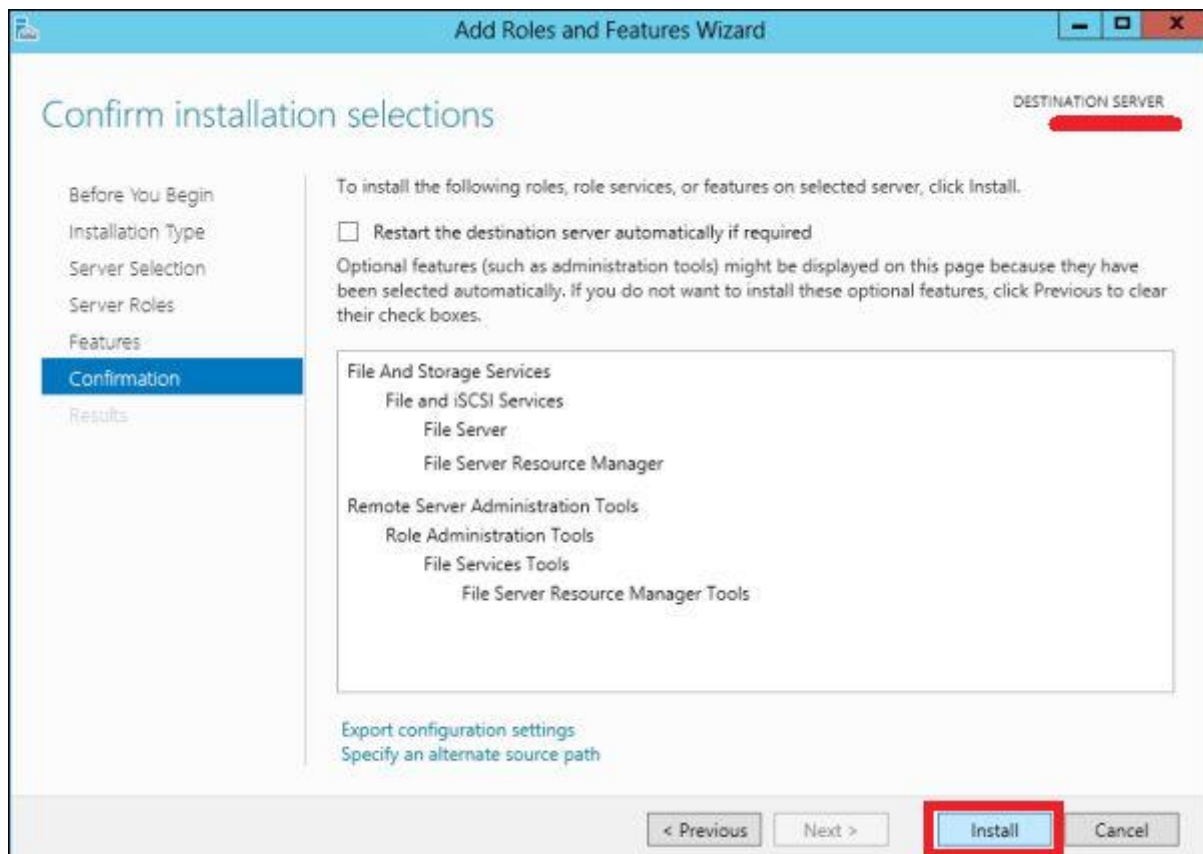
- Click on Add features and then → Next.



- Click the Next button.

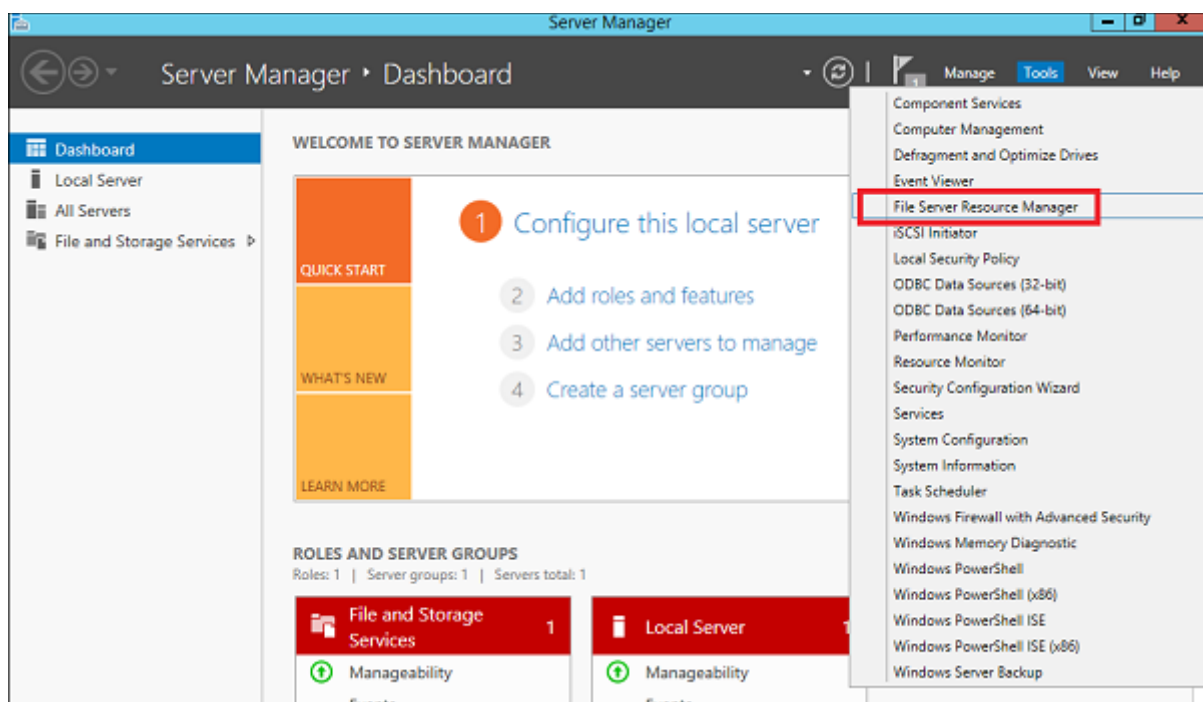


- Click the Install button.



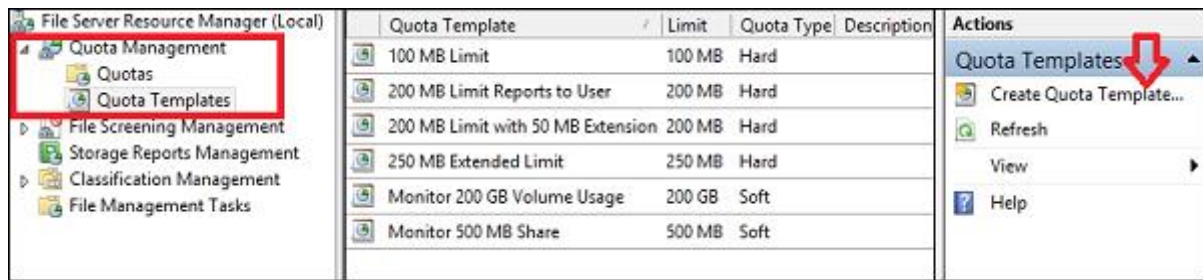
Now that we installed the File Server Resource Manager, let us open it by following the path given below.

- Click on Server Manager → Tools → File Server Resource Manager.



To set quota to a folder, we should follow the steps as below.

- On the left panel click on Quota Management → expand Quota Templates → click on Create Quota Template... on the right-hand side panel as shown in the screenshot given below.



- A new table will be opened, where the most important aspect to put is Space Limit depending on the need. Here, we will put 2GB and then → OK.

The 'Create Quota Template' dialog box is shown. The 'Copy properties from quota template (optional):' section has '100 MB Limit' selected. The 'Settings' tab is active. The 'Template name' is '2 GB' and the 'Description (optional)' is '2 GB'. The 'Space limit' section is highlighted with a red box, showing 'Limit: 2 GB' and 'Hard quota: Do not allow users to exceed limit' selected. The 'Notification thresholds' section is empty. The 'OK' button is highlighted with a red box.

- You must put the threshold for it and once the folder reaches the capacity, it will send you a notification where you will have an option to put the email.

Add Threshold

Generate notifications when usage reaches (%):

E-mail Message | Event Log | Command | Report

☐ Send e-mail to the following administrators:

 Format: account@domain. Use semicolons to separate accounts.

☐ Send e-mail to the user who exceeded the threshold

E-mail message

Type the text to use for the Subject line and message.
 To identify the quota, limit, usage, or other information about the current threshold, you can use Insert Variable to insert a variable in your text.

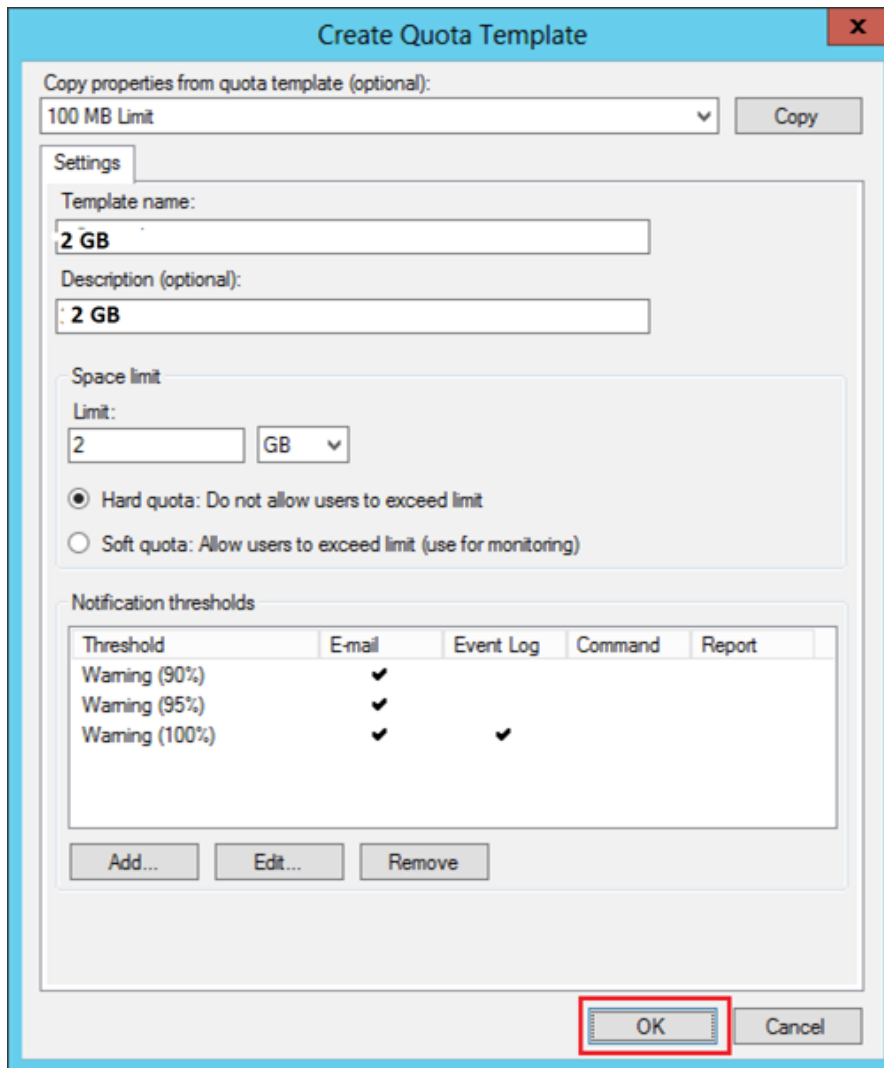
Subject:

Message body:

Select variable to insert:

Inserts the e-mail addresses of the administrators who receive the e-mail.

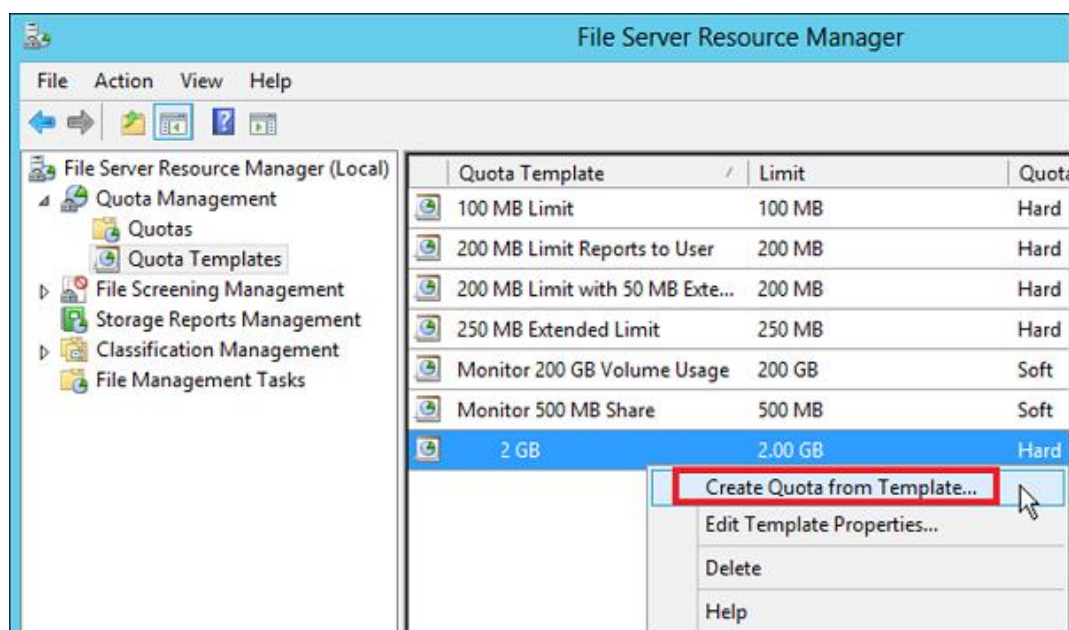
5. Click OK.



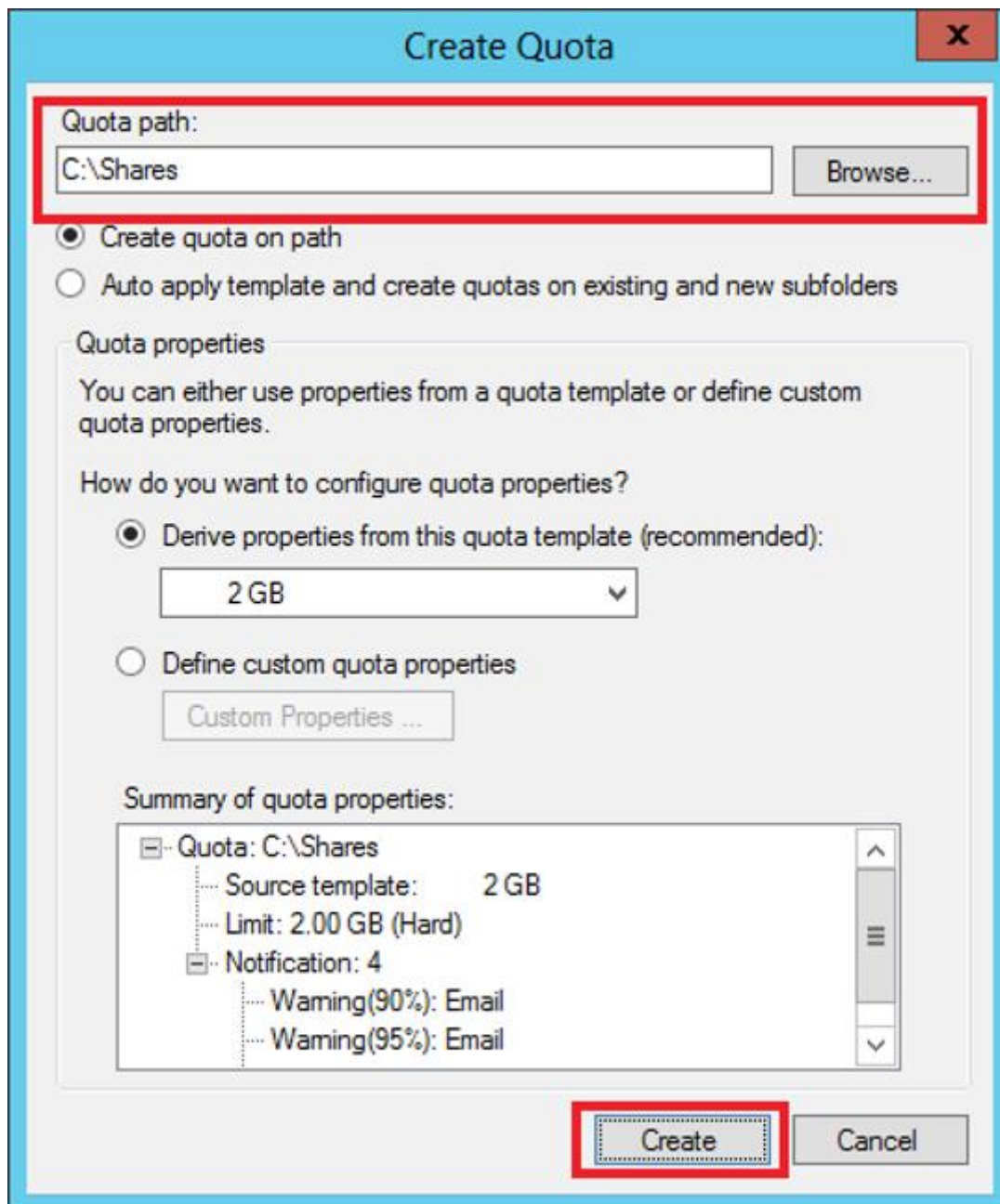
The 'Create Quota Template' dialog box is shown. It has a title bar with a close button. Below the title bar, there is a section 'Copy properties from quota template (optional):' with a dropdown menu set to '100 MB Limit' and a 'Copy' button. The 'Settings' tab is active. It contains a 'Template name:' field with '2 GB' entered, a 'Description (optional):' field with '2 GB' entered, and a 'Space limit' section with a 'Limit:' field set to '2' and a unit dropdown set to 'GB'. Below this, there are two radio buttons: 'Hard quota: Do not allow users to exceed limit' (selected) and 'Soft quota: Allow users to exceed limit (use for monitoring)'. At the bottom, there is a 'Notification thresholds' table with columns 'Threshold', 'E-mail', 'Event Log', 'Command', and 'Report'. The table has three rows: 'Warning (90%)', 'Warning (95%)', and 'Warning (100%)'. The 'E-mail' column has checkmarks for all three rows, and the 'Event Log' column has a checkmark for 'Warning (100%)'. Below the table are 'Add...', 'Edit...', and 'Remove' buttons. At the bottom right, there are 'OK' and 'Cancel' buttons, with the 'OK' button highlighted by a red rectangle.

Threshold	E-mail	Event Log	Command	Report
Warning (90%)	✓			
Warning (95%)	✓			
Warning (100%)	✓	✓		

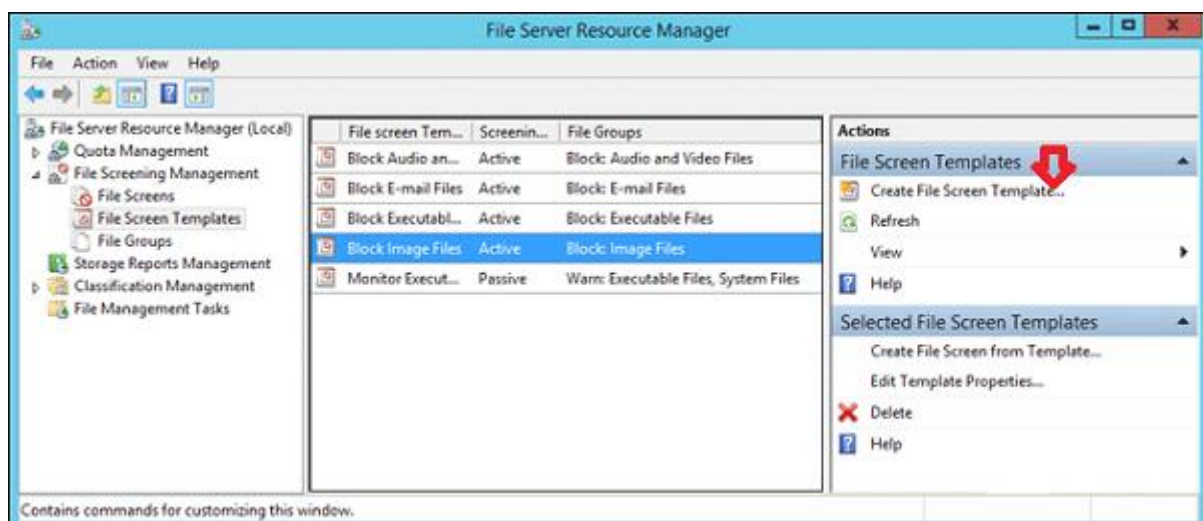
6. Then to attach this quota to a folder – Right Click on the template → click on Create Quota from Template.



7. Click on Browse... and then select your folder → Create.



8. To put a file restriction to your folders, you can go to the left pane File Screening Management → File screening templates → Click on the left panel Create File Screen Template.



- Click on Browse... and find the folder that you want → Create.

Create File Screen

File screen path:

File screen properties

You can either use properties from a file screen template or define custom file screen properties.

How do you want to configure file screen properties?

☒ Derive properties from this file screen template (recommended):

☐ Define custom file screen properties:

Summary of file screen properties:

- File screen:
 - Source template: Block Image Files
 - Screening type: Active
 - File groups: Image Files
 - Notifications: Email, Event log